

Equidistribution and counting of periodic flat tori

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Abstract

Let G be a semisimple Lie group without compact factor and $\Gamma < G$ a torsion-free, cocompact, irreducible lattice. According to Selberg, periodic orbits of regular Weyl chamber flows live on maximal flat periodic tori of the space of Weyl chambers. We prove that these flat periodic tori equidistribute exponentially fast towards the quotient of the Haar measure. From the equidistribution formula, we deduce a higher rank prime geodesic theorem. These counting and equidistribution results also hold in the non cocompact, finite covolume case for $G = \mathrm{SL}(d, \mathbb{R})$ and $\Gamma < \mathrm{SL}(d, \mathbb{Z})$ a finite index subgroup.

1 Introduction

Let G be a semisimple, connected, real linear Lie group without compact factor. Let K be a maximal compact subgroup, A be a maximal $\mathbb{R}$ -split torus, $A^+ \subset A$ a closed positive chamber such that the Cartan decomposition $G = KA^+K$ holds. Denote by $M := Z_K(A)$ the centralizer of A in K , by $\mathfrak{a} := \mathrm{Lie} A$ the Cartan subspace, by $\mathfrak{a}^+$ the closed positive chamber in the Lie algebra and by $\mathfrak{a}^{++}$ its interior.

Let $\Gamma < G$ be a torsion-free, cocompact lattice. The double coset space $\Gamma \backslash G/M$ is called the *space of Weyl chambers* of the symmetric space $\Gamma \backslash G/K$. We count and study the equidistribution of the compact right A -orbits in the space of Weyl chambers.

1.1 Pioneering works on hyperbolic surfaces

In this case, $G = \mathrm{PSL}(2, \mathbb{R})$ is the isometry group of the Poincaré half-plane $\mathbb{H}^2$, the space of Weyl chamber is the unit tangent bundle of the hyperbolic surface $\Gamma \backslash \mathbb{H}^2$ and the right action of A on $\Gamma \backslash G/M$ corresponds to the geodesic flow. Periodic orbits of the geodesic flow projects in the surface to primitive closed geodesics.

Prime geodesic theorems In 1959, Huber [Hub59] proved a prime geodesic theorem for compact hyperbolic surfaces. He obtained an estimate of the number of primitive closed geodesics as their length grows to infinity. More precisely, let $N(T)$ be the number of primitive closed geodesics of length less than T on a hyperbolic surface. He proved that as T tends to infinity,

$$N(T) \sim e^T/T.$$

This term is similar to the asymptotic $x/\log x$ given by the prime number theorem¹ for the number of primes less than x . In 1969, using dynamical methods, Margulis [Mar69] extended the prime geodesic theorem to negatively curved compact manifolds. He proved that the exponential growth rate of $N(T)$ is equal to the topological entropy of the geodesic flow. Later on, relying on Selberg's Trace formula, Hejhal [Hej76] and Randol [Ran77] obtained a complete asymptotic development of the counting function in terms of the spectrum of the Laplace-Beltrami operator. In 1980, Sarnak [Sar80] extended their complete asymptotic development to finite area surfaces.

Let us state one of the various equivalent formulations of the prime geodesic theorem. For a closed geodesic c on $\Gamma \backslash \mathbb{H}^2$, denote by $\ell(c)$ the length of this geodesic. Let c_0 be the primitive closed geodesic underlying c . Then as $T \rightarrow +\infty$

$$\sum_{c_0} \left\lfloor \frac{T}{\ell(c_0)} \right\rfloor \ell(c_0) = \sum_{c, \ell(c) \leq T} \ell(c) \sim C_G \mathrm{vol}(D_t), \quad (1)$$

¹See Pollicott's research statement §1.2 [Pol]

where the first sum is over all primitive closed geodesics, the second sum is over all closed geodesics, C_G is some positive constant, vol is the Haar measure on G induced by the Riemannian metric on $\mathbb{H}^2$ and D_t is the preimage in G of a ball in $\mathbb{H}^2$ of radius T centered in i . This sum is similar to the second Chebyshev function: the weighted sum of the logarithms of primes less than a given number, where the weight is the highest power of the prime that does not exceed the given number. The second Chebyshev function is essentially equivalent to the prime counting function and their asymptotic behavior is the same.

Equidistribution of closed geodesics Margulis in his 1970 thesis² and Bowen [Bow72b], [Bow72a] independently studied the spatial distribution of the closed orbits of the geodesic flow. They proved that they uniformly equidistribute towards a measure of maximal entropy as their period tends to infinity. In the second 1972 paper, Bowen proved the uniqueness of the measure of maximal entropy for the geodesic flow. As a consequence, the measure of maximal entropy of the geodesic flow is equal to the quotient of the Haar measure. Later, Zelditch [Zel92] generalized Bowen's equidistribution theorem to finite area hyperbolic surfaces.

Let us recall Bowen and Margulis' result for a compact hyperbolic surface. For every (primitive) periodic orbit $c \subset \Gamma \backslash \text{PSL}(2, \mathbb{R})$, denote by $\mathcal{P}_c$ the unique probability measure invariant under the geodesic flow supported on c . For every $T > 0$, we denote by $\mathcal{G}_p(T)$ the set of (primitive) periodic orbits of minimal period less than T . Bowen and Margulis proved that for every bounded smooth function f ,

$$\frac{T}{e^T} \sum_{c \in \mathcal{G}_p(T)} \int f \, d\mathcal{P}_c \xrightarrow{T \rightarrow \infty} \int f \, dm_\Gamma,$$

where m_Γ is the measure of maximal entropy, which also corresponds in our case to the quotient measure of the Haar measure on $\Gamma \backslash \text{PSL}(2, \mathbb{R})$.

The following non exhaustive list [DeG77], [GW80], [PP83], [Rob03], [Nau05], [MMO14] provides some of the many subsequent works tackling the counting and equidistribution problem in several different rank one generalisations.

1.2 Main results

In this article, we focus on the higher rank case³ for G , meaning that $\dim_{\mathbb{R}} A \geq 2$.

Definition 1.1 (Maximal flat periodic tori). *For any right A -orbit F in $\Gamma \backslash G/M$, we define the set of periods of F as*

$$\Lambda(F) := \{Y \in \mathfrak{a} \mid ze^Y = z, \forall z \in F\}.$$

A period Y in $\Lambda(F)$ is called regular if $Y \in \mathfrak{a}^{++}$. When $\Lambda(F)$ is a lattice of $\mathfrak{a}$, we say F a maximal flat periodic torus or a compact periodic A -orbit.

Denote by $C(A)$ the set of maximal flat periodic tori in $\Gamma \backslash G/M$. For every $F \in C(A)$, we denote by L_F the quotient measure on F of $\text{Leb}_{\mathfrak{a}}$, the Lebesgue measure on $\mathfrak{a}$. Note that L_F is not a probability measure. Its total mass, denoted by $\text{vol}_{\mathfrak{a}}(F)$, is the Lebesgue measure of any fundamental domain in $\mathfrak{a}$ of the lattice $\Lambda(F)$.

Main counting result Adopting the same notation as above, vol denotes the Haar measure on G whose quotient on the symmetric space $X := G/K$ equals the measure induced by the Riemannian metric. Denote by $\|\cdot\|$ the Euclidean norm on $\mathfrak{a}$ coming from the Killing form on $\mathfrak{g}$ and by $B_{\mathfrak{a}}$ the balls for this norm. For every $T > 0$, set $B_{\mathfrak{a}}^{++}(0, T) := B_{\mathfrak{a}}(0, T) \cap \mathfrak{a}^{++}$ and $D_T := K \exp(B_{\mathfrak{a}}(0, T))K$, which is the preimage by the quotient map $G \rightarrow X$ of the ball of radius T centered at eK in the symmetric space X .

Theorem 1.2. *Let G be a semisimple, connected, real linear Lie group without compact factor and $\Gamma < G$ be a torsion-free, cocompact irreducible lattice or $G = \text{SL}(d, \mathbb{R})$ with $d \geq 2$ and $\Gamma < \text{SL}(d, \mathbb{Z})$ a finite index subgroup. Then there exist constants $C_G > 0$ and $u > 0$ such that for $T > 0$*

$$\sum_{F \in C(A)} |\Lambda(F) \cap B_{\mathfrak{a}}^{++}(0, T)| \, \text{vol}_{\mathfrak{a}}(F) = \text{vol}(D_T)(C_G + O(e^{-uT})). \quad (2)$$

²See Parry's review [Par]

³more precisely, we do not have restrictions on the rank of G

We deduce this counting result from the subsequent equidistribution statement.

Main equidistribution result Denote by $\pi : G \rightarrow \Gamma \backslash G/M$ the projection and by $\tilde{m}_\Gamma$ the quotient measure of the Haar measure vol . We normalise $\tilde{m}_\Gamma$ to obtain a probability measure that we denote by m_Γ . We obtain a higher rank version of the Bowen-Margulis equidistribution formula with an exponential rate of convergence.

Theorem 1.3. *Under the same hypothesis and for the same constants $C_G > 0$ and $u > 0$ as in the previous Theorem 1.2, for all $T > 0$ and every Lipschitz function f on $\Gamma \backslash G/M$ we have*

$$\frac{1}{\text{vol}(D_T)} \sum_{F \in C(A)} |\Lambda(F) \cap B_a^{++}(0, T)| \int_F f \, dL_F = C_G \int_{\Gamma \backslash G/M} f \, dm_\Gamma + O(e^{-uT} |f|_{\text{Lip}}). \quad (3)$$

The asymptotic behavior of the main term as T tends to infinity is $\text{vol}(D_T) \sim C_0 T^{\frac{\dim A - 1}{2}} e^{\delta_0 T}$, where $\delta_0 > 0$ is determined by the root system of $\mathfrak{g}$, the Lie algebra of G and $C_0 > 0$ is given by the Borel–Harish-Chandra formula.

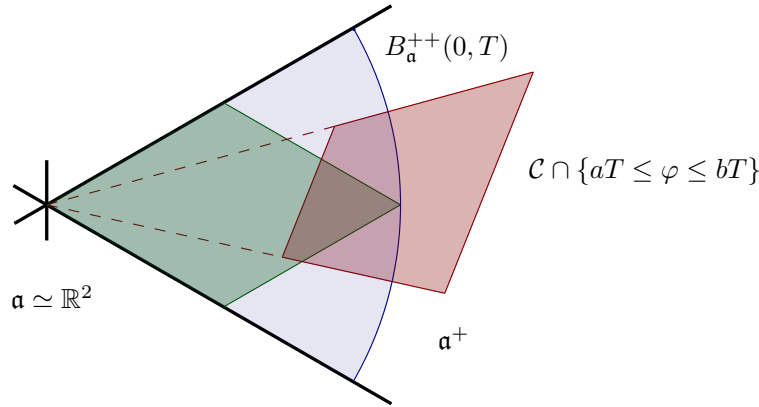


Figure 1: This is a positive Weyl chamber for $\text{SL}(3, \mathbb{R})$ and $T > 0$ is large. In blue, our counting region $B_a^{++}(0, T)$. In green, Deitmar’s [Dei04] counting region. In red, Guedes Bonthonneau–Guillarmou–Weich’s [GBGW21] counting region where $\mathcal{C}$ is a convex cone strictly inside $\mathfrak{a}^{++}$ delimited by the red dashed lines, $0 < a < b$ are real numbers and φ is a linear form strictly positive in $\mathfrak{a}^+$.

Remark 1.4. Note that in the rank one case, any flat periodic torus F corresponds to a primitive closed geodesic. Furthermore, both $\text{vol}_a(F)$ and its smallest regular period correspond to the length of the geodesic. Therefore Theorem 1.2 is a higher rank version of the prime geodesic theorem (1).

1. In the compact case, Spatzier in his thesis [Spa83] computed using the root spaces of the Lie algebra of G , the topological entropy of every regular Weyl chamber flows: right action of $\exp(\mathbb{R}Y)$ on $\Gamma \backslash G/M$, where $Y \in \mathfrak{a}^{++}$ is non zero. Furthermore, δ_0 , the exponential growth rate of $\text{vol}(D_T)$, is a sharp upper bound of the topological entropy of regular Weyl chamber flow. He also proved that δ_0 is equal to the exponential growth rate of the sum over maximal flat periodic tori of smallest regular period less than t of $\text{vol}_a(F)$, as t goes to infinity. Knieper [Kni05] studied the equidistribution of periodic orbits of regular Weyl chamber flows in the same setting. He obtained an equidistribution formula towards the measure of maximal entropy of the most chaotic regular Weyl chamber flow, the one whose topological entropy is δ_0 .

1.1 In the finite volume case, Oh [Oh04] proved that the number of maximal flat periodic tori of bounded volume is always finite.

2. In the compact case, Deitmar [Dei04] used a Selberg trace formula and methods from analytical number theory to prove a similar version of Theorem 1.2 with a different summation region in the Weyl chamber, the one in green in Figure 1. He later on generalized this counting result to the non compact finite volume case $\text{SL}(3, \mathbb{Z}) \backslash \text{SL}(3, \mathbb{R})$, in a joint work with Gon and Spilioti in [DSG19].

3. Recently and for the compact case, Guillarmou–Guedes Bonthonneau–Weich [GBGW21, Theorem 2, equation (0.3)] obtained an equidistribution formula. The region where they count the multiplicity of

periodic tori is defined using any convex non degenerate closed cone $\mathcal{C}$ strictly inside $\mathfrak{a}^{++}$, any choice of positive numbers $0 < a < b$ and any linear form φ that takes positive values in $\mathfrak{a}^+$ as shown in red in Figure 1. They take a different approach, relying on the spectral properties of the A -action via their previous study of Ruelle-Taylor resonances with Hilgert [GBGHW20].

4. Because our counting region is different (shown in blue in Figure 1), our first asymptotic term is new in higher rank. It would be interesting to see whether we could generalize our methods to the regions in red and green in order to recover the results of [Dei04], [GBGW21]. None of the above works provides estimates on the speed of convergence.

5. For the non-compact, finite volume case $\mathrm{SL}(3, \mathbb{Z}) \backslash \mathrm{SL}(3, \mathbb{R})$, Einsiedler–Lindenstrauss–Michel–Venkatesh in [ELMV11] use the classification of diagonal invariant measures and subconvexity estimates to deduce an equidistribution result for the following collection of tori. They take sets of flat periodic tori of the same discriminant and prove that the sum of Lebesgue measures on those tori, normalized by the total mass, equidistributes towards the quotient measure of the Haar measure as the discriminant goes to infinity.

6. Using a dictionary between closed geodesic orbits of hyperbolic surfaces and objects coming from number theory, [Sar82] could deduce counting results on class numbers of totally real orders. Later, [DSG19] did the same for $\mathrm{SL}(3, \mathbb{Z})$. It would be interesting to use a dictionary between compact periodic A -orbits and number theory to deduce a number theoretic version of Theorem 1.2.

7. Since the multiplicity term $|\Lambda(F) \cap B_{\mathfrak{a}^{++}}^+(0, T)|$ is at most polynomial and the asymptotic behavior of $\mathrm{vol}(D_T)$ is dominated by the exponential term, we expect that the equidistribution formula holds if we replace this multiplicity term by 1 and sum over the flat periodic tori whose smallest regular period is of length less than T .

1.3 On the proof of the main theorem

The proof of the equidistribution result in the cocompact case follows Roblin’s proof [Rob03] closely, where he proved counting and equidistribution results for some infinite covolume hyperbolic manifolds. We replace all the ingredients from hyperbolic geometry with their higher rank counterparts, such as Hopf coordinates, Patterson-Sullivan measures, the angular distribution of lattice points. One significant difference in higher rank cases is that we need to carefully treat the boundary of the Weyl chamber, while in the hyperbolic case, it is just a point.

For the non-cocompact case of finite index subgroups $\Gamma < \mathrm{SL}(d, \mathbb{Z})$, we first prove the equidistribution (Theorem 1.3) on compact sets of $\Gamma \backslash G/M$. Then we prove the non-escape of mass for compact periodic A -orbits. The critical observation is that there exist two large compact sets $\Omega_T, \Omega'_T \subset \Gamma \backslash G/M$ depending on the parameter T such that for any compact periodic A -orbit F with a regular period of length less than T , the measure of F outside the compact set, $F \cap \Omega_T^c$, is bounded by its measure inside the compact set, $F \cap (\Omega_T \setminus \Omega'_T)$. Equidistribution is known for functions supported on Ω_T . Therefore we bound the mass outside the compact set Ω_T of the measure in Theorem 1.3 (an average of measures on compact periodic A -orbits with a regular period of length less than T) by the Haar measure of $\Omega_T \setminus \Omega'_T \subset (\Omega'_T)^c$, which decays exponentially fast as T goes to infinity due to the choice of Ω'_T .

Organization of the paper

In Section 2, we gather the basic facts and preliminaries about semisimple real Lie groups, the Furstenberg boundary, Hopf coordinates, higher rank Patterson-Sullivan measure, volume estimates and the angular distribution of lattice points.

In Section 3, we prove a lemma about comparing the angular part of an element in G with its contracting and repelling fixed points in the Furstenberg boundary. In Section 4, we relate loxodromic elements and periodic tori.

In Section 5 and 6, we prove Theorem 1.3 for cocompact lattices and for $\Gamma < \mathrm{SL}_d(\mathbb{Z})$ acting on G/M freely, respectively.

In Appendix A, we introduce the language of orbifolds to treat the general case of finite index subgroups of $\mathrm{SL}_d(\mathbb{Z})$.

In Appendix B, we follow the works of Gorodnik-Nevo [GN12a] [GN12b] and explain why their results work in our setting.

Notation. In the paper, given two real functions f and g , we write $f \ll g$ or $f = O(g)$ if there exists a constant $C > 0$ only depending on G, Γ such that $f \leq Cg$. We write $f \asymp g$ if $f \ll g$ and $g \ll f$.

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2 Background

In the whole article, G is a semisimple, connected, real linear Lie group, without compact factor.

Classical references for this section are [Thi07], [GJT98], [Hel01]. One also may refer to the exposition in [DG21].

Let K be a maximal compact subgroup of G . Then $X = G/K$ is a globally symmetric space of non-compact type and $G = \text{Isom}_0(X)$. We fix a base point $o \in X$ such that $K = \text{Stab}_G(o)$. For every $x \in X$, we denote by $K_x := \text{Stab}_G(x)$. Note that for any $h_x \in G$ such that $h_x o = x$, then $K_x = h_x K h_x^{-1}$, independently of the choice of h_x .

Geometric Weyl chambers Denote by $\mathfrak{g}$ (resp. $\mathfrak{k}$) the Lie algebra of G (resp. K) and consider the Cartan decomposition in the Lie algebra $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$. Let $\mathfrak{a} \subset \mathfrak{p}$ be a *Cartan subspace* of $\mathfrak{g}$. Then $A := \exp(\mathfrak{a})$ is a maximal $\mathbb{R}$ -split torus of G . Denote by $M := Z_K(A)$ the centralizer of A in K . The *real rank* of G , denoted by r_G , is equal to $\dim_{\mathbb{R}} \mathfrak{a}$. We say that G is *higher rank* when $r_G \geq 2$.

For any linear form α on $\mathfrak{a}$, set $\mathfrak{g}_\alpha := \{v \in \mathfrak{g} \mid \forall u \in \mathfrak{a}, [u, v] = \alpha(u)v\}$. The *set of restricted roots* is denoted by $\Sigma := \{\alpha \in \mathfrak{a}^* \setminus \{0\} \mid \mathfrak{g}_\alpha \neq \{0\}\}$. The kernel of each restricted root is a hyperplane of $\mathfrak{a}$. The *Weyl chambers* of $\mathfrak{a}$ are the connected components of $\mathfrak{a} \setminus \bigcup_{\alpha \in \Sigma} \ker(\alpha)$. We choose a *positive Weyl chamber* by fixing such a connected component and denote it (resp. its closure) by $\mathfrak{a}^{++}$ (resp. $\mathfrak{a}^+$). In the Lie group, we denote by $A^{++} := \exp(\mathfrak{a}^{++})$ (resp. $A^+ := \exp(\mathfrak{a}^+)$). Denote by $N_K(A)$ the normalizer of A in K . The group $N_K(A)/M$ is the *Weyl group*, denoted by $\mathcal{W}$. The Weyl group also acts on the Lie algebra $\mathfrak{a}$ by the adjoint action, which acts transitively on the set of connected components of $\mathfrak{a} \setminus \bigcup_{\alpha \in \Sigma} \ker(\alpha)$.

A *geometric Weyl chamber* is a subset of X of the form $g.(A^+o)$, where $g \in G$. The *base point* of the geometric Weyl chamber gA^+o is the point $go \in X$. In [DG21, §2], we obtained the following identifications between the space of Weyl chambers and the set of geometric Weyl chambers of X ,

$$G/M \simeq G.(A^+o). \quad (4)$$

Cartan projection

Definition 2.1. For any $g \in G$, we define, by *Cartan decomposition*, a unique element $\underline{a}(g) \in \mathfrak{a}^+$ such that $g \in K \exp(\underline{a}(g))K$. The map $\underline{a} : G \rightarrow \mathfrak{a}^+$ is called the *Cartan projection*.

The Cartan projection allows to define an $\mathfrak{a}^+$ -valued function on $X \times X$, denoted by $d_{\underline{a}}$. For every $x, y \in X$, any choice $h_x, h_y \in G$ such that $h_x o = x$ and $h_y o = y$, we set

$$d_{\underline{a}}(x, y) := \underline{a}(h_x^{-1}h_y).$$

This function does not depend on the choice of h_x and h_y up to right multiplication by K . By [Hel01, Chapter V, Lemma 5.4], we endow $\mathfrak{a}$ with a scalar product coming from the Killing form on $\mathfrak{g}$. We denote by $\|\cdot\|$ the associated norm on $\mathfrak{a}$ and define the G -invariant riemannian distance on X

$$d_X(x, y) := \|d_{\underline{a}}(x, y)\|.$$

The following fact is standard for symmetric spaces of non-compact type.

Fact 2.2. *For every $x, y \in X$, there is a geometric Weyl chamber based on x containing y . If furthermore, $d_{\underline{a}}(x, y) \in \mathfrak{a}^{++}$, such a geometric Weyl chamber is defined by a unique element $h_{xy}M \in G/M$ such that $h_{xy}o = x$ and $h_{xy}e^{d_{\underline{a}}(x, y)}o = y$.*

Proof. Fix $x, y \in X$ and choose $h_x, h_y \in G$ such that $h_x o = x$ and $h_y o = y$. By Cartan decomposition of $h_x^{-1}h_y$ and Definition 2.1, there exists $k, l \in K$ such that

$$h_x^{-1}h_y = ke^{d_{\underline{a}}(x, y)}l^{-1}.$$

Set $h_{xy} := h_x k$. Since K fixes o , we deduce that $h_{xy}o = x$ and $h_{xy}e^{d_{\underline{a}}(x, y)}o = y$.

For all $k', l' \in K$, the elements $h_x k'$ and $h_y l'$ also respectively send o to x and y . Note that we have another Cartan decomposition of $(h_x k')^{-1}h_y l'$ given by $(k')^{-1}k e^{d_{\underline{a}}(x, y)} l^{-1}l'$. Applying the same construction, we still recognize that $h_x k'(k')^{-1}k = h_{xy}$. Hence h_{xy} does not depend on the choice of representatives h_x and h_y , and it depends on the choice of $k, l \in K$ in the Cartan decomposition.

It remains to show that h_{xy} is unique up to right multiplication by elements of M when $d_{\underline{a}}(x, y) \in \mathfrak{a}^{++}$. In this case, the elements $k, l \in K$ given by Cartan decomposition are defined up to right multiplication by elements in M . Hence the fact. $\square$

Jordan projection Denote by Σ^+ the subset of roots which take positive values in the positive Weyl chamber. It allows to define the following nilpotent subalgebras $\mathfrak{n} := \bigoplus_{\alpha \in \Sigma^+} \mathfrak{g}_{\alpha}$ and $\mathfrak{n}^- = \bigoplus_{\alpha \in \Sigma^+} \mathfrak{g}_{-\alpha}$. Denote by $N := \exp(\mathfrak{n})$ and $N^- := \exp(\mathfrak{n}^-)$ two maximal unipotent subgroups of G .

By Jordan decomposition, every element $g \in G$ admits a unique decomposition $g = g_e g_h g_u$ where g_e, g_h and g_u commute and such that g_e (resp. g_h, g_u) is conjugated to an element in K (resp. A^+, N). The element g_e (resp. g_h, g_u) is called the *elliptic part* (resp. *hyperbolic part, unipotent part*) of g .

Definition 2.3. *For any element $g \in G$, there is a unique element $\lambda(g) \in \mathfrak{a}^+$ such that the hyperbolic part g_h is conjugated to $\exp(\lambda(g)) \in A^+$. The map $\lambda : G \rightarrow \mathfrak{a}^+$ is called the Jordan projection.*

Any element $g \in G$ such that $\lambda(g) \in \mathfrak{a}^{++}$ is called loxodromic.

Denote by G^{lox} the set of loxodromic elements of G and for any subset $S \subset G$, denote by $S^{\text{lox}} := S \cap G^{\text{lox}}$.

Equivalently (Cf. §4 [Dan21]), loxodromic elements are conjugated in G to elements in $A^{++}M$.

Asymptotic Weyl chambers Denote by $P := MAN$ and by $\mathcal{F} := G/P$ the *Furstenberg boundary*. We recall the interpretation of $\mathcal{F}$ in terms of asymptotic Weyl chambers.

Following the exposition in [DG21], we introduce the following equivalence relation between geometric Weyl chambers:

$$g_1 A^+ o \sim g_2 A^+ o \iff \sup_{a \in A^{++}} d_X(g_1 a o, g_2 a o) < +\infty.$$

Equivalence classes for this relation are called *asymptotic Weyl chambers*. Denote by η_0 (resp. ζ_0) the asymptotic Weyl chamber of $A^+ o$ (resp. $(A^+)^{-1} o$). The set of asymptotic Weyl chambers identifies with the Furstenberg boundary (see for instance [DG21, Fact 2.5] for a proof),

$$\mathcal{F} \simeq (G.(A^+ o)/\sim) \simeq K/M \simeq K.\eta_0. \quad (5)$$

Remark that $\zeta_0 = k_l \eta_0$ where $k_l \in N_K(A)$ satisfies $k_l A^+ k_l^{-1} = (A^+)^{-1}$. Furthermore, $\text{Stab}_G(\eta_0) = P$ and $\text{Stab}_G(\zeta_0) = MAN^-$.

In the remainder of the article, we identify $G.(A^+ o)/\sim$ with $\mathcal{F}$ and $G.(A^+ o)$ with G/M . We prove that a geometric Weyl chamber is uniquely determined by its base point in X and the asymptotic Weyl chamber it represents.

Fact 2.4. *The following G -equivariant map is a diffeomorphism:*

$$\begin{aligned} G/M &\xrightarrow{\sim} X \times \mathcal{F} \\ gM &\mapsto (go, g\eta_0). \end{aligned}$$

For every $(x, \xi) \in X \times \mathcal{F}$, we denote by $g_{x,\xi}M \in G/M$ the geometric Weyl chamber of base point x asymptotic to ξ .

Proof. Note that $M \in G/M$ corresponds to the geometric Weyl chamber A^+o , of base point o and asymptotic Weyl chamber η_0 . Since $\text{Stab}_G(o) = K$ and $\text{Stab}_K(\eta_0) = M$, we deduce that the map is injective.

Let us prove that the map is surjective. Fix $x \in X$ and $\xi \in \mathcal{F}$. Choose a representative $h_x \in G$ such that $h_x o = x$. By the identification $\mathcal{F} \simeq K/M \simeq K.\eta_0$ in (5) there exists $k_{x,\xi}M \in K/M$, such that $k_{x,\xi}\eta_0 = h_x^{-1}\xi$. Hence, by G -equivariance and since $k_{x,\xi}o = o$ we deduce that $h_x k_{x,\xi}M \in G/M$ maps to (x, ξ) .

By G -equivariance, we only need to prove that the map is a diffeomorphism at M . By Bruhat decomposition (Cf. [Hel01, Chapter IX, Cor. 1.8, Cor. 1.9]), $g = \mathfrak{n} \oplus \mathfrak{m} \oplus \mathfrak{a} \oplus \mathfrak{n}^-$ and $T_{\eta_0}\mathcal{F} = \mathfrak{n}^-$. By Iwasawa decomposition (Cf. [Hel01, Chapter IX, Thm 1.3]) $G = NAK$, we deduce that $T_o X \simeq \mathfrak{n} \oplus \mathfrak{a}$. Hence $T_M G/M = \mathfrak{n} \oplus \mathfrak{a} \oplus \mathfrak{n}^- \simeq T_o X \times T_{\eta_0}\mathcal{F}$. $\square$

Busemann and Iwasawa cocycle For every $\xi \in \mathcal{F}$ and $g \in G$, consider, by Iwasawa decomposition KAN , the unique element $\sigma(g, \xi) \in \mathfrak{a}$, called the *Iwasawa cocycle*, such that if $k_\xi \in K$ satisfies $k_\xi \eta_0 = \xi$, then

$$gk_\xi \in K \exp(\sigma(g, \xi))N. \quad (6)$$

The *cocycle relation* holds (Cf. [BQ16, Lemma 5.29]) i.e. for all $g_1, g_2 \in G$ and $\xi \in \mathcal{F}$, then

$$\sigma(g_1 g_2, \xi) = \sigma(g_1, g_2 \xi) + \sigma(g_2, \xi). \quad (7)$$

Note that restricted to $K \times \mathcal{F}$, the Iwasawa cocycle is the zero function, i.e. for every $k \in K$ and $\xi \in \mathcal{F}$, then $\sigma(k, \xi) = 0$. This motivates the following Definition of the Busemann cocycle for two points of X and an asymptotic Weyl chamber.

Definition 2.5. *For every $x, y \in X$ and $\xi \in \mathcal{F}$, we define the Busemann cocycle by*

$$\beta_\xi(x, y) := \sigma(h_x^{-1}h_y, h_y^{-1}\xi)$$

independently of the choice of $h_x, h_y \in G$ such that $h_x o = x$ and $h_y o = y$.

Remark that for every $x, y \in X$ and $\xi \in \mathcal{F}$, for all $g \in G$ and all $z \in X$,

$$\beta_\xi(x, y) = \beta_{g\xi}(gx, gy) \quad (8)$$

$$\beta_\xi(x, y) = \beta_\xi(x, z) + \beta_\xi(z, y). \quad (9)$$

The first equation is the G -invariance of the formula, whereas the second is due to the cocycle relation of the Iwasawa cocycle.

Transverse points in $\mathcal{F}$ The subset of *ordered transverse pairs* of $\mathcal{F} \times \mathcal{F}$ is defined by the G -orbit

$$\mathcal{F}^{(2)} := \{(g\eta_0, g\xi_0) \mid g \in G\}. \quad (10)$$

We say that $\xi, \eta \in \mathcal{F}$ are *opposite* or *transverse* if $(\xi, \eta) \in \mathcal{F}^{(2)}$.

In terms of asymptotic Weyl chambers, $\xi, \eta \in \mathcal{F}$ are opposite when there exists a geometric Weyl chamber $g.(A^+o)$ asymptotic to ξ such that $g.((A^+)^{-1}o)$ is asymptotic to η . Note that (Cf. §3.2 [Thi07]) we have the following identifications

$$\mathcal{F}^{(2)} \simeq G/AM.$$

Definition 2.6. For every $(\xi, \eta) \in \mathcal{F}^{(2)}$, for any choice $g_{\xi, \eta} \in G$ such that $g_{\xi, \eta}(\eta_0, \zeta_0) = (\xi, \eta)$, we denote by

$$(\xi\eta)_X := g_{\xi, \eta} \cdot (Ao)$$

the associated maximal flat in the symmetric space X .

For every $(x, \xi) \in X \times \mathcal{F}$, we denote by $\xi_x^\perp \in \mathcal{F}$ the unique opposite point to ξ such that $x \in (\xi\xi_x^\perp)_X$. Equivalently, $\xi_x^\perp := g_{x, \xi}\zeta_0$, where $g_{x, \xi}M \in G/M$ corresponds (Cf. Fact 2.4) to the geometric Weyl chamber of base point x asymptotic to ξ .

Remark 2.7. Note that $(\zeta_0)_o^\perp = \eta_0$ and vice-versa.

Hopf coordinates Let $\mathcal{H}$ be the Hopf coordinate map of G/M (Cf. [Thi07, Chapter 8, §8.G.2] or [DG21])

$$\begin{aligned} \mathcal{H} : G/M &\rightarrow \mathcal{F}^{(2)} \times \mathfrak{a} \\ gM &\mapsto (g\eta_0, g\zeta_0, \sigma(g, \eta_0)). \end{aligned}$$

Hopf coordinates are left G -equivariant and right A -equivariant in the following sense:

- (i) the left action of G on G/M reads in those coordinates equivariantly on $\mathcal{F}^{(2)}$ and using the Iwasawa cocycle as follows. For all $h \in G$ and $(\xi, \eta, Y) \in \mathcal{F}^{(2)} \times \mathfrak{a}$,

$$h(\xi, \eta, Y) = (h\xi, h\eta, Y + \sigma(h, \xi)). \quad (11)$$

- (ii) the right action of A on G/M reads for all $(\xi, \eta, Y) \in \mathcal{F}^{(2)} \times \mathfrak{a}$ and $a \in A$ by keeping the first two coordinates constant and translating the last one by $\log(a)$

$$\mathcal{H}(\mathcal{H}^{-1}(\xi, \eta, Y)a) = (\xi, \eta, Y + \log(a)).$$

Using the geometric Weyl chamber interpretation and the Busemann cocycle notations, the Hopf map reads the same as in Roblin's work [Rob03]:

$$\begin{aligned} X \times \mathcal{F} &\longrightarrow \mathcal{F}^{(2)} \times \mathfrak{a} \\ (x, \xi) &\longmapsto (\xi, \xi_x^\perp, \beta_\xi(o, x)). \end{aligned} \quad (12)$$

This translated map is also left G -equivariant in the sense that for every $g \in G$ and every $(x, \xi) \in X \times \mathcal{F}$, using the cocycle relation (8), the element $(gx, g\xi)$ has Hopf coordinates

$$(g\xi, g\xi_x^\perp, \beta_{g\xi}(o, go) + \beta_\xi(o, x)).$$

Note that $\beta_{g\xi}(o, go) = \sigma(g, \xi)$, therefore the notations are consistent.

2.1 The Furstenberg boundary

Representations of a semisimple Lie group Let us first recall a few facts about representations of a semisimple Lie group. Let (V, ρ) be a representation of G into a real vector space of finite dimension. For every real character $\chi : \mathfrak{a} \rightarrow \mathbb{R}$, we denote by

$$V_\chi := \{v \in V \mid \rho(u)v = \chi(u)v, \forall u \in \mathfrak{a}\}$$

the associated vector space. The set of *restricted weights* is the subset

$$\Sigma(\rho) := \{\chi \mid V_\chi \neq \{0\}\}.$$

They are partially ordered using the positive Weyl chamber as follows.

$$(\chi_1 \leq \chi_2) \Leftrightarrow (\chi_1(u) \leq \chi_2(u), \forall u \in \mathfrak{a}^+).$$

When the representation ρ is irreducible, the set of restricted weights admits a maximum, called the *maximal restricted weight*. The irreducible representation ρ is *proximal* when the subspace of the maximal restricted weight is a line.

Restricted weights of the fundamental representations For the adjoint representation, the set of restricted weights coincides with the set of restricted roots Σ . Denote by Σ^+ the set of positive restricted roots and by $\Pi \subset \Sigma^+$ the set of simple roots. Tits ([BQ16, Lemma 6.32]) proved that for every $\alpha \in \Pi$, there exists an irreducible and proximal representation (ρ_α, V^α) of G such that the restricted weights are in

$$\left\{ \chi^\alpha, \chi^\alpha - \alpha, \chi^\alpha - \alpha - \sum_{\beta \in \Pi} n_\beta \beta \mid n_\beta \in \mathbb{Z}_+ \right\}. \quad (13)$$

Furthermore, the maximal weights $(\chi^\alpha)_{\alpha \in \Pi}$ of these representations form a basis of $\mathfrak{a}^*$.

Distances in the projective space For every $\alpha \in \Pi$, we choose a Euclidean norm $\|\cdot\|$ on V^α such that the elements in $\rho_\alpha(A)$ (resp. $\rho_\alpha(K)$) are symmetric (resp. unitary). Note that $\|\rho_\alpha(a)\| = \exp(\chi^\alpha(\log a))$ for all $a \in A^+$. Abusing notation, we denote by $\|\cdot\|$ the induced Euclidean norm on $V^\alpha \wedge V^\alpha$. Remark that for all $a \in A^+$,

$$\|\wedge_2 \rho_\alpha(a)\| = \exp((2\chi^\alpha - \alpha) \log a). \quad (14)$$

We define the distance in the projective space for all $x, y \in \mathbb{P}(V^\alpha)$ as follows,

$$d(x, y) := \frac{\|v_x \wedge v_y\|}{\|v_x\| \cdot \|v_y\|} \quad (15)$$

independently of the choice of $v_x, v_y \in V$ such that $x = \mathbb{R}v_x$ and $y = \mathbb{R}v_y$. For all $x \in \mathbb{P}(V^\alpha)$ and $\varepsilon \in (0, 1]$, denote by $B(x, \varepsilon)$ the ball centered at x of radius ε for this distance.

Denote by $x_+^\alpha \in \mathbb{P}(V^\alpha)$ the projective point corresponding to the eigenspace for the maximal restricted weight χ^α . Since $\rho_\alpha(A)$ are symmetric endomorphisms for the Euclidean norm on V^α , the orthogonal hyperplane to x_+^α is $\rho_\alpha(A)$ -invariant and abusing notations we write

$$(x_+^\alpha)^\perp = \oplus_{\chi \in \Sigma(\rho_\alpha) \setminus \{\chi^\alpha\}} V_\chi^\alpha.$$

For all projective point $y \in \mathbb{P}(V^\alpha)$, we denote by $y^\perp \subset V^\alpha$ the orthogonal hyperplane and by $\varphi_y \in (V^\alpha)^*$ a linear form such that $\ker \varphi_y = y^\perp$. For all $x, y \in \mathbb{P}(V^\alpha)$, we define (independently of the choice of non-zero $v_x \in x$)

$$\delta(y, x) := \frac{|\varphi_y(v_x)|}{\|\varphi_y\| \cdot \|v_x\|}. \quad (16)$$

By properties of the norms and distances on the projective space, the previous function is symmetric and for all $x, y \in \mathbb{P}(V^\alpha)$,

$$\delta(y, x) = \delta(x, y) = d(y^\perp, x) = d(y, x^\perp). \quad (17)$$

Hence $d(x_+^\alpha, (x_+^\alpha)^\perp) = 1$. For all $\varepsilon > 0$, denote by $\mathcal{V}_\varepsilon((x_+^\alpha)^\perp)^\complement := \{y^\alpha \in \mathbb{P}(V^\alpha) \mid \delta(y^\alpha, x_+^\alpha) \geq \varepsilon\}$. We prove the following dynamical lemma.

Lemma 2.8. *Let $\varepsilon > 0$ and $a \in A^+$. Assume there exists $\alpha \in \Pi$ such that $\alpha(\log a) \geq -2\log(\varepsilon)$. Then $\rho_\alpha(a)\mathcal{V}_\varepsilon((x_+^\alpha)^\perp)^\complement \subset B(x_+^\alpha, \varepsilon)$.*

Proof. We use the notations in §14.1 [BQ16]. Let $\alpha \in \Pi$ such that $\alpha(\log a) \geq -2\log(\varepsilon)$. Recall (14) that $\|\wedge_2 \rho_\alpha(a)\| = \exp((2\chi^\alpha - \alpha) \log a)$ and $\|\rho_\alpha(a)\| = \exp(\chi^\alpha(\log a))$. We compute the gap between the first and second eigenvalues of $\rho_\alpha(a)$,

$$\gamma_{1,2}(\rho_\alpha(a)) := \frac{\|\wedge_2 \rho_\alpha(a)\|}{\|\rho_\alpha(a)\|^2} = e^{-\alpha(\log a)}.$$

By assumption, $e^{-\alpha(\log a)} < \varepsilon^2$, hence $\gamma_{1,2}(\rho_\alpha(a)) < \varepsilon^2$. Then we apply Lemma 14.2 (iii) in [BQ16], for every $y \in \mathcal{V}_\varepsilon((x_+^\alpha)^\perp)^\complement$,

$$d(\rho_\alpha(a)y, x_+^\alpha) \delta(x_+^\alpha, y) < \gamma_{1,2}(\rho_\alpha(a)).$$

By definition $\delta(y, x_+^\alpha) \geq \varepsilon$, hence $d(\rho_\alpha(a)y, x_+^\alpha) < \varepsilon$ and we deduce that $\rho_\alpha(a)\mathcal{V}_\varepsilon((x_+^\alpha)^\perp)^\complement \subset B(x_+^\alpha, \varepsilon)$. $\square$

Distances and balls in $\mathcal{F}$ Using the fundamental representations $(\rho_\alpha)_{\alpha \in \Pi}$, Tits (Cf. [BQ16, Lemma 6.32]) also proved that the following map is an embedding:

$$\begin{aligned} \mathcal{F} &\longrightarrow \prod_{\alpha \in \Pi} \mathbb{P}(V^\alpha) \\ \xi = k\eta_0 &\longmapsto (x^\alpha(\xi))_{\alpha \in \Pi} := (\rho_\alpha(k)x_+^\alpha)_{\alpha \in \Pi}. \end{aligned}$$

We thus define the following distance on $\mathcal{F}$ for all $\xi, \eta \in \mathcal{F}$

$$d(\xi, \eta) := \sup_{\alpha \in \Pi} d(x^\alpha(\xi), x^\alpha(\eta)). \quad (18)$$

For all $\xi \in \mathcal{F}$ and $\varepsilon \in (0, 1)$, we denote the balls for this distance by

$$B(\xi, \varepsilon) := \{\eta \in \mathcal{F} \mid d(\xi, \eta) < \varepsilon\}. \quad (19)$$

Fact 2.9. *The distance d is equivalent to the Riemannian distance on $\mathcal{F}$ induced from the embedding on the product space $\prod_{\alpha \in \Pi} \mathbb{P}(V^\alpha)$.*

Proof. Recall that two distances d, d' on a space X are equivalent if $d \asymp d'$ i.e. there exists $C > 1$ such that for all x, y in X , we have

$$\frac{1}{C}d'(x, y) \leq d(x, y) \leq Cd'(x, y).$$

On the projective space $\mathbb{P}(V^\alpha)$, for each point $\mathbb{R}v$, its tangent space is given by $v^\perp$, the hyperplane orthogonal to v with respect to the Euclidean norm and we obtain a Riemannian metric by restricting the Euclidean norm to $v^\perp$. Denote by d_α the induced Riemannian distance on $\mathbb{P}(V^\alpha)$. The distance d_α between two lines is given by their angle in $[0, \pi/2]$. Since the distance d between two lines defined in (15) is the sine of the angle given by d_α , we deduce that $d \asymp d_\alpha$.

Let us now construct a Riemannian distance d_P on the product space $\prod_{\alpha \in \Pi} \mathbb{P}(V^\alpha)$ using the Riemannian metric of the product space. Recall that on any product space $(X \times Y, g)$ where (X, g_1) and (Y, g_2) are endowed with Riemannian metrics g_1 and g_2 , the product Riemannian metric is given for all $(x, y; v, w) \in T_{(x, y)}X \times Y$ where $(x, v) \in T_x X$ and $(y, w) \in T_y Y$, by

$$g(x, y; v, w) = g_1(x, v) + g_2(y, w).$$

The Riemannian distance d associated to this product Riemannian metric g satisfies

$$\max\{d_1, d_2\} \leq d \leq d_1 + d_2.$$

Since for every $\alpha \in \Pi$, the distances d_α and d are equivalent, we deduce that the Riemannian product distance d_P is equivalent to the maximal metric i.e. $d_P \asymp d := \sup_{\alpha \in \Pi} d_\alpha$. Using Tits' embedding of $\mathcal{F}$ in to the product space $\prod_{\alpha \in \Pi} \mathbb{P}(V^\alpha)$, we deduce that the induced metric is non-degenerate on $\mathcal{F}$. Hence, the Riemannian distance on $\mathcal{F}$ induced by d_P is equivalent to the maximal distance d . $\square$

Similarly, noting that $(\zeta_0)_o^\perp = \eta_0$, we set

$$\delta(\xi, \eta) := \inf_{\alpha \in \Pi} d(x^\alpha(\xi), x^\alpha(\eta_o^\perp)) = \inf_{\alpha \in \Pi} d(x^\alpha(\xi), x^\alpha(\eta_o^\perp)^\perp). \quad (20)$$

For all $\xi \in \mathcal{F}$ and $\varepsilon \in (0, 1)$, we denote the balls for δ by

$$\mathcal{V}_\varepsilon(\xi) := \{\eta \in \mathcal{F} \mid \delta(\eta, \xi) < \varepsilon\}. \quad (21)$$

Using the above notations given for the balls in $\mathcal{F}$ for δ and d and their K -invariance, we upgrade the dynamical Lemma 2.8 to elements in G whose Cartan projection is far from the walls of the Weyl chambers.

Lemma 2.10. *For all $g \in G$, choose $k, l \in K$ by Cartan decomposition such that $g = k \exp(\underline{a}(g)) l^{-1}$. Let $\varepsilon > 0$ and assume that $d(\underline{a}(g), \partial \mathfrak{a}^+) \gg -2 \log \varepsilon$, then $g \mathcal{V}_\varepsilon(l \zeta_0)^\mathbb{G} \subset B(k \eta_0, \varepsilon)$.*

Proof. Note that $\alpha(v) \asymp d(v, \ker \alpha)$ for all $v \in \mathfrak{a}^+$. Hence by taking the infimum over $\alpha \in \Pi$, then using that $\inf_{\alpha \in \Pi} d(v, \ker \alpha) = d(v, \cup_{\alpha \in \Pi} \ker \alpha)$ and finally, because $\mathfrak{a}^+$ is a salient cone, $\partial \mathfrak{a}^+ = \mathfrak{a}^+ \cap (\cup_{\alpha \in \Pi} \ker \alpha)$, we deduce that for all $v \in \mathfrak{a}^+$,

$$d(v, \partial \mathfrak{a}^+) \asymp \inf_{\alpha \in \Pi} \alpha(v).$$

Now using the underlying constant in $\asymp$, we may assume that, $\inf_{\alpha \in \Pi} \alpha(\underline{a}(g)) \geq -2 \log \varepsilon$. Apply the dynamical Lemma 2.8 simultaneously for all $\alpha \in \Pi$, using Remark 2.7 that $(\zeta_0)_o^\perp = \eta_0$, we deduce that $e^{\underline{a}(g)} \mathcal{V}_\varepsilon(\zeta_0)_o^\perp \subset B(\eta_0, \varepsilon)$. Finally, we deduce the Lemma by invariance of left K -action on both d and δ . $\square$

Action of G on $\mathcal{F}$ We want to understand how the left action of G on $\mathcal{F}$ distorts the balls for δ and d . Let $C_a > 1$ be a positive constant such that for all $v \in \mathfrak{a}$,

$$\frac{1}{\sqrt{C_a}} \|v\| \leq \sup_{\alpha \in \Pi} |\chi^\alpha(v)| \leq \sqrt{C_a} \|v\|.$$

This constant gives the comparison of the sup-norm induced by the dual basis $(\chi^\alpha)_{\alpha \in \Pi}$ with the Euclidean norm $\|\cdot\|$ on $\mathfrak{a}$.

Lemma 2.11. *There exist $C_0, C_1 > 0$ such that for all g in G and ξ, η in $\mathcal{F}$, we have the following inequalities:*

- (i) $d(g\xi, g\eta) \leq C_1 e^{C_0 d_X(o, go)} d(\xi, \eta)$,
- (ii) $\delta(g\xi, g\eta) \leq C_1 e^{C_0 d_X(o, go)} \delta(\xi, \eta)$,
- (iii) $\|\sigma(g, \xi) - \sigma(g, \eta)\| \leq C_1 e^{C_0 d_X(o, go)} d(\xi, \eta)$,
- (iv) $\|\sigma(g, \xi)\| \leq C_a d_X(o, go)$.

Furthermore, for every $x, y \in X$ and $\xi \in \mathcal{F}$, (iv) is the same as

$$(iv') \quad \|\beta_\xi(x, y)\| \leq C_a d_X(x, y).$$

In particular, for all $x \in X$ we set $C_x := C_1 e^{C_0 d_X(o, x)}$. Then for all $h_x \in G$ such that $h_x o = x$ and all $\xi \in \mathcal{F}$ and every $r \in (0, C_x^{-1})$, the inequalities given by (i) and (ii) imply

- (i') $B(h_x \xi, C_x^{-1} r) \subset h_x B(\xi, r) \subset B(h_x \xi, C_x r)$,
- (ii') $\mathcal{V}_{C_x^{-1} r}(h_x \xi) \subset h_x \mathcal{V}_r(\xi) \subset \mathcal{V}_{C_x r}(h_x \xi)$.

Proof. For each V^α , by (13.1) in [BQ16], we have

$$d(x^\alpha(g\xi), x^\alpha(g\eta)) \leq \|\rho_\alpha(g)\|^2 \|\rho_\alpha(g^{-1})\|^2 d(x^\alpha(\xi), x^\alpha(\eta)).$$

By (18) and $\|\rho_\alpha(g)\| = \|\rho_\alpha \exp(\underline{a}(g))\| = \exp(\chi^\alpha(\underline{a}(g)))$, we obtain the first inequality for $C_0 = 4C_a$.

For (ii), we first prove that $(x^\alpha((g\eta)_o^\perp))^\perp = \rho_\alpha(g) x^\alpha(\eta_o^\perp)^\perp$. There exist $k_1, k \in K$ such that $\eta = k_1 \eta_0$ and $gk_1 = kan \in KAN$. Then due to k preserving o and the Euclidean metric on V^α , we obtain

$$(x^\alpha((g\eta)_o^\perp))^\perp = (x^\alpha((k\eta_0)_o^\perp))^\perp = \rho_\alpha(k) (x^\alpha((\eta_0)_o^\perp))^\perp.$$

Due to AN preserving $(x^\alpha((\eta_0)_o^\perp))^\perp = (x^\alpha(\zeta_0))^\perp$, we deduce that $\rho_\alpha(k) (x^\alpha((\eta_0)_o^\perp))^\perp = \rho_\alpha(gk_1) (x^\alpha((\eta_0)_o^\perp))^\perp$. Therefore, we obtain $(x^\alpha((g\eta)_o^\perp))^\perp = \rho_\alpha(g) (x^\alpha(\eta_o^\perp))^\perp$. Then for all $\xi, \eta \in \mathcal{F}$,

$$\begin{aligned} \delta(x^\alpha(g\xi), x^\alpha((g\eta)_o^\perp)) &= d(x^\alpha(g\xi), x^\alpha((g\eta)_o^\perp))^\perp = d(\rho_\alpha(g) x^\alpha(\xi)^\perp, \rho_\alpha(g) x^\alpha(\eta_o^\perp)^\perp) \\ &\leq \|\rho_\alpha(g)\|^2 \|\rho_\alpha(g)^{-1}\|^2 d(x^\alpha(\xi), x^\alpha(\eta_o^\perp)^\perp). \end{aligned}$$

Therefore, since $\|\rho_\alpha(g)\| \|\rho_\alpha(g)^{-1}\| \leq \exp(2 \sup(\chi^\alpha(\underline{a}(g)), \chi^\alpha(\underline{a}(g))))$ and $C_0 = 4C_a$, we deduce that

$$\delta(g\xi, g\eta) = \inf_{\alpha \in \Pi} \delta(x^\alpha(g\xi), x^\alpha((g\eta)_o^\perp)) \leq C_1 e^{C_0 \|\underline{a}(g)\|} \delta(\xi, \eta).$$

(iii) is given in [BQ16, Lemma 13.1].

(iv), see [DG21, Lemma 3.12] for a similar statement, and it is also a direct consequence of [BQ16, Lemma 6.33 (ii), Corollary 8.20].

Finally (iv') is a consequence of the formulas $\beta_\xi(x, y) = \sigma(h_x^{-1} h_y, h_y^{-1} \xi)$ and $d_X(x, y) = \|\underline{a}(h_x^{-1} h_y)\|$ independently of the choice of $h_x, h_y \in G$ such that $h_x o = x$ and $h_y o = y$. $\square$

2.2 Disintegration of the Haar measure

Patterson–Sullivan measures were generalized to the higher rank setting in [Alb99], [Qui02]. We follow Thirion’s [Thi07, Chapter 9 §9.e] construction of higher rank Patterson–Sullivan measures on the space of Weyl chambers for $\mathrm{SL}(d, \mathbb{R})$, which also works in our more general setting.

We start by his so-called Patterson densities. For $x \in X$, let K_x be the stabilizer group of x in G . Let μ_x be the unique K_x invariant probability measure on the Furstenberg boundary $\mathcal{F}$. Then we have for $g \in G$ and $x \in X$

$$g_*\mu_x = \mu_{gx}, \quad (22)$$

where $g_*\mu_x$ is the pushforward of μ_x under the g action. This relation holds because the stabilizer of $g_*\mu_x$ is given by $gK_xg^{-1} = K_{gx}$. Let $\rho = \frac{1}{2} \sum_{\alpha \in \Sigma} \alpha$ be the half of the sum of positive roots with multiplicities. By [Qui02, Lemma 6.3] or [Hel00, I 5.1], we have for g in G

$$\frac{dg_*\mu_o}{d\mu_o}(\xi) = e^{-\rho\sigma(g^{-1}, \xi)}, \quad (23)$$

which is a G quasi-invariant measure. Then we will introduce the Gromov product to obtain a G -invariant measure on $\mathcal{F}^{(2)}$.

Definition 2.12. For a pair $(\xi, \eta) \in \mathcal{F}^{(2)}$, we associate it with the unique element in the Lie algebra $\mathfrak{a}$ such that for all weights χ^α

$$\chi^\alpha(\xi|\eta)_o := -\log \delta(x^\alpha(\xi), x^\alpha(\eta_o^\perp)) = -\log \frac{|\varphi(v)|}{\|\varphi\| \|v\|},$$

where $v \in V^\alpha - \{0\}$ is a representative of $x^\alpha(\xi)$ and φ is a non zero linear form such that $\ker \varphi = x^\alpha(\eta_o^\perp)^\perp$.

Since the δ function (17) takes value in $(0, 1]$, then $\chi^\alpha(\xi|\eta)_o \in [0, +\infty)$. This definition already appears in [BPS19, Section 8.10], [Sam15, Section 4] for semisimple Lie groups and [Thi07] for $\mathrm{SL}_d(\mathbb{R})$. Our definition of δ seems different from the one in [BPS19], [Sam15]. By using the correspondence between linear forms and hyperplanes for Euclidean spaces, we can verify that they are the same. An important property is that [Sam15, Lemma 4.12]: for all $g \in G$ and $(\xi, \eta) \in \mathcal{F}^{(2)}$, we have

$$(g\xi|g\eta)_o - (\xi|\eta)_o = \iota\sigma(g, \xi) + \sigma(g, \eta), \quad (24)$$

where ι is the inverse involution on $\mathfrak{a}$. We also define the Gromov product at other points x in X by G -invariance, by setting

$$(\xi|\eta)_x = (h_x^{-1}\xi|h_x^{-1}\eta)_o,$$

where h_x is some element such that $h_x o = x$. Since by (24), the Gromov product at o is left K -invariant, this definition is independent of the choice of h_x . For all $x \in X$ and $(\xi, \eta) \in \mathcal{F}^{(2)}$, we define the $(0, 1]$ -valued function

$$f_x(\xi, \eta) = \exp(-\rho(\xi|\eta)_x).$$

We define measures ν_x on $\mathcal{F}^{(2)}$ by

$$d\nu_x(\xi, \eta) = \frac{d\mu_x(\xi)d\mu_x(\eta)}{f_x(\xi, \eta)}. \quad (25)$$

Proposition 2.13. For all $x \in X$, the measure ν_x is G -invariant and equal to ν_o . We denote it by ν .

In the hyperbolic case, the measures μ_x are called Patterson–Sullivan and $\nu \otimes \mathrm{Leb}_{\mathbb{R}}$ is the Bowen–Margulis–Sullivan measure. In the $\mathrm{SL}_d(\mathbb{R})$ case, Thirion [Thi07] gave a construction of this measure and proved those properties. We include a proof for completeness.

Proof. By (24), for all $x \in X$, all $(\xi, \eta) \in \mathcal{F}^{(2)}$ and every $h_x \in G$ such that $h_x o = x$

$$f_x(\xi, \eta) = f_o(h_x^{-1}\xi, h_x^{-1}\eta) = f_o(\xi, \eta) \exp(-\rho(\iota\sigma(h_x^{-1}, \xi) + \sigma(h_x^{-1}, \eta)))$$

On the other hand,

$$\frac{d\mu_x}{d\mu_o}(\xi) = \frac{d(h_x)_*\mu_o}{d\mu_o}(\xi) = e^{-\rho\sigma(h_x^{-1}, \xi)}.$$

We obtain the same formula for η . Combing the above two equations and using that $\rho\sigma(h_x^{-1}, \xi) = \rho\sigma(h_x^{-1}, \xi)$, we obtain that

$$\nu_x = \nu_o.$$

By definition of the Gromov product, we have for all $g \in G$

$$f_x(g\xi, g\eta) = f_{g^{-1}x}(\xi, \eta).$$

By equation (22) and using that $\nu_{g^{-1}x} = \nu_x$,

$$d\nu_x(g\xi, g\eta) = \frac{d\mu_x(g\xi)d\mu_x(g\eta)}{f_x(g\xi, g\eta)} = \frac{d\mu_{g^{-1}x}(\xi)d\mu_{g^{-1}x}(\eta)}{f_{g^{-1}x}(\xi, \eta)} = d\nu_{g^{-1}x}(\xi, \eta) = d\nu_x(\xi, \eta).$$

Hence ν_x is G -invariant. $\square$

Bochi, Potrie and Sambarino proved that the Gromov product $(\xi|\eta)_o$ in norm is almost the same as the distance between o and the maximal flat $(\xi\eta)_X \subset X$.

Lemma 2.14. [BPS19, Proposition 8.12] *There exist $C_3 > 1, C' > 0$ such that for any $(\xi, \eta) \in \mathcal{F}^{(2)}$, we have*

$$\frac{1}{C_3} \|(\xi|\eta)_o\| \leq d_X(o, (\xi\eta)_X) \leq C_3 \|(\xi|\eta)_o\| + C'.$$

By G -invariance, we deduce that for every $x \in X$ and $(\xi, \eta) \in \mathcal{F}^{(2)}$

$$\frac{1}{C_3} \|(\xi|\eta)_x\| \leq d_X(x, (\xi\eta)_X) \leq C_3 \|(\xi|\eta)_x\| + C'.$$

With this G -invariant measure ν on $\mathcal{F}^{(2)}$, now we can disintegrate the Haar measure on G/M along Hopf coordinates.

Proposition 2.15. *The product measure $\nu \otimes \text{Leb}$ on $\mathcal{F}^{(2)} \times \mathfrak{a}$ is a disintegration in Hopf coordinates of a Haar measure on G/M .*

Proof. The product measure $\nu \otimes \text{Leb}$ is G -invariant by Proposition 2.13 and the Hopf coordinates. So it is a Haar measure on G/M . $\square$

2.3 Cartan regular isometries

Recall that by Cartan decomposition, for every element $g \in G$ there exist $k, l \in K$ and a unique element $\underline{a}(g) \in \mathfrak{a}^+$ such that $g = k \exp(\underline{a}(g)) l^{-1}$. Note that k and l are defined up to right multiplication by elements in $Z_K(\exp(\underline{a}(g)))$.

Definition 2.16. *For all $x \in X$, we denote by $\underline{a}_x : G \rightarrow \mathfrak{a}^+$ the map that assigns to every $g \in G$ the $\mathfrak{a}^+$ -distance between x and gx , i.e. $\underline{a}_x(g) := \underline{d}_x(x, gx)$. We say that g is x -cartan regular if $\underline{a}_x(g) \in \mathfrak{a}^{++}$.*

Let g be an x -cartan regular element, consider $h, h' \in G$ such that $ho = h'o = x$ with $he^{\underline{a}_x(g)}o = gx$ and $h'e^{\underline{a}_x(g^{-1})}o = g^{-1}x$. We set $g_x^+ := h\eta_0$ and $g_x^- := h'\eta_0$. In particular, when $x = o$, we can take $h = k$ and $h' = lk_l$.

Note that every $g \in G$ we have $\underline{a}_x(g) = \underline{a}(h_x^{-1}gh_x)$, independently of the choice of $h_x \in G$ such that $h_x o = x$. Furthermore, provided that g is x -cartan regular,

$$g_x^\pm = h_x(h_x^{-1}gh_x)_o^\pm. \quad (26)$$

Remark that $(x, g_x^+) \in X \times \mathcal{F}$ (resp. (x, g_x^-)) is the unique geometric Weyl chamber based on x containing gx (resp. $g^{-1}x$). In the $\text{PSL}(2, \mathbb{R})$ case, an element g is x -cartan regular when $gx \neq x$, then $g_x^+ \in \partial \mathbb{H}^2$ (resp. g_x^-) is the asymptotic endpoint of the half geodesic based on x going through gx (resp. $g^{-1}x$).

Lemma 2.17. *For all $g \in G$, every $x, y \in X$, the following bound holds:*

$$\|\underline{a}_x(g) - \underline{a}_y(g)\| \leq 2d_X(x, y).$$

Proof. By [Kas08, Lemma 2.3] in our choice of notations: for all $h, h' \in G$ we have the following inequalities, $\|\underline{a}(hh') - \underline{a}(h)\| \leq \|\underline{a}(h')\|$ and $\|\underline{a}(h'h) - \underline{a}(h)\| \leq \|\underline{a}(h')\|$. Let $x, y \in X$ and choose $h_x, h_y \in G$ such that $h_x o = x$ and $h_y o = y$. We compare the Cartan projection of $h = h_y^{-1} g h_y$ to the Cartan projection of its conjugate by $h' = h_y^{-1} h_x$, using that $\|\underline{a}(h')\| = \|\underline{a}(h'^{-1})\|$ we get

$$\|\underline{a}_x(g) - \underline{a}_y(g)\| \leq 2\|\underline{a}(h_y^{-1} h_x)\|.$$

Since $\|\underline{a}(h_y^{-1} h_x)\| = d_X(x, y)$, we deduce the Lemma. $\square$

2.4 Volume growths and decay

We introduce here some subsets on G . They will be used to obtain the main term and the exponentially decaying error term in our main Theorems 1.2, 1.3.

For $t > 1$, let

$$D_t := K \exp(B_{\mathfrak{a}}(0, t)) K,$$

and its subset of Cartan-regular elements

$$D_t^{reg} := K \exp(B_{\mathfrak{a}}(0, t) \cap \mathfrak{a}^{++}) K.$$

For $0 < s < t$, let

$$D_t^s := \{g \in D_t \mid \underline{a}(g) \in \overline{B_{\mathfrak{a}}(\partial \mathfrak{a}^+, s)}\}$$

be the set of elements in D_t whose Cartan projection have distance at most s to the boundary of the Weyl chamber.

For all $x \in X$, we define similar sets

$$\begin{aligned} D_t(x) &:= h_x D_t h_x^{-1}, \\ D_t^{reg}(x) &:= h_x D_t^{reg} h_x^{-1}, \\ D_t^s(x) &:= h_x D_t^s h_x^{-1}. \end{aligned}$$

These sets are independent of the choice of h_x .

For a subset S of G , its volume is defined as its Haar measure $m_G(S)$. Recall volume estimates from [Kni97], [Hel00, Thm 5.8], [GOS09, Thm 6.1]. There exist $C_0 > 0$ and $\delta_0 > 0$ such that as $t \rightarrow \infty$, we have

$$\text{vol}(D_t) \sim C_0 t^{\frac{\dim A - 1}{2}} e^{\delta_0 t}, \quad (27)$$

where $\delta_0 := 2 \max_{Y \in B_{\mathfrak{a}}(0, 1)} \rho(Y)$ and ρ is equal to the half of the sum of positive roots with multiplicities.

Lemma 2.18 (Prop. 7.1 [GN10]). *The function $t \mapsto \log \text{vol}(D_t)$ is uniformly locally Lipschitz for $t > 1$.*

This means that there exists $C > 0$ such that for all $0 < \epsilon < 1$, we have

$$\text{vol}(D_{t+\epsilon}) \leq e^{C\epsilon} \text{vol}(D_t).$$

Lemma 2.19. *There exists $\epsilon_G > 0$ such that for every $0 < \epsilon < \epsilon_G$, there exists $\kappa(\epsilon) > 0$ such that for $t > 1$*

$$\frac{\text{vol}(D_t^{\epsilon t})}{\text{vol}(D_t)} = O(\text{vol}(D_t)^{-\kappa(\epsilon)}). \quad (28)$$

Proof. The proof is similar to Lemma 9.2 and 9.4 in [GW07]. Let $\mathfrak{a}^+(s, t) = \{v \in \mathfrak{a}^+ \cap B_{\mathfrak{a}}(0, t), d(v, \partial \mathfrak{a}^+) \leq s\}$. Then by Harish-Chandra's formula (see [Hel00, Chapter I Theorem 5.8]), we have

$$\text{vol}(D_t^s) = \int_{\mathfrak{a}^+(s, t)} \xi(v) dv,$$

where

$$\xi(v) = \prod_{\alpha \in \Sigma^+} \sinh(\alpha(v))^{m_{\alpha}} \ll e^{2\rho(v)},$$

and $2\rho = \sum_{\alpha \in \Sigma^+} m_\alpha \alpha$ where $m_\alpha = \dim \mathfrak{g}_\alpha$. By Lemma 9.2 in [GW07], if ϵ smaller than some constant ϵ_G , then by the strict convexity of $\mathfrak{a}_1^+$, there exists $\kappa'(\epsilon) > 0$ such that

$$\max_{v \in \mathfrak{a}^+(\epsilon, 1)} 2\rho(v) \leq \delta_0 - \kappa'(\epsilon).$$

So by Harish-Chandra's formula, we have $\text{vol}(D_t^{\epsilon t}) \ll t^{\dim A} e^{(\delta_0 - \kappa'(\epsilon))t}$. Due to the asymptotic of $\text{vol}(D_t)$ (27), the proof is complete. $\square$

Lemma 2.20. *Let Γ be a lattice in G , then for all $t > 1$,*

$$\frac{|\Gamma \cap D_t^{\epsilon t}|}{\text{vol}(D_t)} = O(\text{vol}(D_t)^{-\kappa(\epsilon)}).$$

Proof. Let $\epsilon' > 0$ be a small constant such that the ball centered at e with radius ϵ' satisfies $B(e, \epsilon')^2 \cap \Gamma = \{e\}$. Then we have

$$|\Gamma \cap D_t^{\epsilon t}| \leq \frac{\text{vol}(B(e, \epsilon') D_t^{\epsilon t})}{\text{vol}(B(e, \epsilon'))}.$$

By [Kas08, Lemma 2.3], we have for $h' \in B(e, \epsilon')$ and $h \in D_t^{\epsilon t}$,

$$\|\underline{a}(h'h) - \underline{a}(h)\| \leq \|\underline{a}(h')\| \leq \ell\epsilon',$$

for some $\ell > 0$. Therefore the product set

$$B(e, \epsilon') D_t^{\epsilon t} \subset D_{t+\ell\epsilon'}^{\epsilon t + \ell\epsilon'}.$$

Hence we have

$$|\Gamma \cap D_t^{\epsilon t}| \leq \frac{\text{vol}(D_{t+\ell\epsilon'}^{\epsilon t + \ell\epsilon'})}{\text{vol}(B(e, \epsilon'))},$$

which is $O(\text{vol}(D_t)^{1-\kappa(\epsilon)})$ due to Lemma 2.18 and (2.19). $\square$

As a corollary, we have

Lemma 2.21. *For $0 < \epsilon < \epsilon_G/2$, $t > 1$ and $x \in X$ with $d_X(o, x) < \min\{\frac{\epsilon}{2(1-2\epsilon)}, \frac{\kappa(2\epsilon)}{4(1-\kappa(2\epsilon))}\}t$, we have*

$$\frac{|\Gamma \cap D_t^{\epsilon t}(x)|}{\text{vol}(D_t)} = O(\text{vol}(D_t)^{-\kappa(2\epsilon)/2}).$$

Proof. By Lemma 2.17, we have

$$\|a_o(\gamma) - a_x(\gamma)\| \leq 2d_X(x, o).$$

Therefore by Lemma 2.20 with 2ϵ we obtain

$$|\Gamma \cap D_t^{\epsilon t}(x)| \leq |\Gamma \cap D_{t+2d_X(x,o)}^{\epsilon t + 2d_X(x,o)}| \ll \text{vol}(D_{t+2d_X(x,o)})^{1-\kappa(2\epsilon)},$$

where we use the hypothesis that $\epsilon t + 2d_X(x, o) \leq 2\epsilon(t + 2d_X(x, o))$.

By hypothesis, we have

$$(1 - \kappa(2\epsilon))(t + 2d_X(o, x)) \leq (1 - \kappa(2\epsilon)/2)t.$$

Then by $\text{vol}(D_t) \in [1/C, C]e^{\delta_0 t} t^{\frac{\dim A - 1}{2}}$, we have

$$\text{vol}(D_{t+2d_X(x,o)})^{1-\kappa(2\epsilon)} = O(\text{vol}(D_t)^{1-\kappa(2\epsilon)/2}).$$

The proof is complete. $\square$

2.5 Angular distribution of Lattice points

Theorem 2.22. *Let G be a connected, real linear, semisimple Lie group of non-compact type. Let $\Gamma < G$ be an irreducible lattice. There exist $\kappa > 0$ and $C_4 > 0$. Let $x \in X$ and $(B_t(x))_{t>0}$ be $D_t(x)^{reg}$. Then for all Lipschitz test functions $\psi \in Lip(\mathcal{F} \times \mathcal{F})$, there exists $E(t, \psi, x) = O(Lip(\psi)C_x \text{vol}(D_t)^{-\kappa})$ when $t > C_4 d_X(o, x)$ such that*

$$\frac{1}{\text{vol}(B_t)} \sum_{\gamma \in B_t(x) \cap \Gamma} \psi(\gamma_x^+, \gamma_x^-) = \frac{1}{\text{vol}(\Gamma \backslash G)} \int_{\mathcal{F} \times \mathcal{F}} \psi d\mu_x \otimes \mu_x + E(t, \psi, x).$$

This is due to Gorodnik-Nevo in [GN12a]. We include the proof of this version for Lipschitz functions in the appendix. As a corollary, combined with Lemma 2.21 we have

Lemma 2.23. *There exists $C_5 > 0$ and $C > 0$ such that if $t > C_5 d_X(o, x)$, then*

$$\frac{|\Gamma \cap D_t(x)|}{\text{vol}(D_t)} \leq C.$$

Proof. Due to the definition $C_x = C_1 e^{C_0 d_X(o, x)}$, we know that if $t \gg d_X(o, x)$, then by taking $\psi = 1$ Theorem 2.22 implies that

$$|\Gamma \cap D_t^{reg}(x)| \ll \text{vol}(D_t).$$

For the part $|\Gamma \cap (D_t(x) - D_t^{reg}(x))|$, if $t \gg d_X(o, x)$, then we can use Lemma 2.21 to bound it. Combing these two parts, we obtain the lemma. $\square$

3 A configuration for being loxodromic

Recall Definition 2.3 that the elements in G of Jordan projection in $\mathfrak{a}^{++}$ are called loxodromic. Equivalently, loxodromic elements are conjugated to elements in $A^{++}M$. Let $g \in G^{lox}$ be a loxodromic element, choose $h_g \in G$ such that $h_g^{-1}gh_g \in \exp(\lambda(g))M$. Note that $gh_gM = h_g e^{\lambda(g)}M$. Denote by $g^+ := h_g \eta_0$ (resp. $g^- := h_g \zeta_0$) the attracting (resp. repelling) fixed points in $\mathcal{F}$ for the action of g . They are independent of the choice of h_g . Hence for every $Y \in \mathfrak{a}$, in Hopf coordinates

$$g(g^+, g^-, Y) = (g^+, g^-, Y + \lambda(g)). \quad (29)$$

3.1 Distances on G/M

Denote by d_1 the left G -invariant and right K -invariant Riemannian distance on G/M .

Distance for the Hopf coordinates For every pair $(\xi^+, \xi^-, v), (\eta^+, \eta^-, w) \in \mathcal{F}^{(2)} \times \mathfrak{a}$, we define

$$d_2((\xi^+, \xi^-, v), (\eta^+, \eta^-, w)) := \sup(d(\xi^+, \eta^+), d(\xi^-, \eta^-), \|v - w\|). \quad (30)$$

Due to the Definitions (18), the distance d_2 is not left G -invariant even though it is left K -invariant. Abusing notations, for every $z_1, z_2 \in G$, we also denote by $d_2(z_1M, z_2M) := d_2(\mathcal{H}(z_1M), \mathcal{H}(z_2M))$. For all $(\xi^+, \xi^-, v) \in \mathcal{F}^{(2)} \times \mathfrak{a}$, all $r \in (0, \frac{1}{2}\delta(\xi^+, \xi^-))$, the ball of radius r for d_2 centered in that element is

$$B(\xi^+, r) \times B(\xi^-, r) \times B_{\mathfrak{a}}(v, r).$$

Lemma 3.1. *For $g \in G$ and z_1, z_2 in G , we have*

$$d_2(gz_1M, gz_2M) \leq \sup(C_1 e^{C_0 \|\mathfrak{a}(g)\|}, 1) d_2(z_1M, z_2M).$$

Proof. We write down z_1M and z_2M in Hopf coordinates, we denote by $(\xi_i^+, \xi_i^-, v_i) := \mathcal{H}(z_iM)$ for $i = 1, 2$. By (11) and Lemma 2.11 (i),(ii),(iii), we have

$$\begin{aligned} d_2(gz_1M, gz_2M) &= d_2((g\xi_1^+, g\xi_1^-, v_1 + \sigma(g, \xi_1^+)), (g\xi_2^+, g\xi_2^-, v_2 + \sigma(g, \xi_2^+))) \\ &= \sup(d(g\xi_1^+, g\xi_2^+), d(g\xi_1^-, g\xi_2^-), \|v_1 - v_2 + \sigma(g, \xi_1^+) - \sigma(g, \xi_2^+)\|) \\ &\leq \sup(C_1 e^{C_0 \|\underline{a}(g)\|} d(\xi_1^+, \xi_2^+), C_1 e^{C_0 \|\underline{a}(g)\|} d(\xi_1^-, \xi_2^-), C_1 e^{C_0 \|\underline{a}(g)\|} d(\xi_1^+, \xi_2^+) + \|v_1 - v_2\|) \\ &\leq \sup(C_1 e^{C_0 \|\underline{a}(g)\|}, 1) d_2(z_1M, z_2M). \end{aligned}$$

The proof is complete. $\square$

Local equivalence Denote by $B_1(zM, r) \subset G/M$ the ball of radius r centered on zM , for the distance d_1 .

Lemma 3.2. *There exist a neighbourhood O of eM and $C_2 > 0$ such that for every $z_1, z_2 \in O$,*

$$\frac{1}{C_2} d_2(z_1, z_2) \leq d_1(z_1, z_2) \leq C_2 d_2(z_1, z_2).$$

The main idea is to use the fact that two Riemannian metrics on a manifold are locally equivalent. We have already constructed a Riemannian metric $d_{\mathcal{F}}$ on $\mathcal{F}$ and proved that it is equivalent to the supreme distance d defined in (18).

On the product space $\mathcal{F} \times \mathcal{F} \times \mathfrak{a}$, we have the product distance d_2 from d on $\mathcal{F}$ and $d_{\mathfrak{a}}$ on $\mathfrak{a}$. We also have the product Riemannian distance from $d_{\mathcal{F}}$ on $\mathcal{F}$ and $d_{\mathfrak{a}}$ on $\mathfrak{a}$, which is denoted by d_3 . Due to $d_{\mathcal{F}}$ and d equivalent, d_2 and d_3 are equivalent. Now we can use a lemma about comparing Riemannian distances. We call two distances d, d' locally equivalent if for any $x \in M$, there exists an open set V containing x such that d, d' restricted to V are equivalent.

Lemma 3.3. *Let d and d' be two Riemannian distances on the same open manifold M . Then d and d' is locally equivalent.*

The proof is classic and we skip it here.

Proof of Lemma 3.2. Applying Lemma 3.3 to d_1, d_3 , we obtain Lemma 3.2 by noticing d_2 and d_3 are equivalent. $\square$

We will upgrade Lemma 3.2 to a version with base point eM replaced by any gM . We first obtain an expanding rate estimate of the action of G on G/M with respect to the distance d_2 .

Definition 3.4. *For $x \in X$, let*

$$C_x = 8C_2C_1 \exp(C_0 d_X(o, x)). \quad (31)$$

Fix

$$\epsilon_0 > 0 \quad (32)$$

such that O contains both balls centered at eM of radius ϵ_0 with respect to d_1, d_2 respectively.

Lemma 3.5. *For $x \in X$ and $z_1, z_2 \in G/M$ with $x = \pi(z_1)$, if $d_2(z_1, z_2) < \epsilon_0/C_x$ or $d_1(z_1, z_2) < \epsilon_0$, then*

$$d_1(z_1, z_2) \leq C_x d_2(z_1, z_2)/4.$$

Proof. We take h_x such that $h_x^{-1}z_1 = eM$. Then we have either

$$d_2(h_x^{-1}z_1, h_x^{-1}z_2) \leq C_x d_2(z_1, z_2) < \epsilon_0,$$

(due to Lemma 3.1) or $d_1(h_x^{-1}z_1, h_x^{-1}z_2) = d_1(z_1, z_2) < \epsilon_0$. Due to the choice of ϵ_0 , we can apply Lemma 3.2 and 3.1 to obtain

$$d_1(z_1, z_2) = d_1(h_x^{-1}z_1, h_x^{-1}z_2) \leq C_2 d_2(h_x^{-1}z_1, h_x^{-1}z_2) \leq C_x d_2(z_1, z_2)/4.$$

The proof is complete. $\square$

3.2 Corridors of maximal flats

Recall from Definition 2.6, for every point $y \in X$ and every $\xi \in \mathcal{F}$, we denote by $\xi_y^\perp \in \mathcal{F}$ the opposite element such that $y \in (\xi \xi_y^\perp)_X$.

Definition 3.6. Let $x \in X$ and $r > 0$. We denote by $\mathcal{F}^{(2)}(x, r)$ the open corridor of maximal flats at distance r of x

$$\mathcal{F}^{(2)}(x, r) := \{(\xi, \eta) \in \mathcal{F}^{(2)} \mid d_X(x, (\xi\eta)_X) < r\}. \quad (33)$$

We denote by $\widetilde{\mathcal{F}^{(2)}}(x, r)$ the set of Weyl chambers based in $B_X(x, r)$

$$\widetilde{\mathcal{F}^{(2)}}(x, r) := \left\{ (\xi, \xi_y^\perp, \beta_\xi(o, y)) \in \mathcal{F}^{(2)} \times \mathfrak{a} \mid y \in B_X(x, r) \right\}. \quad (34)$$

By (12), we obtain

Fact 3.7. For all $x \in X$ and $r > 0$, $\widetilde{\mathcal{F}^{(2)}}(x, r)$ is the preimage of $B_X(x, r)$ by the projection $G/M \rightarrow G/K$.

Lemma 3.8. Let $x \in X$ and $\min\{\frac{\epsilon_0}{2}, \frac{\log 2}{C_0}\} > r > 0$. Then for every $\epsilon \in (0, C_x^{-1}r)$, all $(\xi^+, \xi^-) \in \mathcal{F}^{(2)}(x, r)$,

$$B(\xi^+, \epsilon) \times B(\xi^-, \epsilon) \subset \mathcal{F}^{(2)}(x, 2r).$$

Proof. We can find a point $z \in (\xi^+, \xi^-, \mathfrak{a})$ (maximal flat of ξ^+, ξ^-) such that $d_X(\pi(z), x) < r$. For $(\xi, \eta) \in B(\xi^+, \epsilon) \times B(\xi^-, \epsilon)$, we can find $z' \in (\xi, \eta, \mathfrak{a})$ with the same $\mathfrak{a}$ coordinate as z . Then

$$d_2(z, z') = d(\xi, \xi^+) + d(\eta, \xi^-) < 2\epsilon < \epsilon_0/C_x.$$

We can apply Lemma 3.5 to z, z' and we obtain

$$d_X(\pi(z), \pi(z')) \leq d_1(z, z') \leq C_{\pi(z)} d_2(z, z')/4.$$

We have

$$C_{\pi(z)} = 8C_2C_1e^{C_0d_X(o, \pi(z))} \leq C_x e^{C_0r} \leq 2C_x.$$

Therefore

$$d_X(x, \pi(z')) \leq d_X(x, \pi(z)) + d_X(\pi(z), \pi(z')) < r + 2C_x(2\epsilon)/4 \leq 2r.$$

Hence $(\xi, \eta) \in \mathcal{F}^{(2)}(x, 2r)$. □

Lemma 3.9. Let $g \in G$ and $x \in X$. Assume there is a transverse pair $(\xi^+, \xi^-) \in \mathcal{F}^{(2)}$ of fixed points for the action of g on $\mathcal{F}$. Then there exists w in the Weyl group $\mathcal{W}$ such that

$$\|w(\lambda(g)) - \underline{a}_x(g)\| \leq 2d_X(x, (\xi^+\xi^-)_X).$$

Proof. For every transverse pair (ξ^+, ξ^-) , there exists, up to right multiplication by elements of AM , an $h \in G$ such that $h(\eta_0, \zeta_0) = (\xi^+, \xi^-)$.

By assumption, ξ^+ and ξ^- are fixed by g , i.e. $gh \in hAM$. By Cartan decomposition, there exists $w \in \mathcal{W}$ such that for every $p \in hAMo$, we have $\underline{a}_p(g) = w(\lambda(g))$.

Since $hAMo = hAo$, which is equal to the flat $(\xi^+\xi^-)_X$. It then follows from Lemma 2.17 that for every $p \in (\xi^+\xi^-)_X$

$$\|w(\lambda(g)) - \underline{a}_x(g)\| = \|\underline{a}_p(g) - \underline{a}_x(g)\| \leq 2d_X(x, p).$$

Taking the infimum over the points in the flat $(\xi^+\xi^-)_X$ yields the upper bound. □

3.3 The configuration

Recall that for all $x \in X$, we defined the constant $C_x = 8C_2C_1e^{C_0d_X(o,x)}$.

Definition 3.10. Denote by r_0 the unique zero in $(0, 1)$ of the real valued function $r \mapsto -\log r - \max\{C_3, 2\}r$. For all $\varepsilon > 0$ and $x \in X$ we define some function

$$t_0(x, \varepsilon) \gg 2 \log C_x - 2 \log(\varepsilon),$$

where the constant underlying $\gg$ is the same as in Lemma 2.10.

Proposition 3.11. For all $x \in X$ and $r \in (0, r_0)$ and $\varepsilon \in (0, \min\{C_x^{-1}r, \varepsilon_0\})$, every $\gamma \in G$ satisfying the following conditions is loxodromic.

- (i) $\underline{a}_x(\gamma) \in \mathfrak{a}^{++}$ and $d(\underline{a}_x(\gamma), \partial \mathfrak{a}^+) \geq t_0(x, \varepsilon)$,
- (ii) $(\gamma_x^+, \gamma_x^-) \in \mathcal{F}^{(2)}$ are transverse and $d_X(x, (\gamma_x^+ \gamma_x^-)_X) < r$.

Furthermore, its attracting and repelling point satisfy $\gamma^\pm \in B(\gamma_x^\pm, \varepsilon)$.

Proof. There exist $k_{\gamma_x^+}, l_{\gamma_x^-} \in h_x K$ (as h and $h'k_\iota$ in Definition 2.16), defined up to right multiplication by elements of M and independent of the choice of representative $h_x \in G$ such that $\gamma = k_{\gamma_x^+} e^{\underline{a}_x(\gamma)} l_{\gamma_x^-}^{-1}$. Apply Lemma 2.10, to the element $h_x^{-1} \gamma h_x = h_x^{-1} k_{\gamma_x^+} e^{\underline{a}_x(\gamma)} (h_x^{-1} l_{\gamma_x^-})^{-1} \in K A^{++} K$,

$$h_x^{-1} \gamma h_x \mathcal{V}_{C_x^{-1}\varepsilon}(h_x^{-1} \gamma_x^-)^{\mathbb{G}} \subset B(h_x^{-1} \gamma_x^+, C_x^{-1}\varepsilon).$$

We multiply by h_x on the left $\gamma h_x \mathcal{V}_{C_x^{-1}\varepsilon}(h_x^{-1} \gamma_x^-)^{\mathbb{G}} \subset h_x B(h_x^{-1} \gamma_x^+, C_x^{-1}\varepsilon)$. Using the properties of $C_x > 0$ (Lemma 2.11), we deduce the following inclusions

- $h_x B(h_x^{-1} \gamma_x^+, C_x^{-1}\varepsilon) \subset B(\gamma_x^+, \varepsilon)$,
- $\mathcal{V}_\varepsilon(\gamma_x^-)^{\mathbb{G}} \subset h_x \mathcal{V}_{C_x^{-1}\varepsilon}(h_x^{-1} \gamma_x^-)^{\mathbb{G}}$.

Hence $\gamma \mathcal{V}_\varepsilon(\gamma_x^-)^{\mathbb{G}} \subset B(\gamma_x^+, \varepsilon)$. Recall that ι is the opposition involution and $k_\iota \in N_K(A)$ such that $\iota = -Ad(k_\iota)$, then

$$\gamma^{-1} = l_{\gamma_x^-} k_\iota e^{\iota \underline{a}_x(\gamma)} (k_{\gamma_x^+} k_\iota)^{-1}.$$

Since $\iota \underline{a}_x(g)$ is at distance at most t_0 from $\partial \mathfrak{a}^+$ and $(\gamma^{-1})_x^\pm = \gamma_x^\mp$, we deduce that $\gamma^{-1} \mathcal{V}_\varepsilon(\gamma_x^+)^{\mathbb{G}} \subset B(\gamma_x^-, \varepsilon)$.

Due to $d_X(o, ((h_x^{-1} \gamma_x^+)(h_x^{-1} \gamma_x^-))_X) = d_X(x, (\gamma_x^+ \gamma_x^-)_X) < r$, by Lemma 2.14 and Definition 2.12, we obtain

$$\delta(h_x^{-1} \gamma_x^+, h_x^{-1} \gamma_x^-) \geq e^{-C_3 r}.$$

Then by Lemma 2.11, we have

$$\delta(\gamma_x^+, \gamma_x^-) \geq C_x^{-1} \delta(h_x^{-1} \gamma_x^+, h_x^{-1} \gamma_x^-) \geq C_x^{-1} e^{-C_3 r}.$$

Due to the choice of ε, r , we have $C_x^{-1} e^{-C_3 r} > 2\varepsilon$. Hence we have $B(\gamma_x^\pm, \varepsilon) \subset \mathcal{V}_\varepsilon(\gamma_x^\mp)^{\mathbb{G}}$. Then we deduce that γ (resp. γ^{-1}) has an attracting fixed point $\xi^+ \in B(\gamma_x^+, \varepsilon)$ (resp. $\xi^- \in B(\gamma_x^-, \varepsilon)$).

Since γ admits a fixed maximal flat $(\xi^+ \xi^-)_X$, we apply Lemma 3.9,

$$\|w(\lambda(\gamma)) - \underline{a}_x(\gamma)\| \leq 2d_X(x, (\xi^+ \xi^-)_X),$$

for some w in the Weyl group. By hypothesis $\varepsilon < C_x^{-1}r$, Lemma 3.8 implies that $B(\gamma_x^+, \varepsilon) \times B(\gamma_x^-, \varepsilon) \subset \mathcal{F}^{(2)}(x, 2r)$. Hence $w(\lambda(\gamma)) \in B(\underline{a}_x(\gamma), 4r)$. Using that $r < r_0$ and $\varepsilon < C_x^{-1}r$, we get a lower bound $t_0(x, \varepsilon) > -2 \log r > 4r$. We deduce that $B(\underline{a}_x(\gamma), 4r) \subset \mathfrak{a}^{++}$, therefore $w = id$ and γ is loxodromic.

Finally, because the basin of attraction of γ^+ (resp. γ^-) is a dense open set of $\mathcal{F}$, there are points in $B(\gamma_x^+, \varepsilon)$ (resp. $B(\gamma_x^-, \varepsilon)$) that γ (resp. γ^{-1}) will attract to γ^+ (resp. γ^-). Since $\mathcal{F}$ is Hausdorff for d , we deduce that $\gamma^+ = \xi^+$ (resp. $\gamma^- = \xi^-$). $\square$

4 Loxodromic elements and periodic tori

In this part, we give a relation between conjugacy classes of loxodromic elements and periodic tori. We denote in brackets the Γ -conjugacy classes of elements in Γ . Set $\mathcal{G}_{lox}$ the set of Γ -conjugacy classes of loxodromic elements and

$$\mathcal{G}_{lox}(t) := \{[\gamma] \in [\Gamma^{lox}] \mid \lambda(\gamma) \in B_{\mathfrak{a}}(0, t)\}. \quad (35)$$

For every loxodromic element $\gamma \in \Gamma^{lox}$ denote by $\mathcal{L}_\gamma$ the measure of G/M supported on the A -orbit of Hopf coordinates $(\gamma^+, \gamma^-, \mathfrak{a})$ such that its disintegration in Hopf coordinates is given by

$$\mathcal{L}_\gamma := D_{\gamma^+} \otimes D_{\gamma^-} \otimes Leb_{\mathfrak{a}}, \quad (36)$$

where $D_{\gamma^\pm}$ is the Dirac measure at $\gamma^\pm$. Note that the quotient in $\Gamma \backslash G/M$ of the A -orbit $(\gamma^+, \gamma^-, \mathfrak{a})$ only depends on $[\gamma]$. Denote by $F_{[\gamma]}$ the quotient of this A -orbit in $\Gamma \backslash G/M$. By (29), that is $\gamma(\gamma^+, \gamma^-, Y) = (\gamma^+, \gamma^-, Y + \lambda(\gamma))$ for every $Y \in \mathfrak{a}$, we obtain $\lambda(\gamma) \in \Lambda(F_{[\gamma]})$. If we take g_γ an element such that $(\gamma^+, \gamma^-, 0) = g(\eta_0, \zeta_0, 0)$, then the formula also implies $g_\gamma^{-1} \gamma g_\gamma \in \exp(\lambda(\gamma))M$. With this g_γ , the orbit F can be written as $F_{[\gamma]} = \Gamma g_\gamma AM$.

In this subsection, we always suppose that $\Gamma < G$ is a cocompact lattice of G . We have a lemma by Selberg

Lemma 4.1. *Let Γ be a cocompact lattice. Let F be a right A -orbit in $\Gamma \backslash G/M$. If $\Lambda(F) \cap \mathfrak{a}^{++} \neq \emptyset$, then F is a compact periodic A -orbit.*

Proof. We can write $F = \Gamma g AM$. For $Y \in \Lambda(F) \cap \mathfrak{a}^{++}$, by $\Gamma g M = \Gamma g \exp(Y)M$, we know there exists a loxodromic element $\gamma \in \Gamma$ such that $\gamma g = g \exp(Y) m_Y$. By Selberg's lemma in [Sel60] or [PR72], we know that $\Gamma_\gamma \backslash G_\gamma$ is compact with G_γ and Γ_γ the centralizer of γ in G and Γ , respectively. Since γ is loxodromic, so G_γ is a conjugation of a maximal torus. Now $g AM g^{-1}$ commutes with γ , so $G_\gamma = g AM g^{-1}$. Then $\Gamma_\gamma \backslash G_\gamma = (\Gamma \cap G_\gamma) \backslash G_\gamma$ compact implies that $\Gamma g AM = \Gamma G_\gamma g$ is compact in $\Gamma \backslash G$. So F is compact in $\Gamma \backslash G/M$. $\square$

Let $\mathcal{G}(A) := \{(Y, F) \mid F \in C(A), Y \in \Lambda(F) \cap \mathfrak{a}^{++}\}$.

Lemma 4.2. *Let Γ be a cocompact lattice. If the action of Γ on G/M is free, then we have well defined maps*

$$\Psi : \mathcal{G}_{lox} \rightarrow \mathcal{G}(A), [\gamma] \mapsto (\lambda(\gamma), F_{[\gamma]})$$

and

$$\Phi : \mathcal{G}(A) \rightarrow \mathcal{G}_{lox}, (Y, F) \mapsto [\gamma_Y].$$

We also have $\Psi \circ \Phi = Id_{\mathcal{G}(A)}$ and $\Phi \circ \Psi = Id_{\mathcal{G}_{lox}}$.

Proof. For a compact periodic A -orbit F , we can write it as $\Gamma g AM$ with some $g \in G$. For $Y \in \Lambda(F) \cap \mathfrak{a}^{++}$, there exists a $\gamma_Y \in \Gamma$ such that $\gamma_Y g = g \exp(Y) m_Y$ for some $m_Y \in M$. This γ_Y is unique. Otherwise, we have $\gamma'_Y g = g \exp(Y) m'_Y$ with $m'_Y \neq m_Y$, then $\gamma_Y^{-1} \gamma'_Y = g m_Y^{-1} m'_Y g^{-1}$. This element $\gamma_Y^{-1} \gamma'_Y$ in Γ fixes gM in G/M and is not identity, which contradicts that Γ acts on G/M freely.

This g is unique up to left multiplication by Γ and right multiplication by AM . This defines a Γ -conjugacy class $[\gamma_Y]$ in $\mathcal{G}_{lox}$, characterised by $g^{-1} \gamma_Y g \in \exp(Y)M$. So the map Φ is well-defined.

For $[\gamma]$ in $\mathcal{G}_{lox}$, we have already associated it to a unique periodic orbit $F_{[\gamma]}$, that is $F_{[\gamma]} = \Gamma g_\gamma AM$ with g_γ such that $g_\gamma^{-1} \gamma g_\gamma \in \exp(\lambda(\gamma))M$. Due to $\lambda(\gamma) \in F_{[\gamma]}$, by Lemma 4.1, this orbit $F_{[\gamma]}$ is a compact periodic A -orbit.

For $\Psi \circ \Phi$, due to $g^{-1} \gamma_Y g \in \exp(Y)M$, we know that we can take $g_{\gamma_Y} = g$ and then $\Psi \circ \Phi(Y, F) = \Psi([\gamma_Y]) = (Y, F)$.

For $\Phi \circ \Psi$, from $g_\gamma^{-1} \gamma g_\gamma \in \exp(\lambda(\gamma))M$, we know that $\Phi \circ \Psi([\gamma]) = \Phi(\lambda(\gamma), F_{[\gamma]}) = \gamma$. $\square$

5 Equidistribution of flats

For every loxodromic element $\gamma \in \Gamma^{lox}$, denote by L_γ the quotient measure on $\Gamma \backslash G/M$ of $\mathcal{L}_\gamma$ (Cf. (36)). Note that $L_{[\gamma]}$ is supported on $F_{[\gamma]}$ and is equal to the measure $L_{F_{[\gamma]}}$ given in the introduction. It is also given by the following construction: we push on $F_{[\gamma]}$, the restriction of $Leb_{\mathfrak{a}}$ to any fundamental domain in $\mathfrak{a}$ of the periods $\Lambda(F_{[\gamma]})$, by right A -action of the exponential of such a fundamental domain, starting from any base point on $F_{[\gamma]}$. The construction is independent of both the choice of the fundamental domain and the base point on $F_{[\gamma]}$.

By Lemma 4.2, there is a bijection between $\mathcal{G}_{lox}$ and $\mathcal{G}(A)$. By summing over the compact periodic orbits $F \in C(A)$ first, then summing over $Y \in \Lambda(F) \cap B_{\mathfrak{a}}^{++}(0, t)$, we deduce that

$$\frac{1}{\text{vol}(D_t)} \sum_{[\gamma] \in \mathcal{G}_{lox}(t)} L_{[\gamma]} = \frac{1}{\text{vol}(D_t)} \sum_{F \in C(A)} |\Lambda(F) \cap B_{\mathfrak{a}}^{++}(0, t)| L_F, \quad (37)$$

the measure on the right hand side is exactly the measure in the Theorem 1.3. This formula is also a higher rank analogue of the first part of (1). Set

$$\mathcal{M}_\Gamma^t := \frac{\text{vol}(\Gamma \backslash G)}{\text{vol}(D_t)} \sum_{[\gamma] \in \mathcal{G}_{lox}(t)} L_{[\gamma]}.$$

Let $m_{G/M}$ be the Haar measure on G/M , given by $\nu \otimes Leb_{\mathfrak{a}}$ from Proposition 2.15. Let $m_{\Gamma \backslash G/M}$ be the quotient measure on $\Gamma \backslash G/M$. The main theorem 1.3 is equivalent to the following one if Γ is torsion free or if it acts on G/M freely.

Theorem 5.1. *Let $\Gamma < G$ be a cocompact irreducible lattice which acts freely on G/M . Then there exists $u > 0$ such that for any Lipschitz function f on $\Gamma \backslash G/M$, as $t \rightarrow \infty$*

$$\int f \, d\mathcal{M}_\Gamma^t = \int f \, dm_{\Gamma \backslash G/M} + O(e^{-ut} |f|_{Lip}), \quad (38)$$

where the Lipschitz norm is with respect to the Riemannian distance d_1 on $\Gamma \backslash G/M$.

Remark 5.2. The constant C_G equals $\|m_{\Gamma \backslash G/M}\| / \text{vol}(\Gamma \backslash G)$, which comes from the choice of $m_{G/M} = \nu \otimes Leb_{\mathfrak{a}}$ and only depends on G .

We can separate a Lipschitz function as the sum of its positive part and its negative part. So it is sufficient to prove Theorem 5.1 for non negative Lipschitz functions.

We are going to prove Theorem 5.1 in this section. Before starting the argument, we fix the parameters which will be used later. They come from Proposition 3.11. Choose $u_1 > 0$ small than $\min\{\epsilon_G, 1\}/10$, where ϵ_G is the constant from Lemma 2.19. Set

$$\varepsilon := e^{-u_1 t} \text{ and } t_1 := 3u_1 t. \quad (39)$$

Consider the decay rate function $u \mapsto \kappa(u) > 0$ satisfying Lemma 2.19 and the decay coefficient $\kappa > 0$ given in Theorem 2.22. Set

$$u_2 := \frac{1}{2 \dim(G/AM)} \min\{\delta_0 \kappa(6u_1), \delta_0 \kappa, u_1\} \text{ and } r := e^{-u_2 t}. \quad (40)$$

In this part we use Lip_2 to denote Lipschitz norm with respect to the product distance d_2 on G/M or the product distance on $\mathcal{F}^{(2)}$, according to which space the function lives on.

We lift everything to G/M and prove a local version on G/M in Section 5.1 and 5.2. Then in Section 5.3, we use the partition of unity to obtain a global version (Theorem 5.1) on $\Gamma \backslash G/M$.

5.1 Local convergence on corridors

Recall the notation $\underline{a}_x(\gamma) := d_{\underline{a}}(x, \gamma x) = \underline{a}(h_x^{-1} \gamma h_x)$. For every $\gamma \in \Gamma$ such that $\underline{a}_x(\gamma) \in \mathfrak{a}^{++}$, the geometric Weyl chamber based on x containing γx (resp. $\gamma^{-1}x$) determines $\gamma_x^+ \in \mathcal{F}$ (resp. γ_x^-).

For $x \in X$ and $t > 0$, we define the following measures on $\mathcal{F} \times \mathcal{F}$:

$$\nu_{x,1}^t := \frac{\text{vol}(\Gamma \backslash G)}{\text{vol}(D_t)} \sum_{\gamma \in \Gamma \cap D_t^{\text{reg}}(x)} D_{\gamma_x^+} \otimes D_{\gamma_x^-}, \quad (41)$$

$$\nu_{x,2}^t := \frac{\text{vol}(\Gamma \backslash G)}{\text{vol}(D_t)} \sum_{\gamma \in \Gamma^{\text{lox}} \cap D_t^{\text{reg}}(x)} D_{\gamma^+} \otimes D_{\gamma^-}. \quad (42)$$

Recall that $(\mu_x)_{x \in X}$ denotes the Patterson-Sullivan density given in Proposition 2.13 and ν is the associated conformal measure on $\mathcal{F}^{(2)}$. Let $\text{Lip}_c^+(\mathcal{F}^{(2)}(x, r))$ be the space of positive compactly supported Lipschitz functions on $\mathcal{F}^{(2)}(x, r)$.

Lemma 5.3. *Let Γ be an irreducible lattice in G . Fix $x \in X$. Then for every test function $\psi \in \text{Lip}_c^+(\mathcal{F}^{(2)}(x, r))$ for every $t > C_4 d_X(o, x)$, there exists a function $E(t, \psi, x)$ such that*

$$e^{-C_3 r} \int \psi d\nu - E(t, \psi, x) \leq \int \psi d\nu_{x,1}^t \leq \int \psi d\nu + E(t, \psi, x) \quad (43)$$

where $E(x, \psi, t) = O(C_x \text{Lip}(\psi) \text{vol}(D_t)^{-\kappa})$ when $t \rightarrow \infty$.

Proof. By Theorem 2.22, we obtain the main term with the measure $\mu_x \otimes \mu_x$. Since $(\xi, \eta) \in \mathcal{F}^{(2)}(x, r)$, so by Lemma 2.14, we obtain

$$1 \leq f_x(\xi, \eta)^{-1} \leq e^{C_3 r}.$$

Using the relation $d\nu(\xi, \eta) = \frac{d\mu_x(\xi)d\mu_x(\eta)}{f_x(\xi, \eta)}$, we deduce that $\int \psi d\mu_x \otimes \mu_x \leq \int \psi d\nu \leq e^{C_3 r} \int \psi d\mu_x \otimes \mu_x$. Hence the Lemma. $\square$

Lemma 5.4. *Let Γ be a lattice in G . Fix $x \in X$, for every $t \geq \frac{2 \log C_x}{u_1}$, for every test function $\psi \in \text{Lip}_c^+(\mathcal{F}^{(2)}(x, r))$,*

$$\left| \int \psi d\nu_{x,2}^t - \int \psi d\nu_{x,1}^t \right| \leq \varepsilon \text{Lip}_2(\psi) \frac{|\Gamma \cap D_t(x)| \text{vol}(\Gamma \backslash G)}{\text{vol}(D_t)} + 3 \|\psi\|_\infty \frac{|\Gamma \cap D_t^{t_1}(x)| \text{vol}(\Gamma \backslash G)}{\text{vol}(D_t)},$$

where ε and t_1 are given in (39).

Proof. We split the difference between $\frac{\text{vol}(D_t)}{\text{vol}(\Gamma \backslash G)} \int \psi d\nu_{x,1}^t$ and $\frac{\text{vol}(D_t)}{\text{vol}(\Gamma \backslash G)} \int \psi d\nu_{x,2}^t$,

$$\begin{aligned} \sum_{\gamma \in \Gamma \cap D_t^{\text{reg}}(x)} \psi(\gamma_x^+, \gamma_x^-) - \sum_{\gamma \in \Gamma^{\text{lox}} \cap D_t^{\text{reg}}(x)} \psi(\gamma^+, \gamma^-) &= \sum_{\gamma \in \Gamma \cap D_t^{\text{reg}}(x)} \psi(\gamma_x^+, \gamma_x^-) - \sum_{\gamma \in \Gamma^{\text{lox}} \cap D_t^{\text{reg}}(x)} \psi(\gamma_x^+, \gamma_x^-) \\ &\quad + \sum_{\gamma \in \Gamma^{\text{lox}} \cap D_t^{\text{reg}}(x)} \psi(\gamma_x^+, \gamma_x^-) - \psi(\gamma^+, \gamma^-). \end{aligned}$$

For the first term on the right hand side, note that $\Gamma^{\text{lox}} \subset \Gamma$, hence

$$\sum_{\gamma \in \Gamma \cap D_t^{\text{reg}}(x)} \psi(\gamma_x^+, \gamma_x^-) - \sum_{\gamma \in \Gamma^{\text{lox}} \cap D_t^{\text{reg}}(x)} \psi(\gamma_x^+, \gamma_x^-) = \sum_{\gamma \in (\Gamma \setminus \Gamma^{\text{lox}}) \cap D_t^{\text{reg}}(x)} \psi(\gamma_x^+, \gamma_x^-).$$

Note that $t \geq t_1 = 3u_1 t > 0$ since $u_1 \leq 1/10$, hence we have the following inclusion

$$D_t^{\text{reg}}(x) \subset D_t^{t_1}(x) \sqcup (D_t(x) \setminus D_t^{t_1}(x)).$$

Using that $t \geq \frac{2 \log C_x}{u_1}$, we deduce that $t_1 = 3u_1 t \geq t_0 := 2 \log C_x - 2 \log \varepsilon = 2 \log C_x + 2u_1 t$. Apply Proposition 3.11 to every $\gamma \in D_t(x) \setminus D_t^{t_1}(x)$ such that $(\gamma_x^+, \gamma_x^-) \in \mathcal{F}^{(2)}(x, r)$. Any such element is loxodromic i.e. $D_t(x) \setminus D_t^{t_1}(x) \subset G^{\text{lox}}$. Hence $\Gamma \cap (D_t(x) \setminus D_t^{t_1}(x)) \subset \Gamma^{\text{lox}}$ is a set of loxodromic elements. So the non-loxodromic must lie in $(\Gamma \setminus \Gamma^{\text{lox}}) \cap D_t^{\text{reg}}(x) \subset D_t^{t_1}(x)$. We deduce the following upper bound.

$$\left| \sum_{\gamma \in (\Gamma \setminus \Gamma^{\text{lox}}) \cap D_t^{\text{reg}}(x)} \psi(\gamma_x^+, \gamma_x^-) \right| \leq \|\psi\|_\infty |\Gamma \cap D_t^{t_1}(x)|. \quad (44)$$

For the lower term, we split the sum over the partition $\Gamma^{lox} \cap (D_t(x) \setminus D_t^{t_1}(x))$ and $\Gamma^{lox} \cap D_t^{t_1}(x)$.

$$\begin{aligned} \sum_{\gamma \in \Gamma^{lox} \cap D_t^{reg}(x)} \psi(\gamma_x^+, \gamma_x^-) - \psi(\gamma^+, \gamma^-) &= \sum_{\substack{\gamma \in \Gamma^{lox} \\ \gamma \in D_t(x) \setminus D_t^{t_1}(x)}} \psi(\gamma_x^+, \gamma_x^-) - \psi(\gamma^+, \gamma^-) \\ &+ \sum_{\gamma \in \Gamma^{lox} \cap D_t^{t_1}(x) \cap D_t^{reg}(x)} \psi(\gamma_x^+, \gamma_x^-) - \psi(\gamma^+, \gamma^-). \end{aligned}$$

We bound the lower term.

$$\left| \sum_{\gamma \in \Gamma^{lox} \cap D_t^{t_1}(x) \cap D_t^{reg}(x)} \psi(\gamma_x^+, \gamma_x^-) - \psi(\gamma^+, \gamma^-) \right| \leq 2\|\psi\|_\infty |\Gamma \cap D_t^{t_1}(x)|. \quad (45)$$

By Proposition 3.11, the elements $\gamma \in \Gamma \cap (D_t(x) \setminus D_t^{t_1}(x))$ with $(\gamma_x^+, \gamma_x^-) \in \mathcal{F}^{(2)}(x, r)$ are loxodromic and their attractive and repelling points are at distance at most ε of respectively $\gamma_x^\pm$. Using that ψ is Lipschitz and supported on $\mathcal{F}^{(2)}(x, r)$, we bound above the last term.

$$\left| \sum_{\gamma \in \Gamma \cap (D_t(x) \setminus D_t^{t_1}(x))} \psi(\gamma_x^+, \gamma_x^-) - \psi(\gamma^+, \gamma^-) \right| \leq \varepsilon \text{Lip}_2(\psi) |\Gamma \cap D_t(x)|. \quad (46)$$

Finally, we use the triangular inequality, regroup the terms (44), (45) and (46), then multiply everything by $\frac{\text{vol}(\Gamma \setminus G)}{\text{vol}(D_t)}$ to obtain the main upper bound. $\square$

5.2 From corridors to Weyl chambers

Lemma 5.5. *Let $\tilde{\psi} \in \text{Lip}_c^+(\widetilde{\mathcal{F}^{(2)}}(x, r))$ be a compactly supported nonnegative, Lipschitz function and set*

$$\psi := \int_{\mathfrak{a}} \tilde{\psi}(\cdot, \cdot; v) dv.$$

Then $\psi \in \text{Lip}_c^+(\mathcal{F}^{(2)}(x, r))$ and the following norm bounds hold:

- (a) $\text{Lip}_2(\psi) \leq 2(2r)^{\dim \mathfrak{a}} \text{Lip}_2(\tilde{\psi})$.
- (b) $\|\psi\|_\infty \leq (2r)^{\dim \mathfrak{a}} \|\tilde{\psi}\|_\infty$.

For $x \in X$ and $t > 0$, we define the following measure on $\mathcal{F}^{(2)} \times \mathfrak{a}$ by

$$\mathcal{M}_{x,2}^t := \frac{\text{vol}(\Gamma \setminus G)}{\text{vol}(D_t)} \sum_{\gamma \in \Gamma^{lox} \cap D_t^{reg}(x)} \mathcal{L}_\gamma = \nu_{x,2}^t \otimes \text{Leb}_{\mathfrak{a}}. \quad (47)$$

Lemma 5.6. *Let Γ be an irreducible lattice in G . Fix $x \in X$, for every $t \geq \max\{\frac{2 \log C_x}{u_1}, C_4 d_X(o, x)\}$, for every test function $\tilde{\psi} \in \text{Lip}_c^+(\widetilde{\mathcal{F}^{(2)}}(x, r))$,*

$$\begin{aligned} \left| \int \tilde{\psi} d\mathcal{M}_{x,2}^t - \int \tilde{\psi} dm_{G/M} \right| &\leq C_3 r \int \tilde{\psi} dm_{G/M} + \\ &(2r)^{\dim \mathfrak{a}} \left(E(t, \tilde{\psi}, x) + 2\varepsilon \text{Lip}_2(\tilde{\psi}) \frac{|\Gamma \cap D_t(x)| \text{vol}(\Gamma \setminus G)}{\text{vol}(D_t)} + 3\|\tilde{\psi}\|_\infty \frac{|\Gamma \cap D_t^{t_1}(x)| \text{vol}(\Gamma \setminus G)}{\text{vol}(D_t)} \right), \end{aligned}$$

where $E(x, \tilde{\psi}, t) = O(C_x \text{Lip}(\tilde{\psi}) \text{vol}(D_t)^{-\kappa})$ as introduced in Lemma 5.3 and ε, t_1 are given in (39).

Proof. We set $\psi(\xi^+, \xi^-) := \int_{\mathfrak{a}} \tilde{\psi}(\xi^+, \xi^-; v) dv$. Using Fubini's theorem on the $\mathfrak{a}$ coordinate and Proposition 2.15 that $m_{G/M} = \nu \otimes Leb_{\mathfrak{a}}$, we deduce that

$$\int \tilde{\psi} d\mathcal{M}_{x,2}^t = \int \psi d\nu_{x,2}^t \text{ and } \int \tilde{\psi} dm_{G/M} = \int \psi d\nu.$$

We only need to bound $\int \psi d\nu_{x,2}^t - \int \psi d\nu$. By definition of these measures,

$$\int \psi d\nu_{x,2}^t - \int \psi d\nu = \int \psi d\nu_{x,1}^t - \int \psi d\nu + \int \psi d\nu_{x,2}^t - \int \psi d\nu_{x,1}^t.$$

Using Lemma 5.4 on the last term on the right, then Lemma 5.3, the convexity inequality $e^{-r} - 1 \geq -r$ and nonnegativity of ψ to the other term, we deduce the following bound.

$$\begin{aligned} \left| \int \psi d\nu_{x,2}^t - \int \psi d\nu \right| &\leq C_3 r \int \psi d\nu + E(t, \psi, x) \\ &\quad + \varepsilon Lip_2(\psi) \frac{|\Gamma \cap D_t(x)| \text{vol}(\Gamma \backslash G)}{\text{vol}(D_t)} + 3\|\psi\|_{\infty} \frac{|\Gamma \cap D_t^{t_1}(x)| \text{vol}(\Gamma \backslash G)}{\text{vol}(D_t)}. \end{aligned}$$

By Lemma 5.5 (a) (b), the Lipschitz constants and norms between ψ and $\tilde{\psi}$ satisfy $Lip_2(\psi) \leq 2(2r)^{\dim \mathfrak{a}} Lip_2(\tilde{\psi})$ and $\|\psi\|_{\infty} \leq (2r)^{\dim \mathfrak{a}} \|\tilde{\psi}\|_{\infty}$. We deduce the domination $E(t, \psi, x) = (2r)^{\dim \mathfrak{a}} O(Lip_2(\tilde{\psi}) C_x \text{vol}(D_t)^{-\kappa})$ and abusing notation we write

$$E(t, \psi, x) = (2r)^{\dim \mathfrak{a}} E(t, \tilde{\psi}, x).$$

Replacing the Lipschitz constants and norms in the upper bound by abuse of notation on $E(t, \psi, x)$ and lastly applying Fubini on the first term yields

$$\begin{aligned} \left| \int \psi d\nu_{x,2}^t - \int \psi d\nu \right| &\leq C_3 r \int \tilde{\psi} dm_{G/M} + \\ &\quad (2r)^{\dim \mathfrak{a}} \left(E(t, \tilde{\psi}, x) + 2\varepsilon Lip_2(\tilde{\psi}) \frac{|\Gamma \cap D_t(x)| \text{vol}(\Gamma \backslash G)}{\text{vol}(D_t)} + 3\|\tilde{\psi}\|_{\infty} \frac{|\Gamma \cap D_t^{t_1}(x)| \text{vol}(\Gamma \backslash G)}{\text{vol}(D_t)} \right). \end{aligned}$$

□

From now on, to the end of this section, we suppose that Γ is a *cocompact irreducible lattice* in G which acts freely on G/M . The measure in equidistribution is denoted by

$$\mathcal{M}^t := \frac{\text{vol}(\Gamma \backslash G)}{\text{vol}(D_t)} \sum_{\substack{\gamma \in \Gamma^{lox} \\ \|\lambda(\gamma)\| \leq t}} \mathcal{L}_{\gamma}. \quad (48)$$

Lemma 5.7. *There exists $C > 0$. Fix $x \in X$, for every test function $\tilde{\psi} \in Lip_c^+(\tilde{\mathcal{F}}^{(2)}(x, r))$,*

$$(1 - Cr) \int \tilde{\psi} d\mathcal{M}_{x,2}^{t-2r} \leq \int \tilde{\psi} d\mathcal{M}^t \leq (1 + Cr) \int \tilde{\psi} d\mathcal{M}_{x,2}^{t+2r} + \|\tilde{\psi}\|_{\infty} \frac{|\Gamma \cap D_{t+2r}^{2r}(x)| \text{vol}(\Gamma \backslash G)}{\text{vol}(D_t)}. \quad (49)$$

Proof. By Lemma 3.9, for every loxodromic element $g \in G^{lox}$ such that $(g^+, g^-) \in \mathcal{F}^{(2)}(x, r)$ then

$$\|\lambda(g) - \underline{a}_x(g)\| \leq 2r.$$

Hence using triangular inequality we deduce the inclusions

$$\begin{aligned} \Gamma^{lox} \cap D_{t-2r}^{reg}(x) \cap \{\gamma \mid (\gamma^+, \gamma^-) \in \mathcal{F}^{(2)}(x, r)\} &\subset \\ &\subset \{\gamma \in \Gamma^{lox} \mid \|\lambda(\gamma)\| \leq t \text{ and } (\gamma^+, \gamma^-) \in \mathcal{F}^{(2)}(x, r)\} \\ &\subset (\Gamma^{lox} \cap D_{t+2r}^{reg}(x)) \cup (\Gamma \cap D_{t+2r}^{2r}(x)), \end{aligned}$$

here the set $\Gamma \cap D_{t+2r}^{2r}(x)$ is used to contain all the γ in the middle set with $\underline{a}_x(\gamma)$ singular. By integrating $\tilde{\psi}$ over $\mathcal{L}_\gamma$, summing and using that $\tilde{\psi}$ is supported on $\tilde{\mathcal{F}}^{(2)}(x, r)$, we deduce

$$\frac{\text{vol}(D_{t-2r})}{\text{vol}(\Gamma \setminus G)} \int \tilde{\psi} d\mathcal{M}_{x,2}^{t-2r} \leq \frac{\text{vol}(D_t)}{\text{vol}(\Gamma \setminus G)} \int \tilde{\psi} d\mathcal{M}^t \leq \frac{\text{vol}(D_{t+2r})}{\text{vol}(\Gamma \setminus G)} \int \tilde{\psi} d\mathcal{M}_{x,2}^{t+2r} + \|\tilde{\psi}\|_\infty |\Gamma \cap D_{t+2r}^{2r}(x)|. \quad (50)$$

Finally, we multiply by $\frac{\text{vol}(\Gamma \setminus G)}{\text{vol}(D_t)}$, apply the local Lipschitz property of $t \mapsto \log(\text{vol}(D_t))$ (Lemma 2.18). $\square$

5.3 Proof of the equidistribution

Fix a nonnegative test function $\tilde{\psi}_\Gamma \in \text{Lip}_c^+(\Gamma \setminus G/M)$. We want to prove the following convergence and dominate its rate

$$\int \tilde{\psi}_\Gamma d\mathcal{M}_\Gamma^t \xrightarrow{t \rightarrow +\infty} \int \tilde{\psi}_\Gamma dm_{\Gamma \setminus G/M}.$$

For balls $B(z, s)$ with $z \in G/M$ and $s > 0$, they will be balls with respect to the Riemannian distance d_1 .

Lemma 5.8. *Recall ϵ_0 from Lemma 3.5. For $0 < s < \min\{\epsilon_0, (\log 2)/C_0\}$ and any $z \in G/M$ and $x = \pi(z) \in X$, we have*

$$B(z, s) \subset \tilde{\mathcal{F}}^{(2)}(x, s)$$

and for $\tilde{\varphi}$ supported on $B(z, s)$

$$\text{Lip}_2 \tilde{\varphi} \leq C_x \text{Lip} \tilde{\varphi}.$$

Proof. By Lemma 3.7, we have the first part.

By Lemma 3.5, we have for $z_1, z_2 \in B(z, s)$

$$d_1(z_1, z_2) \leq C_{\pi(z_1)} d_2(z_1, z_2)/4.$$

Now due to the definition of C_x , we have $C_{\pi(z_1)} \leq C_{\pi(z)} \exp(C_0 d_X(\pi(z), \pi(z_1))) \leq 2C_{\pi(z)}$. Therefore

$$d_1(z_1, z_2) \leq C_x d_2(z_1, z_2).$$

Then use the definition of Lipschitz norm. $\square$

Partition of unity By applying Vitali's covering lemma to the collection $\{B(y, r/10)\}_{y \in \Gamma \setminus G/M}$, there exists a finite set $\{y_i\}_{i \in I}$ such that $B(y_i, r/10)$ are pairwise disjoint and $\cup_{i \in I} B(y_i, r/2)$ is a covering of $\Gamma \setminus G/M$. By disjointness, we know $|I| \ll r^{-\dim(G/M)}$. Fix a partition of unity of $\frac{1}{r}$ -Lipschitz functions associated to the open cover $\cup_{i \in I} B(y_i, r)$. For the function $\tilde{\psi}_\Gamma$ on $\Gamma \setminus G/M$, we can write it as $\tilde{\psi}_\Gamma = \sum_{i \in I} \tilde{\psi}_{\Gamma, i}$ using the partition of unity. For each y_i , we can find a lift z_i in G/M such that $d(o, z_i)$ is less than the diameter of $\Gamma \setminus G/M$. By Lemma 5.8, we know that for $x_i = \pi(z_i) \in X$

$$B(z_i, r) \subset \tilde{\mathcal{F}}^{(2)}(x_i, r).$$

We can take t large such that $r = e^{-u_2 t}$ is smaller than the injectivity radius of $\Gamma \setminus G/M$. Then the two balls $B(z_i, r)$ and $B(y_i, r)$ are homeomorphic. Let $\tilde{\psi}_i$ be the lift of $\tilde{\psi}_{\Gamma, i}$ on $B(z_i, r)$.

Furthermore, for every $i \in I$, the function $\tilde{\psi}_i$ is Lipschitz and satisfies the following norm bounds:

$$(p1) \quad \text{Lip}_2(\tilde{\psi}_i) \leq C_x \text{Lip} \tilde{\psi}_i \leq C_x (\text{Lip} \tilde{\psi}_\Gamma + \frac{1}{r} \|\tilde{\psi}_\Gamma\|_\infty) \leq \frac{C_x}{r} \|\tilde{\psi}_\Gamma\|_{\text{Lip}},$$

$$(p2) \quad \|\tilde{\psi}_i\|_\infty \leq \|\tilde{\psi}_\Gamma\|_\infty,$$

$$(p3) \quad \sum_{i \in I} \|\tilde{\psi}_i\|_1 \leq \|\tilde{\psi}_\Gamma\|_1,$$

where the first inequality is due to Lemma 5.8.

Local domination For every $i \in I$, due to Lemma 5.6 and 5.7, we have

$$\pm \left(\int \tilde{\psi}_i d\mathcal{M}^t - \int \tilde{\psi}_i dm_{G/M} \right) \leq r(C_3 + C) \int \tilde{\psi}_i dm_{G/M} + \\ (2r)^{\dim a} \left(E(t \pm 2r, \tilde{\psi}_i, x_i) + 2\varepsilon Lip_2(\tilde{\psi}_i) \frac{|\Gamma \cap D_{t \pm 2r}(x_i)| \text{vol}(\Gamma \setminus G)}{\text{vol}(D_{t \pm 2r})} + 4\|\tilde{\psi}_i\|_\infty \frac{|\Gamma \cap D_{t \pm 2r}^{t_1}(x_i)| \text{vol}(\Gamma \setminus G)}{\text{vol}(D_{t \pm 2r})} \right).$$

Let's estimate the error term in the lower part. By Lemma 5.3, (p1) and Lemma 2.18, we have

$$E(t \pm 2r, \tilde{\psi}_i, x_i) = O(C_{x_i} Lip_2(\tilde{\psi}_i) \text{vol}(D_t)^{-\kappa}) = O\left(\frac{C_{x_i}^2}{r} \text{vol}(D_t)^{-\kappa} |\tilde{\psi}_\Gamma|_{Lip}\right).$$

By compactness, the x_i are in a bounded set, therefore the $\{C_{x_i}\}_{i \in I}$ are uniformly bounded. Hence

$$E(t \pm 2r, \tilde{\psi}_i, x_i) = O\left(\frac{\text{vol}(D_t)^{-\kappa}}{r} |\tilde{\psi}_\Gamma|_{Lip}\right). \quad (51)$$

By Lemma 2.23 and (p1), we have

$$2\varepsilon Lip_2(\tilde{\psi}_i) \frac{|\Gamma \cap D_{t \pm 2r}(x_i)| \text{vol}(\Gamma \setminus G)}{\text{vol}(D_{t \pm 2r})} = O\left(\frac{\varepsilon}{r} |\tilde{\psi}_\Gamma|_{Lip}\right). \quad (52)$$

Using that $t_1 = 3u_1 t$, we get by applying Lemma 2.21 and (p2),

$$3\|\tilde{\psi}_i\|_\infty \frac{|\Gamma \cap D_{t \pm 2r}^{t_1}(x_i)| \text{vol}(\Gamma \setminus G)}{\text{vol}(D_{t \pm 2r})} = O(\|\tilde{\psi}_\Gamma\|_\infty \text{vol}(D_t)^{-\kappa(6u_1)}) = O\left(\frac{\text{vol}(D_t)^{-\kappa(6u_1)}}{r} |\tilde{\psi}_\Gamma|_{Lip}\right). \quad (53)$$

Global domination By the partition of unity, we have

$$\int \tilde{\psi}_\Gamma d\mathcal{M}_\Gamma^t = \sum_i \int \tilde{\psi}_{\Gamma,i} d\mathcal{M}_\Gamma^t = \sum_i \int \tilde{\psi}_i d\mathcal{M}^t$$

and

$$\int \tilde{\psi}_\Gamma dm_{\Gamma \setminus G/M} = \sum_i \int \tilde{\psi}_{\Gamma,i} dm_{\Gamma \setminus G/M} = \sum_i \int \tilde{\psi}_i dm_{G/M}.$$

Therefore, by local dominations, $|I| \ll r^{-\dim(G/M)}$ and (51)-(53), we obtain

$$\int \tilde{\psi}_\Gamma d\mathcal{M}_\Gamma^t - \int \tilde{\psi}_\Gamma dm_{\Gamma \setminus G/M} = O\left(r \sum_{i \in I} \|\tilde{\psi}_i\|_1 \right. \\ \left. + r^{-\dim(G/M)} \left(\frac{\text{vol}(D_t)^{-\kappa}}{r} |\tilde{\psi}_\Gamma|_{Lip} + \frac{\varepsilon}{r} |\tilde{\psi}_\Gamma|_{Lip} + \frac{\text{vol}(D_t)^{-\kappa(6u_1)}}{r} |\tilde{\psi}_\Gamma|_{Lip} \right) \right).$$

Using (p3) and $\|\tilde{\psi}_\Gamma\|_1 \leq \|m_{\Gamma \setminus G/M}\| |\tilde{\psi}_\Gamma|_{Lip}$, we deduce that

$$\int \tilde{\psi}_\Gamma d\mathcal{M}_\Gamma^t - \int \tilde{\psi}_\Gamma dm_{\Gamma \setminus G/M} = O\left(\left(r + \frac{\text{vol}(D_t)^{-\kappa} + \varepsilon + \text{vol}(D_t)^{-\kappa(6u_1)}}{r^{\dim(G/M)+1}}\right) |\tilde{\psi}_\Gamma|_{Lip}\right).$$

Recall the choice of parameter in (40) where $\varepsilon = e^{-u_1 t}$ and $r = e^{-u_2 t}$. Collecting all the error terms together, we obtain that there exists $u > 0$ such that

$$\left| \int \tilde{\psi}_\Gamma d\mathcal{M}_\Gamma^t - \int \tilde{\psi}_\Gamma dm_{\Gamma \setminus G/M} \right| = O(e^{-ut} |\tilde{\psi}_\Gamma|_{Lip}).$$

6 Finite index subgroups of $\mathrm{SL}_d(\mathbb{Z})$

In this section, $G = \mathrm{SL}(d, \mathbb{R})$ where $d \geq 2$ and $\Gamma_1 < \Gamma_0 = \mathrm{SL}(d, \mathbb{Z})$ is a finite index subgroup of Γ_0 which acts freely on G/M . We use Γ to denote both Γ_1 and Γ_0 before section 6.2. Starting from Section 6.2, we only use Γ to denote Γ_1 .

Let us start with examples of finite index subgroups of Γ_0 that act freely on G/M . For every prime number $p \geq 3$, we claim that the finite index subgroup $\Gamma_1 := \ker(\Gamma_0 \rightarrow \mathrm{SL}(d, \mathbb{Z}/p\mathbb{Z}))$ acts freely on G/M . Indeed, assume γ_1 fixes an element G/M , then γ_1 is conjugated in G to an element m in the sign group M . Its projection to $\mathrm{SL}(d, \mathbb{Z}/p\mathbb{Z})$ has the same eigenvalues given by the projection of m . Since γ_1 projects to the identity and $p \geq 3$, we deduce that m is trivial.

Torus in linear algebraic groups We recall some concepts from linear algebraic groups. For more details please see [Bor91] and [BH62]. A subgroup T of $\mathrm{GL}(d, \mathbb{C})$ is a *torus*, if T is diagonalizable over $\mathbb{C}$ and isomorphic to $(\mathbb{C}^*)^n$. Let k be a subfield of $\mathbb{C}$. We say that T is a *k-torus* if it is defined over k i.e. if T as an algebraic subvariety of $\mathrm{GL}(d, \mathbb{C})$ is defined by polynomial equations with coefficients in k . Denote by T_k the k -points of T . A k torus T is *k-split* (here we only need $k = \mathbb{Q}$ or $\mathbb{R}$) if T can be diagonalized to $(\mathbb{C}^*)^n$ by a matrix with coefficients in k . Let T be a $\mathbb{Q}$ -torus T , then by [BH62, Thm 9.4, Lem 8.4] the following conditions are equivalent:

- $T_{\mathbb{Z}} \backslash T_{\mathbb{R}}$ is compact.
- T is $\mathbb{Q}$ -anisotropic i.e. all the $\mathbb{Q}$ characters from $T_{\mathbb{Q}}$ to $\mathbb{Q}^*$ are trivial.
- T contains no non-trivial $\mathbb{Q}$ -split subtorus.

Systole of elements in $\Gamma \backslash G$ For $g \in G$, let $s(g)$ be the systole of the lattice $\mathbb{Z}^d g$ in $\mathbb{R}^d$ i.e.

$$s(g) := \min_{v \in \mathbb{Z}^d g \setminus \{0\}} \{\|v\|\}.$$

Note that the systole is preserved by right multiplication by K since the norm on $\mathbb{R}^d$ is Euclidean. Now Γ preserves $\mathbb{Z}^d$ and the right action of the sign group M also preserves any lattice $\mathbb{Z}^d g$ for all $g \in G$. Hence, this definition extends to $\Gamma g M$ in $\Gamma \backslash G/M$. For $R > 0$, let

$$\Omega(R) := \{\Gamma g M \in \Gamma \backslash G/M \mid 1/s(g) \leq R\}.$$

Then the Mahler criteria implies that $\Omega(R)$ is compact. The union of $\Omega(R)$ for $R > 0$ is the full space $\Gamma \backslash G/M$ and $\{\Omega(R), R > 0\}$ is an increasing family of compact sets.

Siegel domains In [BH62, Section 4], Borel and Harish-Chandra define Siegel domains for the KAN decomposition. We take the inverse of groups in their statement.

Let $G = NAK$ be the Iwasawa decomposition, where N is the upper triangular maximal unipotent subgroup.

Definition 6.1. [BH62] For all $s > 0$ and $u > 0$, set $N_s := \{n \in N \mid \|n\| \leq s\}$ and $A_u := \{a \in A \mid a_j/a_{j+1} > u\}$. A Siegel domain is a subset of G of the form $N_s A_u K$, it is a standard Siegel domains if $s > 1/2$ and $0 < u < \sqrt{3}/2$.

In [BH62, Proposition 4.5], they prove that when $N_s A_u K$ is standard, then

$$G = \Gamma_0 N_s A_u K,$$

which in some sense means that a standard Siegel domain is almost a fundamental domain for the left action of Γ_0 on G . Furthermore, for any standard Siegel domain, the number of elements $\gamma \in \Gamma_0$ such that $\gamma N_s A_u K \cap N_s A_u K \neq \emptyset$ is finite.

From now on, we will denote by $n(g), a(g), k(g)$ the N, A, K components of g in the Iwasawa decomposition NAK . Note that $a(g) = \exp(-\sigma(g^{-1}, \eta_0))$.

We give a relation between the systole and the Iwasawa cocycle in Siegel domains.

Lemma 6.2. *For all $0 < u \leq 1$ and $g \in NA_u K$, we have*

$$a_d(g)u^{d-1} \leq s(g) \leq a_d(g).$$

Proof. Using first the definition of the systole, then that the row vector e_d is right N -invariant and finally that the norm on $\mathbb{R}^d$ is K -invariant, we deduce the upper bound of the systole

$$s(g) \leq \|e_d g\| = \|e_d a(g)\| = a_d(g).$$

For the lower bound, it suffices to prove that for every $v \in \mathbb{Z}^d \setminus \{0\}$,

$$a_d(g)u^{d-1} \leq \|vg\|.$$

First by K -invariance, $\|vg\| = \|v n(g)a(g)\|$. Let us write the coefficients of the row vector $v = (v_1, \dots, v_d)$. Assume that the j -th coefficient v_j is the first non-zero coordinate, where $1 \leq j \leq d$. Then $v n(g)a(g)$ is a row vector with all its first $j-1$ coefficients equal to zero and its j -th coefficient is $(v n(g)a(g))_j = v_j n_{jj}(g)a_j(g)$. Using first that $n_{jj}(g) = 1$ and $\|v n(g)a(g)\| \geq |(v n(g)a(g))_j|$, then that $a(g) \in A_u$ and $|v_j|$ is a non-zero positive integer, we deduce that

$$\|vg\| \geq |v_j|a_j(g) \geq a_d(g)u^{d-j}.$$

Finally for every $v \in \mathbb{Z}^d \setminus \{0\}$, then $d-j \leq d-1$ and since $u \in (0, 1)$, we deduce that $a_d(g)u^{d-1} \leq \|vg\|$, hence the lower bound for the systole. $\square$

Injectivity radius We find a lower bound of the injectivity radius in every point of $\Omega(R) \subset \Gamma_1 \backslash G/M$. For every point $z \in \Gamma_1 \backslash G/M$, denote by $\text{inj}(z)$, the injectivity radius with respect to the Riemannian metric d_1 , i.e. the largest radius for which the exponential map at z is a diffeomorphism.

Lemma 6.3. *There exists $C_7 > 1$ such that for all large enough $R > 2$, every $z \in \Omega(R)$,*

$$\text{inj}(z) \geq R^{-C_7}.$$

Furthermore, there exists a representative $h \in G$ such that $z = \Gamma_1 h M$ and

$$d_X(o, ho) \leq C_7 \log R.$$

Proof. We first construct h . Let $z \in \Omega(R)$ and we start with a representative $g \in G$ such that $z = \Gamma_1 g M$. We choose $h_0 \in N_s A_u K$ a representative in the coset $\Gamma_0 g$, where $N_s A_u K$ is a standard Siegel domain (Definition 6.1) with $u \in (0, 1)$. Note that $s(h_0) = s(z) > 1/R$ by hypothesis, then by the above Lemma 6.2, we deduce that $a_d(h_0) \geq 1/R$. Since $a(h_0) \in A_u$, then $a_j(h_0) \geq \frac{u^{d-j}}{R}$ for all $1 \leq j \leq d$. Hence

$$\frac{u^{d-1}}{R} \leq a_1(h_0) = a_2(h_0)^{-1} \dots a_d(h_0)^{-1} \leq R^{d-2} u^{-(d-2)(d-1)/2}$$

from which we deduce that $a_j(h_0) \leq R^{d-2} u^{-(j-1)-(d-2)(d-1)/2}$ for all $1 \leq j \leq d$. Since $N_s A_u K$ is standard, with $u \in (0, 1)$, one can write it as some negative power of R and deduce the following upper bound for $a(h_0)$ that there is a positive constant $C > 0$ such that

$$\|a(h_0^{-1})\|, \|a(h_0)\| \leq R^C.$$

Now since N_s is bounded and for the operator norm $\|\cdot\|$ of the action on row vectors induced by the Euclidean norm on $\mathbb{R}^d$, we deduce that $\|h_0\| = \|n(h_0)a(h_0)k(h_0)\| \ll R^C$, similarly for $\|h_0^{-1}\|$. Since Γ_1 is a finite index subgroup of Γ_0 , there exists a finite set $\{\gamma_j\}_{j \in J}$ such that $\Gamma_0 = \cup_{j \in J} \Gamma_1 \gamma_j$. Therefore there exists γ_j such that $\Gamma_1 g = \Gamma_1 \gamma_j h_0$. We set $h := \gamma_j h_0$ and deduce that

$$\|h^{-1}\|, \|h\| \ll R^C. \tag{54}$$

Let us compute the Cartan projection of h , using [Kas08, Lemma 2.3] and the compactness of N_s and finiteness of $\{\gamma_j\}_{j \in J}$,

$$d_X(o, ho) \ll d_X(o, h_0 o) = \|\underline{a}_o(h_0)\| \ll \|\underline{a}_o(a(h_0))\| \ll \log R.$$

Denote by $|\cdot|_1$ the Riemannian metric at e associated with the Riemannian distance d_1 . We choose $r_0 > 0$ such that for all $Y \in \mathfrak{g}$ of norm smaller than r_0 , the exponential map is a local diffeomorphism, so that we have

$$\|\exp(Y) - e\| \asymp |Y|_1 \asymp d_1(\exp(Y), e).$$

We prove that if the exponential map for the ball of radius $r \in (0, r_0)$ centered at $z = \Gamma_1 h M$ is not injective, then $r \gg R^{-C'}$ for some positive constant C' . Assume there exist $h_1 \neq h_2 \in G$ such that $\Gamma_1 h_1 M = \Gamma_1 h_2 M$ and $h_1 M, h_2 M \in B(h M, r)$. Abusing notations, since d_1 comes from the left G -invariant and right K -invariant Riemannian metric on G , we can assume that $h_1, h_2 \in B(h, r)$. Then there exists $(\gamma, m) \in \Gamma_1 \times M$, with $\gamma \neq e$ such that $\gamma h_1 = h_2 m$ i.e.

$$\gamma = h_2 m h_1^{-1}.$$

Note that because Γ_1 acts freely on G/M , then γ cannot be conjugated to an element in the sign group, therefore $\gamma^2 \neq e$. Since γ^2 is a matrix with integer coefficient, we deduce on one hand the lower bound

$$\|h^{-1} \gamma^2 h - e\| = \|h^{-1} (\gamma^2 - e) h\| \geq \frac{1}{\|h\| \|h^{-1}\|}.$$

On the other hand, set $g_0 := h^{-1} \gamma h - m$, so that $h^{-1} \gamma h = g_0 + m$ and deduce the upper bound

$$\|h^{-1} \gamma^2 h - e\| = \|g_0^2 + g_0 m + m g_0\| \leq \|g_0^2\| + 2\|g_0\|.$$

By triangle inequality, $\|g_0\| = \|h^{-1} h_2 m h_1^{-1} h - m\| \leq \|h^{-1} h_2 - e\| \|h_1^{-1} h\| + \|e - h_1^{-1} h\|$. Now $h_1, h_2 \in B(h, r)$, therefore $\|g_0\| \ll r$ and

$$\frac{1}{\|h\| \|h^{-1}\|} \leq \|g_0\|^2 + 2\|g_0\| \ll r.$$

Finally, by (54), therefore $r \gg R^{-C'}$ for some constant $C' > 0$ and we deduce the lower bound for the injectivity radius at z . $\square$

Action of the Weyl group

Lemma 6.4. *There exists $c > 0$, such that for any $\eta \in \mathcal{F}$, there exists $w \in \mathcal{W}$ such that*

$$\delta(w\eta, \eta_0) > c.$$

Proof. In the $\mathrm{SL}(d, \mathbb{R})$ case, the Furstenberg boundary $\mathcal{F}$ is the space of complete flats of $\mathbb{R}^d$. Therefore, there exists a basis $(v^j)_{1 \leq j \leq d}$ of $\mathbb{R}^d$ such that $\eta \in \mathcal{F}$ is represented by $(\mathbb{R}v^1, \mathbb{R}v^1 \wedge v^2, \dots, \mathbb{R}v^1 \wedge \dots \wedge v^{d-1})$. The Weyl group in the $\mathrm{SL}(d, \mathbb{R})$ case is isomorphic to the permutation group $\mathfrak{S}_d$. It consists in square matrices $(w_{ij}) \in K$ of coefficients $w_{ij} = \delta_{\tau(i), j}$ where $\tau \in \mathfrak{S}_d$. Left multiplication of $(v_i^j)_{1 \leq i, j \leq d}$ by an element of the Weyl group permutes the columns, right multiplication by the transvection matrices in the upper triangular unipotent group N correspond to operations on the lines of $(v_i^j)_{1 \leq i, j \leq d}$. By Gaussian elimination, one can assume that $(v_i^j)_{1 \leq i, j \leq d}$, representative of $w\eta$ for some $w \in \mathcal{W}$, is lower triangular and the coefficient in the diagonal is the highest in norm of the whole column i.e.

$$v_l^j = 0, \text{ for all } l < j, \text{ and } |v_j^j| = \max_{j \leq l \leq d} \{|v_l^j|\} \text{ for all } 1 \leq j \leq d. \quad (55)$$

On one hand, using that $\wedge^j \mathbb{R}^d$ are the Tits representations for $\mathrm{SL}(d, \mathbb{R})$ and $(\zeta_0)_o^\perp = \eta_0$ in (20), we compute

$$\delta(w\eta, \zeta_0) = \inf_{1 \leq j \leq d} d(\mathbb{R}v^1 \wedge \dots \wedge v^j, (\mathbb{R}e_1 \wedge \dots \wedge e_j)^\perp) = \inf_j \frac{|v_1^1 \dots v_j^j|}{\|v^1 \wedge \dots \wedge v^j\|}.$$

On the other hand, $v^1 \wedge \dots \wedge v^j = \sum_{1 \leq l_1 < \dots < l_j \leq d} \sum_{\tau \in \mathfrak{S}_j} \mathrm{sign}(\tau) v_{l_1}^{\tau(1)} \dots v_{l_j}^{\tau(j)} e_{l_1} \wedge \dots \wedge e_{l_j}$ where $\mathrm{sign}(\tau) \in \{\pm 1\}$ is the signature of the permutation τ . Hence for all $1 \leq j \leq d$, by triangle inequality and (55)

$$\|v^1 \wedge \dots \wedge v^j\| \leq \binom{d}{j} j! |v_1^1 \dots v_j^j| \leq d! |v_1^1 \dots v_j^j|.$$

We deduce that $\delta(w\eta, \zeta_0) \geq (d!)^{-1}$. The Lemma then follows by left multiplication by k_ι of $w\eta$ and ζ_0 , which by K -invariance of δ does not change the inequality. $\square$

Lemma 6.5. *For any $g \in G$, there exists $w \in \mathcal{W}$ such that for any $b = wb'w^{-1}$ with $b' \in \exp(-\mathfrak{a}^{++})$, we have*

$$\log a(gb) = \log b' + \log a(g) + v, \quad (56)$$

where v is a vector of bounded length in $\mathfrak{a}$ with the bound only depending on c in Lemma 6.4.

Proof. Since NAK is a Iwasawa decomposition, we can compute the A part by the Iwasawa cocycle. We have

$$\log a(g) = -\sigma(g^{-1}, \eta_o).$$

Then if we multiple on the right of g by an element $b \in A$, we obtain

$$\log a(gb) = -\sigma(b^{-1}g^{-1}, \eta_o) = -\sigma(b^{-1}, g^{-1}\eta_o) + \log a(g).$$

Due to $b = wb'w^{-1}$ with w in the Weyl group and b' in the negative Weyl chamber, then

$$\log a(gb) = -\sigma((b')^{-1}, w^{-1}g^{-1}\eta_o) + \log a(g).$$

By Lemma 6.4, there exists w such that $\delta(w^{-1}g^{-1}\eta_o, \zeta_o) > c$. By Lemma 14.2(i) and Lemma 6.33 in [BQ16], we finish the proof. $\square$

6.1 Compact periodic diagonal orbits

The first difference with the cocompact case is that not every loxodromic element gives a periodic A -orbit in the quotient $\Gamma \backslash G/M$. So Selberg's lemma is not true. There is a general sufficient condition in [PR72]. For $\mathrm{SL}_d(\mathbb{Z})$, we know exactly when it fails. Recall for γ loxodromic, we have defined an A -orbit $F_{[\gamma]}$ on $\Gamma \backslash G/M$.

Lemma 6.6. *Let $\gamma \in \Gamma$ be a loxodromic element. Then for the following conditions:*

- 1 The A -orbit $F_{[\gamma]}$ is compact periodic;
- 2 The characteristic polynomial $p_\gamma(x) = \det(x - \gamma)$ of γ is irreducible on $\mathbb{Q}[x]$;
- 3 There exists no non-trivial subset I of $\{1, \dots, d\}$ such that

$$\sum_{i \in I} t_i = 0,$$

where $(t_1, \dots, t_d)$ is the Jordan projection of γ ;

we have that (1), (2) are equivalent and (3) implies (2).

Remark 6.7. Here we give an example when $d = 4$ that (1), (2) holds but (3) fails. We can find γ in $\mathrm{SL}_4(\mathbb{Z})$ by using the companion matrix such that $p_\gamma(x) = (x^2 + (5 - \sqrt{2})x + 1)(x^2 + (5 + \sqrt{2})x + 1)$. This polynomial $p_\gamma(x)$ is irreducible on $\mathbb{Q}[x]$ and has four different real roots. We can number them by their absolute values as λ_1 to λ_4 . Then its roots satisfy that $\log |\lambda_1| + \log |\lambda_4| = \log |\lambda_2| + \log |\lambda_3| = 0$.

Before proving Lemma 6.6, we need another lemma. Let G_γ be the centralizer of γ in G .

Lemma 6.8. *Let γ be an element in Γ such that its characteristic polynomial p_γ is irreducible. If β is an element in G_γ with all eigenvalues rational, then β is identity or minus identity.*

Proof. The element γ is diagonalizable in the splitting field of p_γ , a Galois extension K of $\mathbb{Q}$. There exists a vector $v_1 \in K^d$ such that $\gamma v_1 = \lambda_1 v_1$ with $\lambda_1 \in K$. Since p_γ is irreducible, the Galois group $\mathrm{Gal}(K/\mathbb{Q})$ acts transitively on the roots of p_γ . We can get eigenvalues $\lambda_2, \dots, \lambda_d$ and eigenvectors $v_2, \dots, v_d$ as Galois conjugates of λ_1 and v_1 with $\gamma v_j = \lambda_j v_j$. The numbers λ_j are distinct, hence $v_1, \dots, v_d$ form a basis.

Due to β commutes with γ , we have

$$\beta v_j = \mu_j v_j$$

for some μ_j rational. Take σ in the Galois group $\mathrm{Gal}(K/\mathbb{Q})$, then $\beta \sigma(v_j) = \mu_j \sigma(v_j)$. The Galois group $\mathrm{Gal}(K/\mathbb{Q})$ acts on the set $\{v_1, \dots, v_d\}$ transitively (p_γ irreducible), which implies that μ_j 's are equal. Since we are in $\mathrm{SL}_d(\mathbb{R})$, we obtain the lemma. $\square$

Proof of Lemma 6.6. We first prove (3) implies (2): If $p_\gamma(x)$ is reducible then $p_\gamma(x) = p_1(x)p_2(x)$ with p_1, p_2 monic and constant terms of p_1, p_2 equaling ± 1 . Suppose the absolute values of roots of p_1 are $\exp(t_i)$ for $i \in I \subset \{1, \dots, d\}$. Then we obtain $\sum_{i \in I} t_i = 0$ with I non-trivial.

Now we prove that (1) is equivalent to (1'), a condition about the centralizer of γ . Let T_γ be the centralizer of γ in $\mathrm{SL}(d, \mathbb{C})$. The A -orbit $F_{[\gamma]}$ can be written as $F_{[\gamma]} = \Gamma gAM$, and $\gamma \in gAMg^{-1}$ due to the definition of $F_{[\gamma]}$. Because γ is loxodromic, the centralizer G_γ of γ in G equals $gAMg^{-1}$, the real points of the maximal $\mathbb{R}$ -split $\mathbb{Q}$ -torus T_γ . Now $F_{[\gamma]}$ is compact in $\Gamma \backslash G/M$ is equivalent to $\Gamma \cap G_\gamma \backslash G_\gamma = \Gamma_\gamma \backslash G_\gamma$ compact, where Γ_γ is the centralizer of γ in Γ . Notice that $\Gamma_\gamma \backslash G_\gamma$ is a finite cover of $(\Gamma_0)_\gamma \backslash G_\gamma = (T_\gamma)_\mathbb{Z} \backslash (T_\gamma)_\mathbb{R}$ for the $\mathbb{Q}$ -torus T_γ . Then due to [BH62, Thm 9.4], (1) is equivalent to

1' T_γ is a $\mathbb{Q}$ -anisotropic $\mathbb{Q}$ -torus.

Then we prove (2) implies (1'). Take a γ satisfying (2). If T_γ is not $\mathbb{Q}$ -anisotropic, then there exists a $\mathbb{Q}$ -split subtorus [BH62, Lem 8.4]. Take β in the $\mathbb{Q}$ -points of this $\mathbb{Q}$ -split torus, then all the eigenvalues of β is rational. Hence by Lemma 6.8, the element β must be $\pm \mathrm{Id}_d$. There is no nontrivial $\mathbb{Q}$ -split subtorus of T_γ . We obtain a contradiction. So T_γ is $\mathbb{Q}$ -anisotropic.

Finally, we prove (1') implies (2). If p_γ is reducible, suppose $\lambda_1, \dots, \lambda_\ell$ with $1 \leq \ell < d$ is an orbit of the Galois group $\mathrm{Gal}(K/\mathbb{Q})$ on the roots of p_γ . Here K is the splitting field of p_γ . Set $v_j \in K^d$ the corresponding eigenvectors of λ_j , which is also an orbit of the Galois group $\mathrm{Gal}(K/\mathbb{Q})$. For any β in the $\mathbb{Q}$ points of T_γ , since λ_j 's are different, we have for $1 \leq j \leq \ell$

$$\beta v_j = \mu_j v_j.$$

On the symmetric power $\mathrm{Sym}^\ell \mathbb{R}^d$, we have

$$\beta v_1 \cdots v_\ell = \mu_1 \cdots \mu_\ell (v_1 \cdots v_\ell).$$

Now the vector $v_1 \cdots v_\ell$ is fixed under the Galois group, so it is rational, hence $\mu_1 \cdots \mu_\ell$ is also rational. We can define a $\mathbb{Q}$ character by $\chi(\beta) = \mu_1 \cdots \mu_\ell$. Due to $1 \leq \ell < d$, this $\mathbb{Q}$ character is non-trivial. So T_γ is not $\mathbb{Q}$ -anisotropic. $\square$

Sparse set of loxodromic elements Let Γ_c^{lox} be the subset of Γ^{lox} whose elements also satisfy the condition (1) or (2) in Lemma 6.6.

Lemma 6.9. *There exists $1 > \kappa_1 > 0$ such that for $t > 1$,*

$$|(\Gamma^{lox} \setminus \Gamma_c^{lox}) \cap D_t| \ll \mathrm{vol}(D_t)^{1-\kappa_1}.$$

Before proving this lemma, we need a result similar to Theorem 1.8 in [GN12b]. The proof is given in the appendix 8.2.

Proposition 6.10. *Let h be a polynomial on $\mathrm{SL}_d(\mathbb{R})$ with $\mathbb{Z}$ coefficients and not vanishing identically on $\mathrm{SL}_d(\mathbb{R})$. Then there exists $\kappa_h > 0$ such that for $t > 1$*

$$|\{\gamma \in \Gamma \cap D_t, h(\gamma) = 0\}| \ll \mathrm{vol}(D_t)^{1-\kappa_h}.$$

Proof of Lemma 6.9. By Lemma 6.6, the number of elements γ 's not satisfying condition (1) is less than that not satisfying condition (3). The condition (3) in Lemma 6.6 can be translate to equations: $h_i(\gamma) := \det(1 - \wedge^i \gamma) \det(1 + \wedge^i \gamma) = 0$. Then by power saving of integer points in subvarieties (Proposition 6.10), we obtain the result. $\square$

As a corollary, we can replace o by another point x in X , similar to Lemma 2.21.

Lemma 6.11. *For $x \in X$ with $d_X(x, o) \leq \frac{\kappa_1 t}{4(1-\kappa_1)}$, we have*

$$|(\Gamma^{lox} \setminus \Gamma_c^{lox}) \cap D_t(x)| \ll \mathrm{vol}(D_t)^{1-\kappa_1/2}.$$

Proof. By Lemma 2.17, we have

$$\|a_o(\gamma) - a_x(\gamma)\| \leq 2d_X(x, o) \leq \frac{\kappa_1 t}{2(1 - \kappa_1)}.$$

Therefore by Lemma 6.9 and $\text{vol}(D_t) \in [1/C, C]e^{\delta_0 t} t^{\frac{\dim A - 1}{2}}$, we obtain

$$\begin{aligned} |\{\gamma \in \Gamma^{lox} - \Gamma_c^{lox}, a_x(\gamma) \in B_a(0, t)\}| &\leq |\{\gamma \in \Gamma^{lox} - \Gamma_c^{lox}, a_o(\gamma) \in B_a(0, t + \frac{\kappa_1 t}{2(1 - \kappa_1)})\}| \\ &\ll \text{vol}(D_{t + \frac{\kappa_1 t}{2(1 - \kappa_1)}})^{1 - \kappa_1} \ll \text{vol}(D_t)^{1 - \kappa_1/2}. \end{aligned}$$

The proof is complete. $\square$

6.2 Equidistribution for compactly supported functions

In order to make $\Gamma \backslash G/M$ a manifold, from now on we only consider $\Gamma = \Gamma_1$. Similar to the cocompact case, we also need to change the formulation to conjugacy classes of loxodromic elements. Let $\mathcal{G}_c^{lox}$ be the set of Γ conjugacy classes of Γ_c^{lox} .

Lemma 6.12. *There is a bijection between $\mathcal{G}_c^{lox}$ and $\mathcal{G}(A)$.*

Proof. The proof is almost the same as the proof of Lemma 4.2. We replace the use of Lemma 4.1 by Lemma 6.6. The only difference is that we obtain γ_Y from (Y, F) , but we only know that γ_Y is in Γ^{lox} . Due to $F = F_{\gamma_Y}$ compact, from Lemma 6.6, we know that indeed γ_Y is in Γ_c^{lox} . $\square$

By the previous lemma, we obtain

$$\mathcal{M}_\Gamma^t := \frac{\text{vol}(\Gamma \backslash G)}{\text{vol}(D_t)} \sum_{F \in C(A)} |\Lambda(F) \cap B_a^{++}(0, t)| L_F = \frac{\text{vol}(\Gamma \backslash G)}{\text{vol}(D_t)} \sum_{\substack{[\gamma] \in \mathcal{G}_c^{lox} \\ \|\lambda(\gamma)\| \leq t}} L_\gamma.$$

We consider the lift of the measure $\mathcal{M}_\Gamma^t$ to G/M ,

$$\mathcal{M}^t := \frac{\text{vol}(\Gamma \backslash G)}{\text{vol}(D_t)} \sum_{\substack{\gamma \in \Gamma_c^{lox} \\ \|\lambda(\gamma)\| \leq t}} \mathcal{L}_\gamma. \quad (57)$$

The main result of this part is the equidistribution on large compact sets.

Proposition 6.13. *There exist $\zeta > 0$ and $u > 0$ such that for all $t > 0$ and all $f \in \text{Lip}_c(\Omega(e^{\zeta t}))$,*

$$\left| \mathcal{M}_\Gamma^t(f) - \int f dm_{\Gamma \backslash G/M} \right| \ll e^{-ut} |f|_{\text{Lip}}. \quad (58)$$

Before starting the argument, we fix the parameters which will be used later. Choose $u_1 > 0$ smaller than $\min\{\epsilon_G, 1\}/10$, where ϵ_G is the constant from Lemma 2.19. Set

$$\varepsilon := e^{-u_1 t} \text{ and } t_1 := 3u_1 t.$$

Consider the decay rate function $u \mapsto \kappa(u) > 0$ satisfying (2.19), the decay coefficient $\kappa > 0$ given in Theorem 2.22 and κ_1 given in Lemma 6.9. We set

$$u_2 := \frac{1}{4 \dim(G/M)} \min \left\{ \delta_0 \kappa(6u_1), \frac{\delta_0 \kappa}{3}, u_1, \delta_0 \kappa_1 \right\}, \quad r := e^{-u_2 t}. \quad (59)$$

Consider the constant C_7 coming from the injectivity radius Lemma 6.3, the constant C_5 from the counting Lemma 2.23 and C_0 coming from the growth rate of C_x given in (31). Set the exponential decay rate of the systole

$$\zeta := \frac{1}{C_7} \min \left\{ u_2, \frac{\kappa_1}{4(1 - \kappa_1)}, \frac{1}{C_5}, \frac{3u_1}{2(1 - 6u_1)}, \frac{\kappa(6u_1)}{4(1 - \kappa(6u_1))}, \frac{\kappa \delta_0}{6C_0}, \frac{u_1}{4C_0} \right\}. \quad (60)$$

Equidistribution of Weyl chambers We define

$$\mathcal{M}_{x,3}^t := \frac{\text{vol}(\Gamma \backslash G)}{\text{vol}(D_t)} \sum_{\gamma \in \Gamma_c^{lox} \cap D_t^{reg}(x)} \mathcal{L}_\gamma. \quad (61)$$

The following Lemma is a direct consequence of the definition of $\mathcal{M}_{x,2}^t$ given in (47).

Lemma 6.14. *For all $t > 1$, all $x \in X$ and for every test function $\tilde{\psi} \in \text{Lip}_c^+(\tilde{\mathcal{F}}^{(2)}(x, r))$,*

$$\left| \int \tilde{\psi} d\mathcal{M}_{x,3}^t - \int \tilde{\psi} d\mathcal{M}_{x,2}^t \right| \leq \frac{|\Gamma^{lox} \setminus \Gamma_c^{lox} \cap D_t(x)| \text{vol}(\Gamma \backslash G)}{\text{vol}(D_t)} \|\tilde{\psi}\|_\infty.$$

The following statement and its proof is the same as Lemma 5.7 provided one replaces $\mathcal{M}_{x,2}^t$ with $\mathcal{M}_{x,3}^t$.

Lemma 6.15. *There exists $C > 0$. Fix $x \in X$, for every test function $\tilde{\psi} \in \text{Lip}_c^+(\tilde{\mathcal{F}}^{(2)}(x, r))$,*

$$(1 - Cr) \int \tilde{\psi} d\mathcal{M}_{x,3}^{t-2r} \leq \int \tilde{\psi} d\mathcal{M}^t \leq (1 + Cr) \int \tilde{\psi} d\mathcal{M}_{x,3}^{t+2r} + \|\tilde{\psi}\|_\infty \frac{|\Gamma \cap D_{t+2r}^{2r}(x)| \text{vol}(\Gamma \backslash G)}{\text{vol}(D_t)}. \quad (62)$$

Partition of unity Then we only need to use a partition of unity to obtain the global version as in Section 5.3. On the compact set $\Omega(e^{\zeta t})$, by Vitali's covering lemma, there exists a finite set $\{y_i\}_{i \in I}$ in $\Omega(e^{\zeta t})$ such that the $B(y_i, r/10)$ are pairwise disjoint and $\cup_{i \in I} B(y_i, r/2)$ covers $\Omega(e^{\zeta t})$. By disjointness, $|I| \ll r^{-\dim(G/M)}$. By the injectivity radius Lemma 6.3 and choice of ζ in (60) such that $C_7\zeta \leq u_2$, the balls $B(y_i, r)$ are diffeomorphic to balls of radius r in G/M . We can take a partition of unity of $\frac{1}{r}$ -Lipschitz functions associated to the open cover $B(y_i, r)$. For each y_i , by Lemma 6.3 we can find a lift $z_i \in G/M$ such that

$$d_X(o, x_i) \leq C_7\zeta t. \quad (63)$$

where $x_i = \pi_X(z_i)$. We have the same Lipschitz bounds on $\tilde{\psi}_i$. By Lemma 5.8 and (p1)

$$\text{supp} \tilde{\psi}_i \subset B(z_i, r) \subset \tilde{\mathcal{F}}^{(2)}(x, r) \text{ and } \text{Lip}_2(\tilde{\psi}_i) \leq C_{x_i} \text{Lip}(\tilde{\psi}_i) \leq \frac{C_{x_i}}{r} |\tilde{\psi}_\Gamma|_{\text{Lip}}.$$

Hence, by the above equation (63), then $\text{Lip}_2(\tilde{\psi}_i) \ll \frac{e^{C_0 C_7 \zeta t}}{r} |\tilde{\psi}_\Gamma|_{\text{Lip}}$.

Local domination By using Lemma 5.6, 6.14 and 6.15, we obtain similar local domination:

$$\begin{aligned} \pm \left(\int \tilde{\psi}_i d\mathcal{M}^t - \int \tilde{\psi}_i d\mathcal{M}_{G/M} \right) &\leq r(C_3 + C) \int \tilde{\psi}_i d\mathcal{M}_{G/M} + \frac{|\Gamma^{lox} \setminus \Gamma_c^{lox} \cap D_t(x_i)| \text{vol}(\Gamma \backslash G)}{\text{vol}(D_t)} \|\tilde{\psi}_i\|_\infty \\ (2r)^{\dim a} \left(E(t \pm 2r, \tilde{\psi}_i, x_i) + 2\varepsilon \text{Lip}_2(\tilde{\psi}_i) \frac{|\Gamma \cap D_{t \pm 2r}(x_i)| \text{vol}(\Gamma \backslash G)}{\text{vol}(D_{t \pm 2r})} + 4\|\tilde{\psi}_i\|_\infty \frac{|\Gamma \cap D_{t \pm 2r}^{t_1}(x_i)| \text{vol}(\Gamma \backslash G)}{\text{vol}(D_{t \pm 2r})} \right). \end{aligned}$$

For the right term in the first line, by choice of ζ in (60) and Lemma 6.11, we deduce

$$\frac{|\Gamma^{lox} \setminus \Gamma_c^{lox} \cap D_t(x_i)| \text{vol}(\Gamma \backslash G)}{\text{vol}(D_t)} \ll (\text{vol}(D_t))^{-\kappa_1/2}.$$

The lower line contains terms similar to (51) (52) (53) that appear in the cocompact case. Now that x_i can be far from o , the constant C_{x_i} can be big. However, by (63), this distance is bounded above by $C_7\zeta t$, which implies the following upper bound of $C_{x_i}^2$:

$$C_{x_i}^2 = (8C_2C_1)^2 e^{2C_0 d_X(x_i, o)} \ll e^{2C_0 C_7 \zeta t}. \quad (64)$$

Hence for the left-lower term, we deduce that

$$E(t \pm 2r, \tilde{\psi}_i, x_i) = O\left(\frac{C_{x_i}^2}{r} \text{vol}(D_t)^{-\kappa} |\tilde{\psi}_\Gamma|_{\text{Lip}}\right) = O\left(\frac{e^{2C_0 C_7 \zeta t}}{r} \text{vol}(D_t)^{-\kappa} |\tilde{\psi}_\Gamma|_{\text{Lip}}\right).$$

For the mid-lower term similar to (52), by Lemma 2.23 and since $C_5 C_7 \zeta < 1$ according to the choice of ζ in (60), we deduce that $\frac{|\Gamma \cap D_{t \pm 2r}(x_i)|}{\text{vol}(D_{t \pm 2r})} \leq C$.

For the right-lower term similar to (53), using that $C_7 \zeta \leq \min\{\frac{3u_1}{2(1-6u_1)}, \frac{\kappa(6u_1)}{4(1-\kappa(6u_1))}\}$ as given in (60), the hypothesis of Lemma 2.21 are satisfied. Hence $\frac{|\Gamma \cap D_{t \pm 2r}^{t_1}(x_i)| \text{vol}(\Gamma \setminus G)}{\text{vol}(D_{t \pm 2r})} \ll \text{vol}(D_t)^{-\kappa(6u_1)}$.

Global domination Finally, by summing over the partition of unity and by $|I| \ll r^{-\dim(G/M)}$, then collecting the above estimates together, we deduce that

$$\begin{aligned} \left| \int \tilde{\psi}_\Gamma d\mathcal{M}_\Gamma^t - \int \tilde{\psi}_\Gamma dm_{\Gamma \setminus G/M} \right| &\ll r \|\tilde{\psi}_\Gamma\|_1 + \frac{|\tilde{\psi}_\Gamma|_{Lip}}{r^{\dim(G/M)}} \text{vol}(D_t)^{-\frac{\kappa_1}{2}} \\ &+ \frac{|\tilde{\psi}_\Gamma|_{Lip}}{r^{\dim(G/M)+1}} \left(e^{2C_0 C_7 \zeta t} \text{vol}(D_t)^{-\kappa} + e^{C_0 C_7 \zeta t} \varepsilon + \text{vol}(D_t)^{-\kappa(6u_1)} \right). \end{aligned}$$

The proof of Proposition 6.13 is complete due to the choices of ζ and $r = e^{-u_2 t}$ in (59) and (60), where $\varepsilon = e^{-u_1 t}$.

6.3 Non-escape of mass

In order to prove the equidistribution for all bounded Lipschitz functions, we only remains to prove that $\mathcal{M}_\Gamma^t(\Omega(e^{\vartheta t})^c)$ tends to zero as t tends to infinity, where we set

$$\vartheta = \zeta/2.$$

Lemma 6.16 (non-escape of mass). *There exists $c_4 > 0$ such that*

$$\mathcal{M}_\Gamma^t(\Omega(e^{\vartheta t})^c) \ll e^{-c_4 t}.$$

Proof the main theorem for $\Gamma < \text{SL}_d(\mathbb{Z})$. Take a Lipschitz cutoff function ϕ supported on $\Omega(e^{\zeta t})$ and equals 1 on $\Omega(e^{\vartheta t})$. Let $f_1 = \phi f$ and $f_2 = (1 - \phi)f$. Then f_1 is supported on $\Omega(e^{\zeta t})$ and f_2 is supported on $\Omega(e^{\vartheta t})^c$ and with the same Lipschitz bound as f . By applying Proposition 6.13 to f_1 and Lemma 6.16, we obtain

$$\begin{aligned} \left| \int f d\mathcal{M}_\Gamma^t - \int f dm_{\Gamma \setminus G/M} \right| &\leq \left| \int f_1 d\mathcal{M}_\Gamma^t - \int f_1 dm_{\Gamma \setminus G/M} \right| + \left| \int f_2 d\mathcal{M}_\Gamma^t - \int f_2 dm_{\Gamma \setminus G/M} \right| \\ &\ll e^{-ut} |f_1|_{Lip} + m_{\Gamma \setminus G/M}((\Omega(e^{\vartheta t})^c) |f_2|_\infty + \mathcal{M}_\Gamma^t((\Omega(e^{\vartheta t})^c) |f_2|_\infty) \ll e^{-u't} |f|_{Lip}, \end{aligned}$$

here we need a volume estimate (see for example Proposition 7.1 in [KM99]), that is

$$m_{\Gamma \setminus G/M}(\Omega(e^{\vartheta t})^c) \ll e^{-ct}.$$

The proof is complete. □

For $0 < t_1 < t_2$ We define

$$\Omega(t_1, t_2) := \{\Gamma g \in \Gamma \setminus G/M, t_1 < 1/s(g) \leq t_2\} = \Omega(t_2) \setminus \Omega(t_1).$$

Let's state our key observation.

Theorem 6.17. *There exists $C > 0$. For $t > C$ and $\gamma \in \Gamma_c^{lox}$ with $\|\lambda(\gamma)\| \leq t$, then*

$$\text{Leb}(F_{[\gamma]} \cap \Omega(e^{\vartheta t})^c) \ll \text{Leb}(F_{[\gamma]} \cap \Omega(e^{\vartheta t/8}, e^{\vartheta t})).$$

From Theorem 6.17 to Lemma 6.16. Take a Lipschitz function f such that f takes value in $[0, 1]$, f equals 1 on $\Omega(e^{\vartheta t/8}, e^{\vartheta t})$, the support of f is contained in the 1 neighbourhood of $\Omega(e^{\vartheta t/8}, e^{\vartheta t})$ and $|Lip(f)| \leq 2$. Then we obtain

$$\mathcal{M}_\Gamma^t(\Omega(e^{\vartheta t})^c) \ll \mathcal{M}_\Gamma^t(\Omega(e^{\vartheta t/8}, e^{\vartheta t})) \leq \mathcal{M}_\Gamma^t(f).$$

For an element in the support of f , we can write it as Γgh with $g \in \Omega(e^{\vartheta t/8}, e^{\vartheta t})$ and $h \in B(e, 1)$. Then due to the region of h we have for $v \in \mathbb{Z}^d$

$$\|vgh\| = \|(vg)h\| \in \|vg\|[1/C', C']$$

for some $C' > 1$. Therefore the 1 neighbourhood of $\Omega(e^{\vartheta t/8}, e^{\vartheta t})$ is contained in $\Omega(e^{\zeta t})$ if t large. Applying Proposition 6.13, we have

$$\left| \mathcal{M}_\Gamma^t(f) - \int f \, dm_{\Gamma \backslash G/M} \right| \ll e^{-ut} |f|_{Lip}.$$

Then by the choice of f , we obtain

$$\mathcal{M}_\Gamma^t(\Omega(e^{\vartheta t})^c) \ll m_{\Gamma \backslash G/M}(\Omega(e^{\vartheta t/8}/C', C' e^{\vartheta t})) + e^{-ut} \ll e^{-ct},$$

here we need a volume estimate $m_{\Gamma \backslash G/M}(\Omega(e^{\vartheta t/8}/C')^c) \ll e^{-ct}$. $\square$

In order to prove Theorem 6.17, we start with a lemma between the systole of $\mathbb{Z}^d g$ for $\Gamma g \in F$ and the length of an element in $\Lambda(F)$. Since F itself is a torus, we can also interpret it as the relation between the systole of the torus F with the cusp excursion of F on $\Gamma \backslash G$.

Lemma 6.18. *There exists $C_d > 0$. For $\gamma \in \Gamma_e^{lox}$ and $F_{[\gamma]}$ a compact periodic A -orbit in $\Gamma \backslash G/M$, we have*

$$F_{[\gamma]} \subset \Omega(\exp(C_d \|\lambda(\gamma)\|)).$$

Remark 6.19. This lemma is inspired by the discriminant of compact A -orbit defined in [ELMV11]. Here we give a direct relation without using the discriminant.

Proof. Take a point $\Gamma gM \in F_{[\gamma]}$, then there exists $a \in AM$ such that $a = g^{-1}\gamma g$. The Jordan projection of a is the same as γ , that is $\lambda(a) = \lambda(\gamma)$.

Take a nonzero vector $x \in \mathbb{Z}^d g$. Then by $\mathbb{Z}^d ga = \mathbb{Z}^d \gamma g = \mathbb{Z}^d g$, we obtain

$$xa \in \mathbb{Z}^d g, \dots, xa^{d-1} \in \mathbb{Z}^d g.$$

Now $x, xa, \dots, xa^{d-1}$ generates a sublattice in $\mathbb{Z}^d g$. There is no j such that $x_j = 0$, otherwise the length of xb for $b \in A$ can be arbitrarily small, which contradicts the fact that $F_{[\gamma]}$ is compact. Hence its covolume satisfies

$$\text{vol}(\mathbb{R}^d / \langle x, xa, \dots, xa^{d-1} \rangle) = \left| \prod_{1 \leq j \leq d} x_j \det(1, a, \dots, a^{d-1}) \right|,$$

where in $\det(1, a, \dots, a^{d-1})$, the element a_j is seen as a column vector. Now different coordinates of a are different, so the determinant of the Vandermonde matrix in the above formula is nonzero. Hence the lattice generated by $x, xa, \dots, xa^{d-1}$ has rank d and its covolume is greater than 1. Hence

$$\frac{1}{|\prod_{1 \leq j \leq d} x_j|} \leq |\det(1, a, \dots, a^{d-1})| \leq \exp(C \|\lambda(a)\|) = \exp(C \|\lambda(\gamma)\|).$$

Therefore by the inequality of arithmetic and geometric means

$$\max_{b \in A} \frac{1}{\|xb\|} \ll \max_{b \in A} \frac{1}{|\prod_{1 \leq j \leq d} (xb)_j|^{1/d}} = \frac{1}{|\prod_{1 \leq j \leq d} x_j|^{1/d}} \leq \exp(C \|\lambda(\gamma)\|/d).$$

Finally, we obtain

$$\min_{b \in A} s(\mathbb{Z}^d gb) \geq \min_{x \in \mathbb{Z}^d g - \{0\}, b \in A} \|xb\| \gg \exp(-C \|\lambda(\gamma)\|/d).$$

$\square$

This lemma tells us that the compact periodic A -orbit appearing in $\mathcal{M}_\Gamma^t$ is always contained in $\Omega(\exp(C_d t))$. In order to prove Theorem 6.17, we need another lemma describing the growth of systole under the A action.

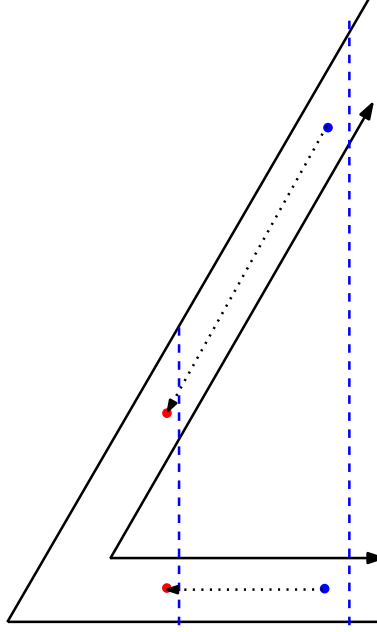


Figure 2: This is a neighbourhood of the positive Weyl chamber. Two blue dashed lines are lines with the fundamental weight $\chi(a) = \log a_1 + \log a_2$ equal to t and ϑt . Two blue points are $\log a(g)$ and two red points are $\log a(g) + \log b'$. The point $\log a(gb)$ has bounded distance to the red point.

Lemma 6.20. *There exist $C, C_8 > 0$. Let y be an element in $\Gamma \backslash G/M$ with $1/s(y) \in [e^{\vartheta t}, \exp(C_d t)]$ and $t > C$, there exists $b \in A$ such that $1/s(yb) \in (e^{\vartheta t/4}, e^{\vartheta t/2})$ and $\|\log b\| \leq C_8 t$.*

Remark 6.21. This lemma is similar to Proposition 4.1 in [TW03], where they also study the growth of systole to prove that there exists a compact set which intersects each orbit of some maximal $\mathbb{R}$ -split torus.

The idea of the proof is that in the Siegel domain, the systole and a cocycle is comparable. Since this cocycle is additive, we can estimate its value after A action, which in turn gives the estimate of systole.

Proof. We only give the proof for Γ_0 , since our definition of Siegel domain only works for Γ_0 . If we have an element y in $\Gamma \backslash G/M$, we can project it to y' in $\Gamma_0 \backslash G/M$ and apply the lemma there to find a b . Then due to the invariance of the systole under covering $s(yb) = s(y'b)$, this b also works for y in the lemma.

For y in $\Gamma_0 \backslash G/M$ with $1/s(y) \in [e^{\vartheta t}, \exp(C_d t)]$, with out loss of generality, we can find a g in a standard Siegel domain $N_{s_0} A_{u_0} K$ such that $y = \Gamma_0 g M$. Then by Lemma 6.2

$$a_d(g) u_0^{d-1} \leq s(g) \leq a_d(g). \quad (65)$$

For $\log a \in \mathfrak{a}$, we define a character

$$\chi(\log a) = \log(a_1 \cdots a_{d-1}) = -\log a_d.$$

By (65), we obtain that $a(g)$ is in

$$\{a \in A \mid \log a_j - \log a_{j+1} \geq \log u_0, 1 \leq j \leq d-1, \chi(\log a) \in [\vartheta t, C_d t - (d-1) \log u_0]\}.$$

Using the affine coordinate, this is equivalent to say that $\log a(g) = v^0 + Y$, with $v^0 \in \mathfrak{a}$, $v_j^0 - v_{j+1}^0 = \log u_0$, $Y \in \mathfrak{a}^{++}$ and

$$\chi(Y) \in [\vartheta t, C_d t - (d-1) \log u_0] - (d \log u_0)/2.$$

Applying (56) to this g , there exists w in $\mathcal{W}$ such that for all b' in the negative Weyl chamber, the equation (56) holds. We can take a $b' = \exp(-sY)$ with $0 < s < 1$ such that

$$\chi(Y - sY) = \vartheta t/3.$$

Then $\|\log b'\| = \|sY\| \leq C_8 t$ if t is large with respect to $\log u_0$. Therefore, We have

$$\log a(gb) = v^0 + (1-s)Y + v = (v^0 + v) + (1-s)Y$$

is in

$$\{a \in A \mid \log a_j - \log a_{j+1} \geq \log u, \ 1 \leq j \leq d-1, \ \chi(\log a) \in [\vartheta t/3 - C, \vartheta t/3 + C]\},$$

with some $1 > u > 0$ and $C > 0$ only depending on c in Lemma 6.4 and u_0 . By Lemma 6.2, we obtain that

$$a_d(gb)u^{d-1} \leq s(gb) \leq a_d(gb).$$

When t is large, we obtain gb with $1/s(gb) \in (e^{\vartheta t/4}, e^{\vartheta t/2})$. $\square$

Proof of Theorem 6.17. The compact periodic A -orbit $F_{[\gamma]}$ is isometric to the flat torus $\mathfrak{a}/\Lambda(F_{[\gamma]})$. We use the quotient Euclidean distance on $F_{[\gamma]}$ and a ball $B(x, r)$ for $x \in F_{[\gamma]}$ and $r > 0$ will be a ball in $F_{[\gamma]}$ with respect to this distance. We are working on the flat torus with a height function given by the inverse of the systole, $1/s(x)$, which tells us how this flat torus is embedded in the large non compact space $\Gamma \backslash G/M$.

We can find a maximal family of points $\{x_j\}_{j \in J} \subset F_{[\gamma]}$ in $\Omega(e^{\vartheta t/4}, e^{\vartheta t/2})$ such that $d(x_j, x_{j'}) \geq C_8 t$ for $j \neq j'$ and the union of balls $\cup_{j \in J} B(x_j, 2C_8 t)$ covers $F_{[\gamma]} \cap \Omega(e^{\vartheta t})^c$. This is always possible, because if the union of balls doesn't cover $F_{[\gamma]} \cap \Omega(e^{\vartheta t})^c$. Then take a point x not covered by the union, so x has distance greater than $2C_8 t$ to $\{x_j\}_{j \in J}$. Using Lemma 6.18 and 6.20, we can find xb with $xb \in \Omega(e^{\vartheta t/4}, e^{\vartheta t/2})$ and $d(x, xb) \leq \|\log b\| \leq C_8 t$. This point xb has distance greater than $C_8 t$ to $\{x_j\}_{j \in J}$. So we can add this point. By this way we can find the desired family of points.

The Lebesgue measure is the quotient measure on the flat torus $F_{[\gamma]} \simeq \mathfrak{a}/\Lambda(F_{[\gamma]})$. Then due to covering,

$$\text{Leb}(F_{[\gamma]} \cap \Omega(e^{\vartheta t})^c) \leq \sum_{j \in J} \text{Leb}(B(x_j, 2C_8 t)) \ll \sum_{j \in J} \text{Leb}(B(x_j, \vartheta t/8)). \quad (66)$$

The last inequality $\text{Leb}(B(x_j, 2C_8 t)) \ll \text{Leb}(B(x_j, \vartheta t/8))$ is due to that we can use a finite number of balls of radius $\vartheta t/8$ to cover a ball of radius $2C_8 t$ in the flat torus and the number of balls needed doesn't depend on t . For any $b \in A$ and $v \in \mathbb{R}^d$, by $\|vb\| \in \|v\| \{\min_i |b_i|, \max_i |b_i|\}$, we obtain that for any $x \in \Gamma \backslash G/M$

$$s(xb)/s(x) \in [e^{-\|\log b\|}, e^{\|\log b\|}].$$

For any $y \in B(x_j, \vartheta t/8)$, we can write it as $y = x_j b$ with $b \in A$ and $\|\log b\| \leq \vartheta t/8$. Therefore we know that $B(x_j, \vartheta t/8) \subset \Omega(e^{\vartheta t/8}, e^{\vartheta t})$. The balls $B(x_j, \vartheta t/8)$ are disjoint by hypothesis that $d(x_j, x_{j'}) \geq C_8 t$. Therefore

$$\sum_{j \in J} \text{Leb}(B(x_j, \vartheta t/8)) = \text{Leb}(\cup_{j \in J} B(x_j, \vartheta t/8)) \leq \text{Leb}(F_{[\gamma]} \cap \Omega(e^{\vartheta t/8}, e^{\vartheta t})). \quad (67)$$

The proof is complete by (66) and (67). $\square$

7 Appendix A: Orbifolds and partitions of unity

Let Γ be any finite index subgroup of $\text{SL}_d(\mathbb{Z})$. Its left action on G/M is no longer assumed to be free and the space $\Gamma \backslash G/M$ is now an orbifold. In Section 6, we used in the partition of unity argument that $\Gamma \backslash G/M$ is a manifold. Our main purpose is now to find a covering compatible with the orbifold structure (Lemma 7.6). The partition of unity constructed from this covering will allow us to complete the proof in this case.

We start with the definition of orbifolds, following [Thu97, Chap 13]. An orbifold O consists of the underlying Hausdorff space X_O and an orbifold atlas $\{(U_i, \tilde{U}_i, \Gamma_i, \varphi_i)\}_{i \in I}$ such that different atlas' should be compatible and where

- each U_i is an open set of X_O and their union covers X_O ,
- the family of sets $\{U_i\}_{i \in I}$ is closed under finite intersection,
- each set $\tilde{U}_i$ is an open subset of $\mathbb{R}^n$,

- Γ_i is a finite group of linear transformations which fixes $\tilde{U}_i$,
- the map φ_i is a homeomorphism from U_i to the quotient $\tilde{U}_i/\Gamma_i$.

Remark 7.1. If d is even, let p be the projection from $\mathrm{SL}_d(\mathbb{R})$ to $\mathrm{PSL}_d(\mathbb{R}) = \mathrm{SL}_d(\mathbb{R})/\{\pm id\}$. Then due to $-id \in M$, as double cosets, the space $\Gamma \backslash \mathrm{SL}_d(\mathbb{R})/M$ is isomorphic to $p(\Gamma) \backslash \mathrm{PSL}_d(\mathbb{R})/p(M)$ and periodic compact A -orbits are also isomorphic under the projection p . It is sufficient to prove equidistribution on $p(\Gamma) \backslash \mathrm{PSL}_d(\mathbb{R})/p(M)$. We make the convention that if d is even, then we consider $G = \mathrm{PSL}_d(\mathbb{R})$.

By [Thu97, Prop 13.2.1], since the finite group M acts properly discontinuous on $\Gamma \backslash G$, the space $O := \Gamma \backslash G/M$ is an orbifold. Since the metric d_1 on G is left G invariant and right K invariant, we use the quotient metric on $\Gamma \backslash G/M$ denoted by d . For this orbifold, its singular locus is defined as

$$\Sigma(O) = \{x \in O \mid M_x \neq \{id\}\},$$

where the isotropy group M_x is the stabilizer of the group action M at a lift of x . Since M is abelian, the isotropy group M_x is independent of the choice of the lift of x . This singular locus is the union of closed subvarieties of O , which has zero measure (Newman's theorem, see [Dre69] for a proof). If we are outside this singular locus, we are in the normal covering situation. Similarly, we have another orbifold structure on $\Gamma \backslash G/M$ coming from the action of Γ on G/M . The singular locus of the Γ action is the same as the singular locus of M action due to the structure of the double coset. We use the same notation $\Sigma(O)$.

Since the group M is much simpler, we will first find a good covering for the action of M on $\Gamma \backslash G$, then pass to Γ on G/M .

Lemma 7.2. *Let $F \in C(A)$ be a compact periodic A -orbit in O , then $F \cap \Sigma(O) = \emptyset$.*

Proof. Let $x = \Gamma g M$ be a point in F , choose a lift Γg in $\Gamma \backslash G$. By definition, we have

$$M_x := \{m \in M, \Gamma g m = \Gamma g\}.$$

For every $m \in M_x$, there exists $\gamma \in \Gamma$ such that $gm = \gamma g$, i.e. $\gamma = gm g^{-1}$. Using that $F \in C(A)$, we choose for any period $Y \in \Lambda(F) \cap \mathfrak{a}^{++}$ an element $\gamma_Y \in \Gamma_c^{lox}$ such that $\gamma_Y = g \exp(Y) m_Y g^{-1}$. Now since M is abelian, $\gamma \in G_{\gamma_Y}$ the centralizer of γ_Y in G and all its eigenvalues are rational. By Lemma 6.8, we deduce that $m = \pm e$. If d is even, then $-e \in M$ and we use the convention in Remark 7.1 to deduce that M_x is trivial for every $x \in F$. $\square$

To construct a covering of balls of radius $\asymp r$, which is compatible with the orbifold structure, we study the right action of the discrete abelian group M on the manifold $\Gamma \backslash G$. The argument used to prove Lemma 6.3 gives a lower bound for the injectivity radius of $\Omega(R)$, seen as a compact subset of $\Gamma \backslash G$.

Lemma 7.3. *There exists $C_9 > 1$ such that for all $R > 2$ and every $z \in \Omega(R) \subset \Gamma \backslash G$,*

$$inj(z) \geq R^{-C_9}.$$

Furthermore, there exists a representative $h \in G$ such that $z = \Gamma h$ and

$$d_X(o, ho) \leq C_9 \log R, \text{ and } \sup(\|h\|, \|h^{-1}\|) \leq R^{C_9}.$$

We start with a general lemma about the action of $M \simeq (\mathbb{Z}/2\mathbb{Z})^{d-1}$ on any manifold. For any point x in the manifold, we denote by $M_x := \mathrm{Stab}_M(x)$.

Lemma 7.4. *Let N be a complete Riemannian manifold such that M act on N isometrically. For any point $y \in N$, if there exists $m \in M \setminus M_y$ and $s \in (0, inj(y)/5)$ such that $B(y, s)m \cap B(y, s) \neq \emptyset$, then there exists $z \in B(y, s)$ such that $M_z \supset \langle M_y, m \rangle$.*

Proof. Consider $y \in N$, an element $m \in M \setminus M_y$ and $s > 0$ as in the statement. Since $B(y, s)m \cap B(y, s) \neq \emptyset$, we deduce that $d(y, ym) < 2s$. By choice of s , the exponential at $T_y N$ identifies $B(0, 5s)$ with $B(y, 5s)$, hence in this local chart, $B(y, 2s) \subset B(y, 5s)$ identifies with some ball of radius $2s$ in $B(0, 5s)$. Now m is an isometry of order 2 of N , hence $(ym, y)m = (y, ym)$. The action of m preserves geodesics on N , hence in the local chart induced by the exponential at $T_y N$, the element m reads as an isometry that preserves the

geodesic segment $[y, ym]$ and flips the endpoints. Therefore, the middle point z of this geodesic segment is fixed by m i.e. $m \in M_z$. Furthermore, $z \in B(y, s)$ since $d(y, z) = \frac{1}{2}d(y, ym) < s$.

Similarly, M_y acts isometrically in particular on the geodesic segment $[y, ym]$. Since M is abelian, the action of M_y fixes the endpoints. Due to the hypothesis of injectivity radius, it fixes the whole geodesic segment. In particular $M_z \supset M_y$. $\square$

Since the group structure is simple, we use the action of M on $\Gamma \backslash G$ to obtain a covering that is compatible with this action. Denote by $C_{10} := 11^{d-1}$, which only depends on d .

Lemma 7.5. *For all $R > 2$ and $s \in (0, R^{-C_9}/C_{10})$, there exists a covering of $\Omega(R) \subset O$ consisting of balls $\{B(x_i, \frac{s_i}{10})\}_{i \in I}$ such that $s_i \in [s, C_{10}s]$ and each larger ball $B_i := B(x_i, s_i)$ is compatible with the orbifold structure. Meaning that for each ball B_i , there exists $(\tilde{B}_i, M_i) \subset \Gamma \backslash G \times M$ where M_i is a subgroup of M , such that $\tilde{B}_i$ is M_i invariant and B_i is homeomorphic to $\tilde{B}_i/M_i$.*

Proof. In $\Gamma \backslash G$, the injectivity radius can also be defined as: for $y = \Gamma g \in \Gamma \backslash G$,

$$inj(y) = \sup_{r_0 > r > 0} \{ \text{for any } \gamma \neq e, \gamma B(g, r) \cap B(g, r) = \emptyset \},$$

where r_0 is the injectivity radius of the group G . For all $x \in \Omega(R)$, by the lower bound for the injectivity radius at x given in Lemma 6.3 and the above characterisation of injectivity radius,

$$\text{for every } y \in B(x, C_{10}s/2), \text{ } inj(y) \geq C_{10}s/2. \quad (68)$$

Fix a point $x \in \Omega(R) \subset \Gamma \backslash G/M$ and $s \in (0, R^{-C_9}/C_{10})$. We define a family of radii $r_j = 11^{j-1} \times 10s \in [s, C_{10}s]$ for $1 \leq j \leq d-1$ and $r_0 = s$. We construct, by induction, a compatible ball containing x . Denote by $y_0 \in \Gamma \backslash G$ a lift of $x = y_0M$. If $B(x, s)M \cap \Sigma(O) = \emptyset$, i.e. $B(y_0, r_0)m \cap B(y_0, r_0) = \emptyset$ for all $m \in M$, we add the ball $B(x, r_0/10)$ to the covering of O , with $B(y_0, r_0) \in \Gamma \backslash G$ and $M_0 := M_{y_0}$ is the trivial group of M .

Assume we have constructed for some $0 \leq k \leq d-1$ a family of points $y_0, \dots, y_k \in \Gamma \backslash G$, a strictly increasing family of sign subgroups $M_0 := M_{y_0} \subset \dots \subset M_k := M_{y_k}$, such that

$$d(y_j, y_0) \leq r_j/10, \text{ for every } j = 0 \dots, k-1. \quad (69)$$

Assume $B(y_k, r_k)m \cap B(y_k, r_k/10) = \emptyset$ for all $m \in M \setminus M_k$ and we add the ball $B(y_kM, r_k)$ to the covering of O , with $B(y_k, r_k) \in \Gamma \backslash G$ and isotopy group M_k . Otherwise, due to (68), we can apply the previous Lemma 7.4 to y_k and $r_k > 0$. We find $y_{k+1} \in B(y_k, r_k)$ with $M_{k+1} = M_{y_{k+1}}$ strictly containing M_k . By hypothesis,

$$d(y_{k+1}, y_0) \leq d(y_{k+1}, y_k) + d(y_k, y_0) \leq r_k + r_k/10 = r_{k+1}/10.$$

By this way, we construct y_{k+1} and M_{k+1} satisfying the hypothesis of induction.

Since M is finite, we must stop at some k , which means that $B(y_k, r_k)m \cap B(y_k, r_k) = \emptyset$ for all $m \in M \setminus M_k$, and we can add this ball to the covering.

We can do this for any $x \in \Omega(R)$, and the proof is complete. $\square$

Now we use the orbifold structure from the action of Γ on G/M . We use the double coset relation to do this step.

Lemma 7.6. *For all $R > 2$ and $s \in (0, R^{-C_9}/2C_{10})$, there exists a covering of $\Omega(R) \subset O$ consisting of balls $\{B(x_i, \frac{s_i}{10})\}_{i \in I}$ such that $s_i \in [s, C_{10}s]$ and each larger ball $B_i := B(x_i, s_i)$ is compatible with the orbifold structure. That is for any $B = B_i$, there exists $(\tilde{B}, \Gamma_B) \subset G/M \times \Gamma$, where Γ_B is a finite subgroup of Γ , such that $\tilde{B}$ is Γ_B invariant and B homeomorphic to $\Gamma_B \backslash \tilde{B}$.*

Proof. Once we have a ball $B(w, r) = B(\Gamma g, r)$ and M_w by Lemma 7.5. Since this ball intersects $\Omega(R)$ and s small, we have $\Gamma g \in \Omega(2R)$. By Lemma 7.3, we can take this g such that $\|g\|, \|g^{-1}\| \leq (2R)^{C_9}$. For each $m \in M_w$, there exists γ_m such that $\gamma_m g = gm$. Let Γ_g be the group generated by γ_m . Through the conjugate action g , the group Γ_g is isomorphic to M_w . Consider the pair $B(gM, r)$ and Γ_g .

If there exists $\gamma \notin \Gamma_g$ such that $\gamma B(gM, r) \cap B(gM, r) \neq \emptyset$, then there exist $g_1, g_2 \in B(g, r)$ such that $\gamma g_1 M = g_2 M \in B(gM, r)$. We must have $\gamma g_1 = g_2 m$. But this means $B(\Gamma g, r) \cap B(\Gamma g, r)m \neq \emptyset$. So m is in M_w and $B(\Gamma g, r)m = B(\Gamma g, r)$. But now, by similar computation as in Lemma 6.3

$$\|\gamma_m - \gamma\| = \|gm g^{-1} - g_2 m g_1^{-1}\| = \|(g - g_2)m g^{-1} + g_2 m(g^{-1} - g_1^{-1})\| \leq 2r\|g\|\|g^{-1}\| \leq 2C_{10}sR^{2C_9} < 1$$

is small. By discreteness, we have $\gamma = \gamma_m$, contradicts to $\gamma \notin \Gamma_m$.

Therefore for any $\gamma \notin \Gamma_g$, we have $\gamma B(gM, r) \cap B(gM, r) = \emptyset$ and $\gamma \in \Gamma_g$ preserves $B(gM, r)$, due to γ preserving the metric. $\square$

Once we have a family of balls $B(x_i, s_i/10)$ as in Lemma 7.6 which covers $\Omega(R)$, by Vitali's covering theorem for metric spaces, we can find a subcollection $B(y_j, s_j/10)$ which are disjoint and the union of larger balls $B(y_j, s_j/2)$ covers the union of $B(x_i, s_i/10)$. In particular, the union of $B(y_j, s_j/2)$ covers $\Omega(R)$ and $B(y_j, s_j)$ is compatible with orbifold structure.

Lemma 7.7. *There exists a constant $C > 0$. With the same assumption of s, R as in Lemma 7.6 There exists a partition of unity $\{\rho_j\}$ subordinated to the open cover $\{B(y_j, s_j)\}$ of $\Omega(R)$ with $s_j \in [s, C_{10}s]$ such that for any $x, y \in B(y_j, s_j) \cap \Omega(R)$, we have*

$$|\rho_j(x) - \rho_j(y)|/d(x, y) \leq \frac{C}{s}.$$

The number of balls $B(y_j, s_j)$ is less than $Cs^{-\dim G}$.

The construction of a partition of unity subordinated to a covering is classic. We add the proof for completeness.

Proof. On each ball $B(y_j, s_j)$ we take the function

$$\tilde{\rho}_j(x) = \max\{0, 1 - 3d(x, B(y_j, s_j/2))/s_j\},$$

which takes value 1 on the ball $B(y_j, s_j/2)$ and vanish outside of $B(y_j, s_j)$ with Lipschitz norm bounded by $3/s$. Let $\rho_j = \tilde{\rho}_j / \sum_j \tilde{\rho}_j$. This is a partition of unity with respect the covering $B(y_j, s_j)$. Let us compute their Lipschitz norms, for x, y in $B(y_j, s_j) \cap \Omega(R)$

$$\begin{aligned} |\rho_j(x) - \rho_j(y)|/d(x, y) &= \left| \frac{\tilde{\rho}_j(x) - \tilde{\rho}_j(y)}{\sum \tilde{\rho}_l(x)} - \tilde{\rho}_j(y) \frac{\sum_l \tilde{\rho}_l(x) - \tilde{\rho}_l(y)}{(\sum \tilde{\rho}_l(x))(\sum \tilde{\rho}_l(y))} \right|/d(x, y) \\ &\leq \frac{3}{s} \frac{1}{\sum \tilde{\rho}_l(x)} + \frac{3\#\{y_l, x \text{ or } y \in B(y_l, s_l)\}}{s} \frac{1}{(\sum \tilde{\rho}_l(x))(\sum \tilde{\rho}_l(y))}. \end{aligned}$$

Due to $x, y \in \Omega(R) \subset \cup_l B(y_l, s_l/2)$, we obtain that $\sum_l \tilde{\rho}_l(x), \sum_l \tilde{\rho}_l(y) \geq 1$. Since different y_l 's have distance at least $s/10$ to each other, by homogeneity of the space, the number of y_l such $x \in B(y_l, s_l)$ is uniformly bounded. Therefore we obtain the lemma. $\square$

For any ψ_Γ Lipschitz function supported on $\Omega(R)$, let $\psi_j = \psi_\Gamma \rho_j$. Then $\psi_\Gamma = \sum \psi_j$. For each ψ_j , by Lemma 7.7, we obtain $|\psi_j|_{Lip} \ll 1/s |\psi_\Gamma|_{Lip}$. Take its lift $\tilde{\psi}_j$ on $\tilde{B}_j$. Then since singular locus has zero measure, and outside the singular locus $\Sigma(O)$ it is a regular covering and $B_j \simeq \tilde{B}_j/\Gamma_{B_j}$, we obtain

$$\int \psi_j dm_{\Gamma \backslash G/M} = \frac{1}{|\Gamma_B|} \int \tilde{\psi}_j dm_{G/M}. \quad (70)$$

Due to $F \in C(A)$ not intersecting singular locus (Lemma 7.2), $\mathcal{M}^t$ being Γ invariant and $B_j \simeq \tilde{B}_j/\Gamma_{B_j}$, we obtain

$$\int \psi_j d\mathcal{M}_\Gamma^t = \frac{1}{|\Gamma_B|} \int \tilde{\psi}_j d\mathcal{M}^t. \quad (71)$$

Proof of Theorem 1.3 for general $\Gamma < \text{SL}_d(\mathbb{Z})$. With (70), (71) and Lemma 7.3, we can redo the argument as in Section 6.2 to obtain the equidistribution for compactly supported functions on the orbifold $\Gamma \backslash G/M$. Then we use this equidistribution result to do the same argument as in Section 6.3 to finish the proof. $\square$

We also need a version of Lemma 6.12 in this case

Lemma 7.8. *There is a bijection between $\mathcal{G}_c^{lox}$ and $\mathcal{G}(A)$.*

Proof. The proof is almost the same as the proof of Lemma 4.2 and 6.12. The main difference is that we may have two different γ_Y, γ'_Y satisfying

$$\gamma_Y g = g e^Y m_Y, \quad \gamma'_Y g = g e^Y m'_Y.$$

Take $\gamma = \gamma_Y^{-1} \gamma'_Y$, then γ commutes with γ_Y and all its eigenvalues are ± 1 . By Lemma 6.8, we know that $\gamma = \pm id$. By convention in Remark 7.1, we obtain that γ is trivial. $\square$

Remark 7.9. The same construction of the covering and the partition of unity for orbifolds $\Gamma \backslash G/M$ also works if G is $\mathbb{R}$ -split, in which case the group M is isomorphic to $\{\pm 1\}^r$ for some $r \in \mathbb{N}$. But we need also prove Lemma 7.2 and Lemma 7.8, which relies on Lemma 6.8. If we have a similar version of Lemma 6.8 for a cocompact irreducible lattice Γ in a $\mathbb{R}$ -split group G , then Theorem 1.3 also works without the torsion-free assumption.

8 Appendix B

8.1 Proof of Theorem 2.22

We give a proof of Theorem 2.22 by redoing the proof of Theorem 7.1 in [GN12a] for Lipschitz functions. Here we have one notation issue, the quotient Γ is on the right G/Γ to be consistent with [GN12a]. Fix notation m_G and $dm_{G/\Gamma} = dm_G/V(\Gamma)$, which is a probability measure.

Recall the quantitative mean ergodic theorem on $L^2(G/\Gamma)$, which is the main engine to obtain equidistribution. For an absolutely continuous probability measure β on G , let $\pi(\beta)f = \int \pi(g)f d\beta(g)$. By Theorem 4.5 in [GN12a], we have

$$\left\| \pi(\beta)f - \int f \right\|_2 \leq C_q \|\beta\|_q^{1/n(G,\Gamma)} \|f\|_2, \quad (72)$$

where $n(G, \Gamma)$ is an integer depending on G, Γ and q is any constant in $[1, 2)$ such that $\|\beta\|_q < \infty$.

Let

$$\epsilon_{inj} > 0$$

be a constant such that if $\epsilon < \epsilon_{inj}$, then the map $\mathcal{O}_\epsilon$ to $\mathcal{O}_\epsilon \Gamma$ is injective from G to G/Γ .

We will prove this version

Theorem 8.1. *Let G be a connected, real linear, semisimple Lie group of non-compact type. Let $\Gamma < G$ be an irreducible lattice. There exists $C_6 > 0$ only depending on $n(G, \Gamma)$ and G . Let $x \in X$ and $(B_t)_{t>0}$ be D_t^{++} . Then for all Lipschitz test functions $\psi \in Lip(\mathcal{F}^{(2)})$, there exists $E(t, \psi) = O(Lip(\psi) \text{vol}(D_t)^{-\kappa})$ when $t > C_6 |\log \epsilon_{inj}|$ such that*

$$\frac{1}{\text{vol}(B_t)} \sum_{\gamma \in B_t \cap \Gamma} \psi(\gamma_o^+, \gamma_o^-) = \frac{1}{\text{vol}(\Gamma \backslash G)} \int_{\mathcal{F} \times \mathcal{F}} \psi d\mu_o \otimes \mu_o + E(t, \psi),$$

where all the implied constants $\ll$ only depending on G and $n(G, \Gamma)$.

Proof of Theorem 2.22. Due to (26) $\gamma_x^+ = h_x(h_x^{-1}\gamma h_x)_o^+$, we apply Theorem 8.1 to the lattice $h_x^{-1}\Gamma h_x$ and the Lipschitz function $\psi'(\cdot, \cdot) := \psi(h_{x\cdot}, h_{x\cdot})$. This is the reason that we need a uniformed version for lattices $h_x^{-1}\Gamma h_x$ and we made dependence of constants in Theorem 8.1 more transparent. The constant $n(G, h_x^{-1}\Gamma h_x)$ is the same as $n(G, \Gamma)$ due to invariance of the Haar measure. For ϵ_{inj} of $h_x^{-1}\Gamma h_x$, we have

$$\inf_{\gamma \in \Gamma - \{e\}} d_G(o, h_x^{-1}\gamma h_x) \geq e^{-Cd_X(o, x)} \inf_{\gamma \in \Gamma - \{e\}} d_G(o, \gamma).$$

By Lemma 2.11, the action of h_x on $\mathcal{F}$ is C_x Lipschitz. From these, we obtain Theorem 2.22. $\square$

Step 1: The first step is to transfer the counting problem to integrals, which can be treated by the mean ergodic theorem.

Let $\mathcal{O}_\epsilon$ be a neighborhood of identity in G with radius ϵ . Let $\tilde{A}^\delta = \{\exp(a), a \in \mathfrak{a}^{++}, d(a, \partial\mathfrak{a}^{++}) \geq \delta\}$.

Lemma 8.2 (Effective Cartan decomposition, Proposition 7.3 in [GN12a], first appeared in [GOS10]). *There exist $\delta > 0$ and $l_0, \epsilon_1 > 0$. If $\epsilon < \epsilon_1$, then for $g = k_1 a k_2 \in K \tilde{A}^\delta K$, we have*

$$\mathcal{O}_\epsilon g \mathcal{O}_\epsilon \subset (\mathcal{O}_{l_0\epsilon} \cap K) k_1 M (\mathcal{O}_{l_0\epsilon} \cap A) a k_2 (\mathcal{O}_{l_0\epsilon} K).$$

For ease of notation, when there is no confusion, we will use k_1, a, k_2 to denote elements come from the Cartan decomposition $g = k_1 a k_2$. Let $\tilde{D}_t^\delta = \{g \in G, a \in \tilde{A}^\delta, d(gK, K) \leq t\}$. Notice that by identifying $\mathcal{F}$ with K/M , we have $k_1 M = \gamma_o^+$ and $k_2^{-1} M = \gamma_o^-$. Let

$$\rho_t(g) = \mathbb{1}_{\tilde{D}_t^\delta}(a) \phi(k_1, k_2),$$

where $\phi(k_1, k_2) = \psi(k_1 M, k_2^{-1} M) = \psi(g_o^+, g_o^-)$.

We introduce two auxiliary functions, which is the replacement of Lipschitz well-roundness of sets in [GN12a]. Recall

$$\text{Lip } \phi = \max \left\{ |\phi|_\infty, \sup_{x \neq y} \frac{|\phi(x) - \phi(y)|}{d(x, y)} \right\}.$$

Let

$$\begin{aligned} \rho_{t,\epsilon}^+(g) &= \mathbb{1}_{\tilde{D}_{t+2\epsilon}^{\delta-l_0\epsilon}}(g) (\phi(k_1, k_2) + (\text{Lip } \phi) l_0 \epsilon) \\ \rho_{t,\epsilon}^-(g) &= \mathbb{1}_{\tilde{D}_{t-2\epsilon}^{\delta+l_0\epsilon}}(g) \max\{\phi(k_1, k_2) - (\text{Lip } \phi) l_0 \epsilon, 0\}. \end{aligned}$$

From the definition, we know $\rho_{t,\epsilon}^- \leq \rho_t \leq \rho_{t,\epsilon}^+$.

Lemma 8.3. *For $g \in \mathcal{O}_\epsilon \gamma \mathcal{O}_\epsilon$ with $\epsilon \leq \epsilon_1$ we obtain*

$$\rho_{t,\epsilon}^-(g) \leq \rho_t(\gamma) \leq \rho_{t,\epsilon}^+(g). \quad (73)$$

Proof. If $\rho_{t,\epsilon}^-(g) \neq 0$, then $g \in \tilde{D}_{t-2\epsilon}^{\delta+l_0\epsilon}$. By $\gamma \in \mathcal{O}_\epsilon g \mathcal{O}_\epsilon$ and Lemma 8.2, we obtain

$$a(\gamma) \in (\mathcal{O}_{l_0\epsilon} \cap A) a(g) \cap D_t \subset \tilde{D}_t^\delta.$$

So $\mathbb{1}_{\tilde{D}_t^\delta}(a(\gamma)) = 1$. By Lemma 8.2 and Lipschitz property of ϕ , we obtain

$$\phi(k_1(\gamma), k_2(\gamma)) \geq \phi(k_1(g), k_2(g)) - (\text{Lip } \phi) l_0 \epsilon.$$

This proves the left hand side. For the other side, the proof is similar. $\square$

Take $\mathbb{1}_\epsilon = \frac{1}{m_G(\mathcal{O}_\epsilon)} \mathbb{1}_{\mathcal{O}_\epsilon}$ be the normalized characteristic function of $\mathcal{O}_\epsilon$. Let $\varphi_\epsilon(g\Gamma) = \sum_{\gamma \in \Gamma} \mathbb{1}_\epsilon(g\gamma)$. The counting is connected to integral by the following.

Lemma 8.4. *For h in $\mathcal{O}_\epsilon$ with $\epsilon \leq \epsilon_1$, we have*

$$\int \varphi_\epsilon(g^{-1} h \Gamma) \rho_{t,\epsilon}^-(g) dm_G(g) \leq \sum_{\gamma \in \Gamma} \rho_t(\gamma) \leq \int \varphi_\epsilon(g^{-1} h \Gamma) \rho_{t,\epsilon}^+(g) dm_G(g). \quad (74)$$

Proof. By using (73), the proof is almost the same as Lemma 2.1 in [GN12a]. $\square$

Step 2: This step will estimate the error terms in the mean ergodic theorem.

We want to apply the mean ergodic theorem to probability measures $\frac{\rho_{t,\epsilon}^\pm}{\int \rho_{t,\epsilon}^\pm}$. Before doing so, we need to compute some integrals. The computation is a bit tedious. **This step is to verify similar stable mean ergodic theorems, the main consequence is (76) and (78).**

Let's first compute the difference.

Lemma 8.5. *We have*

$$\int \rho_{t,\epsilon}^+ dm_G - \int \rho_{t,\epsilon}^- dm_G \ll \left(\epsilon \int \phi + l_0 \epsilon (Lip \phi) \right) m_G(D_t). \quad (75)$$

Proof.

$$\begin{aligned} & \int \rho_{t,\epsilon}^+ dm_G - \int \rho_{t,\epsilon}^- dm_G \\ & \leq m_G(\tilde{D}_{t+2\epsilon}^{\delta-l_0\epsilon}) \left(\int \phi + l_0 \epsilon (Lip \phi) \right) - m_G(\tilde{D}_{t-2\epsilon}^{\delta+l_0\epsilon}) \left(\int \phi - l_0 \epsilon (Lip \phi) \right) \\ & = \left(m_G(\tilde{D}_{t+2\epsilon}^{\delta-l_0\epsilon}) - m_G(\tilde{D}_{t-2\epsilon}^{\delta+l_0\epsilon}) \right) \int \phi + l_0 \epsilon (Lip \phi) \left(m_G(\tilde{D}_{t+2\epsilon}^{\delta-l_0\epsilon}) + m_G(\tilde{D}_{t-2\epsilon}^{\delta+l_0\epsilon}) \right) \\ & \ll \left(\epsilon \int \phi + l_0 \epsilon (Lip \phi) \right) m_G(D_t), \end{aligned}$$

where the last inequality is from the proof of Proposition 7.4 in [GN12a] about volume estimate and Lemma 2.18. $\square$

Let $\tilde{\rho}_{t,\epsilon}^\pm = \rho_{t,\epsilon}^\pm / \int \rho_{t,\epsilon}^\pm$.

Lemma 8.6. *There exists $t_1 > 0$ which only depends on G, δ, ϵ_1 such that the following holds. For $\epsilon < \min\{\int \phi / 2l_0 Lip \phi, \epsilon_1\}$, $t > t_1$ and $f \in L^2(G/\Gamma)$*

$$\|\pi(\tilde{\rho}_{t,\epsilon}^-)f - \int f\|_2 \leq E(t)\|f\|_2, \quad (76)$$

with

$$E(t) = \left(\frac{C}{m_G(D_t)^{q-1}} \frac{(Lip \phi)^q}{(\int \phi)^q} \right)^\kappa, \quad (77)$$

$\kappa = 1/qn(G, \Gamma)$ and $C > 0$ only depending on G .

For $\epsilon \leq \epsilon_1$, $t > t_1$ and $f \in L^2(G/\Gamma)$

$$\|\pi(\tilde{\rho}_{t,\epsilon}^+)f - \int f\|_2 \leq E(t)\|f\|_2. \quad (78)$$

The main difference is that for $\rho_{t,\epsilon}^+$, we don't need an extra condition of ϵ depending on ϕ .

Proof. We compute the integral of $\rho_{t,\epsilon}^-$. We have

$$\int \rho_{t,\epsilon}^- dm_G \geq m_G(\tilde{D}_{t-2\epsilon}^{\delta+l_0\epsilon}) \left(\int \phi - (Lip \phi) l_0 \epsilon \right).$$

Due to Lemma 2.19, we have $m_G(D_{t-2\epsilon}^{\delta+l_0\epsilon}) = O(m_G(D_{t-2\epsilon})^{1-\zeta_1})$ for some $\zeta_1 > 0$, and Lemma 2.18, we obtain

$$m_G(\tilde{D}_{t-2\epsilon}^{\delta+l_0\epsilon}) \geq e^{-C\epsilon} (1 - C m_G(D_t)^{-\zeta_1}) m_G(D_t).$$

Hence if $t > t_1$ depending on C, δ, ϵ_1 , then

$$\int \rho_{t,\epsilon}^- dm_G \gg m_G(D_t) \left(\int \phi - (Lip \phi) l_0 \epsilon \right).$$

Therefore if $\epsilon \leq \int \phi / 2l_0 (Lip \phi), \epsilon_1$, we obtain

$$\int \rho_{t,\epsilon}^- dm_G \gg m_G(D_t) \int \phi. \quad (79)$$

Similarly for the integral of $\rho_{t,\epsilon}^+$, we obtain if $t > t_1$ and $\epsilon \leq \epsilon_1$, then by Lemma 2.19

$$m_G(\tilde{D}_{t+2\epsilon}^{\delta-l_0\epsilon}) \geq m_G(\tilde{D}_t^\delta) \geq m_G(D_t) - Cm_G(D_t)^{1-\zeta_1}. \quad (80)$$

Therefore

$$\int \rho_{t,\epsilon}^+ dm_G \geq \int \rho_t dm_G \geq (m_G(D_t) - Cm_G(D_t)^{1-\zeta_1}) \int \phi \gg m_G(D_t) \int \phi. \quad (81)$$

After these preparation, we can start to compute the integral appears in error term of mean ergodic theorem. By (79), we obtain when $t > t_1$ and $\epsilon \leq \int \phi/2l_0(Lip\phi)$

$$\|\rho_{t,\epsilon}^-\|_q^q / (\int \rho_{t,\epsilon}^-)^q \ll \int |\rho_t|^q / (m_G(D_t) \int \phi)^q \leq \frac{1}{m_G(D_t)^{q-1}} (Lip\phi)^q / (\int \phi)^q.$$

For $\rho_{t,\epsilon}^+$, by (81) and (80) we have

$$\|\rho_{t,\epsilon}^+\|_q^q / (\int \rho_{t,\epsilon}^+)^q \ll \frac{1}{m_G(D_t)^{q-1}} \int (\phi + (Lip\phi)l_0\epsilon)^q / (\int \phi)^q,$$

We obtain if $t > t_1$,

$$\|\rho_{t,\epsilon}^+\|_q^q / (\int \rho_{t,\epsilon}^+)^q \ll \frac{1}{m_G(D_t)^{q-1}} (Lip\phi)^q / (\int \phi)^q. \quad (82)$$

Applying the above formulas for $\tilde{\rho}_{t,\epsilon}^\pm$, combined with mean ergodic estimate (72), we obtain the lemma. $\square$

Step 3: The mean ergodic theorem only gives an estimate of L^2 norm, but what we need is an estimate at some points. So we need to use the Chebyshev inequality. The remaining work is to collect the error terms. This part is similar to the proof of Theorem 1.9 in [GN12a].

Proof of Theorem 2.22. Applying (78) to $f = \varphi_\epsilon$, by Chebyshev's inequality, we obtain for any $\eta > 0$

$$m_{G/\Gamma}\{h|\pi(\tilde{\rho}_{t,\epsilon}^+)(\varphi_\epsilon)(h\Gamma) - \int \varphi_\epsilon| > \eta\} \leq (\frac{E(t)\|\varphi_\epsilon\|_{L^2}}{\eta})^2. \quad (83)$$

If $(E(t)\|\varphi_\epsilon\|_{L^2}/\eta)^2 < m_{G/\Gamma}(\mathcal{O}_\epsilon)/2 = m_G(\mathcal{O}_\epsilon)/2V(\Gamma)$, (here we need $\|\varphi_\epsilon\|_{L^2(G/\Gamma)}^2 = m_G(\mathcal{O}_\epsilon)/V(\Gamma)$.) we will take $\eta = \frac{2E(t)}{m_G(\mathcal{O}_\epsilon)}$, there exists $h \in \mathcal{O}_\epsilon$, such that

$$\pi(\tilde{\rho}_{t,\epsilon}^+)(\varphi_\epsilon)(h\Gamma) < \eta + \int \varphi_\epsilon.$$

Then by Lemma 8.4,

$$\begin{aligned} \sum_{\gamma \in \tilde{D}_t^\delta} \phi(k_1(\gamma), k_2(\gamma)) &\leq \pi(\tilde{\rho}_{t,\epsilon}^+)(\varphi_\epsilon)(h\Gamma) \int \rho_{t,\epsilon}^+ \leq (\eta + \frac{1}{V(\Gamma)}) \int \rho_{t,\epsilon}^+ \\ &= \frac{\int \rho_t}{V(\Gamma)} (1 + \eta V(\Gamma)) + O(\epsilon(Lip\phi)m_G(D_t)), \end{aligned}$$

where the last inequality is due to (75). Therefore

$$\frac{\sum_{\gamma \in \tilde{D}_t^\delta} \phi(k_1(\gamma), k_2(\gamma))}{\int \rho_t} - \frac{1}{V(\Gamma)} \leq \frac{E(t)}{2m_G(\mathcal{O}_\epsilon)} + \epsilon \frac{Lip\phi}{\int \phi} \frac{1}{V(\Gamma)} \ll \frac{E(t)}{\epsilon^\rho} + \epsilon \frac{Lip\phi}{\int \phi},$$

where ρ is the dimension of group G . We also have

$$|\int \rho_t - m_G(D_t) \int \phi| = \int \phi |m_G(\tilde{D}_t^\delta) - m_G(D_t)| \leq \delta m_G(D_t)^{1-\zeta_1} \int \phi,$$

and the trivial bound

$$|\Gamma \cap D_t^\delta| \leq \frac{m_G(\mathcal{O}_{\epsilon_{inj}} D_t^\delta)}{m_G(\mathcal{O}_{\epsilon_{inj}})} \ll m_G(D_{t+\epsilon_{inj}}^{\delta+l_0\epsilon_{inj}}) \epsilon_{inj}^{-dim G} \ll m_G(D_t)^{1-\zeta_1} \epsilon_{inj}^{-dim G},$$

therefore

$$\begin{aligned} & \sum_{\gamma \cap D_t} \phi(k_1(\gamma), k_2(\gamma)) - \frac{m_G(D_t)}{V(\Gamma)} \int \phi \\ & \ll m_G(D_t) (Lip\phi m_G(D_t)^{-\zeta_1} \epsilon_{inj}^{-dim G} + \int \phi \left(m_G(D_t)^{-\zeta_1} + \frac{E(t)}{\epsilon^\rho} + \epsilon \frac{Lip\phi}{\int \phi} \right)). \end{aligned} \quad (84)$$

We can take C_6 large enough such that the term $m_G(D_t)^{-\zeta_1} \epsilon_{inj}^{-dim G}$ is exponentially small on t .

In order to optimize the error term, we take

$$\epsilon = (E(t) \int \phi / Lip\phi)^{1/(1+\rho)},$$

then the error term in the above formula is

$$E(t)^{1/(1+\rho)} \left(\frac{Lip\phi}{\int \phi} \right)^{\rho/(1+\rho)} \ll m_G(D_t)^{-\zeta} \left(\frac{Lip\phi}{\int \phi} \right)^{(\rho+q\kappa)/(1+\rho)} \leq m_G(D_t)^{-\zeta} \left(\frac{Lip\phi}{\int \phi} \right),$$

where the last equality is due to (77) and $q\kappa = 1/n(G, \Gamma) \leq 1$, and where $\zeta = (q-1)\kappa/(1+\rho)$. Here ϵ should be less than $\epsilon_1, \epsilon_{inj}$, but

$$\epsilon \leq \left(\frac{C}{m_G(D_t)^{(q-1)\kappa}} \frac{\int \phi}{Lip\phi} \right)^{1/(1+\rho)} \leq \left(\frac{C}{m_G(D_t)^{(q-1)\kappa}} \right)^{1/(1+\rho)}. \quad (85)$$

The condition on ϵ is satisfied if t is greater than some $t_2 > 0$ and $C_6 |\log \epsilon_{inj}|$. Therefore by (84), we obtain one part of Theorem 2.22 for $t > t_0 = \max\{t_1, t_2\}$, with t_0 not depending on ϕ .

For $\rho_{t,\epsilon}^-$, we can obtain the same bound with extra condition that $\epsilon < \int \phi / 2l_0 Lip\phi$, that is if t is large. Otherwise, we have $\epsilon \geq \int \phi / 2l_0 Lip\phi$, by (85), which implies

$$Lip\phi \gg m_G(D_t)^{\zeta_2} \int \phi, \quad (86)$$

with $\zeta_2 = (q-1)\kappa/\rho$. Therefore by non-negativeness of ϕ

$$\frac{m_G(D_t)}{V(\Gamma)} \left(\int \phi - C m_G(D_t)^{-\zeta_2} Lip\phi \right) \leq 0 \leq \sum_{\gamma \cap D_t} \phi(k_1(\gamma), k_2(\gamma)).$$

By taking $\min\{\zeta, \zeta_2\}$, the proof is complete. $\square$

Remark 8.7. We need to check the condition in Theorem 4.5 in [GN12a]. For real linear algebraic semisimple Lie groups, we don't need that the group is simply connected. This condition is only needed for p -adic case if we look into the proof of Theorem 4.5. Then the crucial condition is that the representation of G on $L_0^2(G/\Gamma)$ is L^{p+} . In [Oh02], an explicit estimate on p is given for some cases. If G is a connected real linear algebraic semisimple Lie group and Γ is an irreducible lattice, then the condition should be true.

In Kelmer-Sarnak [KS09], they explained this for $G = G_1 \times \cdots \times G_r$ with each G_j simple Lie groups and centre free. There are two steps for proving this. If $r = 1$ and the real rank of G is 1, then the spectral gap is true. Otherwise, due to Margulis superrigidity theorem, Γ is commensurate to a congruence lattice, for which we strong spectral gap (deep result from number theory), that is, each simple factor G_j has a spectral gap on $L_0^2(G/\Gamma)$. Once we have the spectral gap, we can compute the matrix coefficients for each irreducible subrepresentation, which will be in L^p for some bounded p , due to the works of many people. From congruence lattice to commensurable lattice, need the Lemma 3.1 of Kleinbock-Margulis [KM99]. We still need to prove $L_0^2(G/\Gamma)$ is L^{p+} from the property of its subrepresentations.

In [KM99], Theorem 3.4, they use a strong spectral gap to obtain an estimate of matrix coefficients for smooth vectors, which implies that $L_0^2(G/\Gamma)$ is L^p for some p . The idea is that if we know an irreducible representation is L^p , then Howe's work tells us the matrix coefficients decay exponentially, and the constants only depend on the group and p for K -finite vectors. Then since we know a strong spectral gap, we can obtain

each irreducible subrepresentation of $L_0^2(G/\Gamma)$ is strong L^p for some finite p , the Howe's work gives a uniform estimate of matrix coefficients of all the irreducible representations. We can obtain a matrix coefficients estimate of $L_0^2(G/\Gamma)$ from its irreducible subrepresentations.

For specialists, they know well. But for us, the step from subrepresentations in L^p to $L_0^2(G/\Gamma)$ in L^p is highly nontrivial.

Now the condition in Corollary 3.5 in [KM99] is that G is a connected algebraic semisimple Lie group centre free without compact factor and Γ is an irreducible lattice. If we can remove centre-free, then we are satisfied.

If we have another group G_1 with non trivial center. Then we consider $G := G_1/Z_1 = \pi(G_1)$, which is centre-free, where Z_1 the center of G_1 . Let Γ_1 be an irreducible lattice of G_1 . Let $\Gamma_2 = \Gamma_1 Z_1$ and $\Gamma = \pi(\Gamma_2)$. Then $G_1/\Gamma_2 \simeq G/\Gamma$ and $L_0^2(G_1/\Gamma_2) \simeq L_0^2(G/\Gamma)$. Now for each simple factor of G , by Theorem 1.12 in [KM99], it has no almost invariant vector on $L_0^2(G/\Gamma)$. So for simple factors of G_1 , it will also have no almost invariant vector on $L_0^2(G_1/\Gamma_2)$. Since Γ_1 is a finite index subgroup of Γ_2 , by Lemma 3.1 in [KM99], for each simple factor of G_1 , it has also no almost invariant vector in $L_0^2(G_1/\Gamma_1)$. Therefore, we can use Theorem 3.4 in [KM99] to deduce the desired version.

Remark 8.8. Theorem 2.22 is exactly Theorem 7.2 in [GN12a] with an explicit error term, where no proof of Theorem 7.2 is given. But we cannot obtain this Theorem directly from Theorem 7.1 for Lipschitz well-rounded sets in [GN12a] by approximating Lipschitz functions by level sets because the level sets of a Lipschitz function may not be uniformly Lipschitz well rounded. For one-dimensional cases, (i.e. $SL_2(R)$, Lipschitz function on $SO(2)$), we can take a Lipschitz function ϕ as the distance to a Cantor set. Then the level sets $\{\phi < 1/n\}$ approximate the Cantor set. Each set is Lipschitz well-rounded, but the constant in Lipschitz well-rounded blow up as n tends to infinity because the number of intervals in $\{\phi < 1/n\}$ goes to infinite.

8.2 Integer points on subvarieties

For the proof of Proposition 6.10, we need Corollary 1.11 from [GN12a].

Lemma 8.9. *Let $\Gamma(p) = \{\gamma \in \Gamma, \gamma \equiv Id \pmod{p}\}$ for prime p . There exists $\epsilon > 0$ such that for all primes $p, \gamma \in \Gamma$ and $t > 1$, we have*

$$|\{\gamma\Gamma(p) \cap D_t\}| = \frac{\text{vol}(D_t)}{[\Gamma : \Gamma(p)] \text{vol}(\Gamma \backslash G)} + O(\text{vol}(D_t)e^{-\epsilon t}).$$

Proof. Recall Lipschitz well-roundness in [GN12a, Definiton 1.1]: there exist $C > 0, \epsilon_1 > 0$ and $t_1 > 0$ such that for $0 < \epsilon < \epsilon_1$ and $t > t_1$, we have

$$\text{vol}(\mathcal{O}_\epsilon D_t \mathcal{O}_\epsilon) \leq (1 + C\epsilon) \text{vol}(\cap_{u,v \in \mathcal{O}_\epsilon} u D_t v),$$

where $\mathcal{O}_\epsilon$ is the ball $B(e, \epsilon)$ in G . This is true for D_t . Because of Lemma 2.18 we have

$$\text{vol}(\mathcal{O}_\epsilon D_t \mathcal{O}_\epsilon) \leq \text{vol}(D_{t+\epsilon}) \leq (1 + C\epsilon) \text{vol}(D_{t-\epsilon}) \leq (1 + C\epsilon) \text{vol}(\cap_{u,v \in \mathcal{O}_\epsilon} u D_t v).$$

We can use Theorem 4.5 [GN12a] and Lipschitz well-roundness of D_t to verify conditions in Corollary 1.11 [GN12a]. Then Corollary 1.11 [GN12a] implies the result. $\square$

Proof of Proposition 6.10. If we replace D_t by the ball $B_t = \{\gamma \in \text{SL}_d(\mathbb{Z}), |tr(\gamma^t \gamma)| < e^t\}$, this proposition is Theorem 1.8 in [GN12b]. Since there is no detailed proof of Theorem 1.8 in [GN12b], we give a proof for D_t for completeness. The idea of proof is similar, the main difference is in the estimate of the number of fibres.

Let p be a prime number to be chosen later depending on t . Let π_p be the map from $\text{SL}_d(\mathbb{Z})$ to $\text{SL}_d(\mathbb{Z}/p\mathbb{Z})$. Then $\{\gamma \in \text{SL}_d(\mathbb{Z}) \cap D_t, h(\gamma) = 0\}$ is a subset in the preimage $\pi_p^{-1}\{\gamma \in \text{SL}_d(\mathbb{Z}/p\mathbb{Z}), h(\gamma) = 0\}$. This can be seen as a fibre space, each fibre is given by $\gamma\Gamma(p)$ for some $\gamma \in \Gamma$ with $h(\pi_p(\gamma)) = 0$. We only need to estimate the number of fibres and the size of each fibre.

For the size of each fibre, Lemma 8.9 gives us an asymptotic.

For the number of fibre, we have

$$|\{\gamma \in \text{SL}_d(\mathbb{Z}/p\mathbb{Z}), h(\gamma) = 0\}| \ll p^{\dim-1}, \quad (87)$$

if h does not vanish on $\mathrm{SL}_d(\mathbb{Z}/p\mathbb{Z})$, which is true if p is greater than the coefficients of h . Here $\dim$ is the dimension of SL_d . This bound can be obtained from [LW54, Lemma 1] or Lemma 1 [Tao]. The constant in the upper bound only depends on the degree of h and d .

Therefore, by Lemma 8.9 and (87), we obtain

$$\begin{aligned} |\{\gamma \in \mathrm{SL}_d(\mathbb{Z}) \cap D_t, h(\gamma) = 0\}| &\ll p^{\dim-1} \left(\frac{\mathrm{vol}(D_t)}{[\Gamma : \Gamma(p)] \mathrm{vol}(\Gamma \backslash G)} + O(\mathrm{vol}(D_t)e^{-\epsilon t}) \right) \\ &\leq \mathrm{vol}(D_t) \left(\frac{1}{p \mathrm{vol}(\Gamma \backslash G)} + O(p^{\dim-1}e^{-\epsilon t}) \right). \end{aligned}$$

By the Bertrand–Chebyshev Theorem, there is always a prime p in the interval $(n, 2n)$ with $n > 1$. So we can take a prime p of size $e^{\epsilon t/\dim}$. The proof is complete. $\square$

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